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Chapter 1

Introduction

1.1 From sound law to derivation

Historical linguists ordinarily test an etymology by carrying a reconstructed form through the sound changes that separate it from its alleged reflex. Most such derivations remain implicit. A scholar knows that Proto-Germanic **p* yields Old English *f*, that West Germanic **z* became *r* under the appropriate conditions, and that one change must have preceded another because the reverse order produces the wrong form. For a single word this mental arithmetic presents little difficulty. Across hundreds of words and scores of interacting changes it becomes treacherous. Each step may look familiar while the derivation as a whole is false.

In computer-assisted phonological reconstruction (CAPR), I represent each sound law as a finite-state transducer and compose the transducers in the order argued for below. The resulting cascade applies every change without discretion: the same rule, with the same environment, reaches every eligible form. A proposed reconstruction either yields the Old English comparator or it does not. Moving a rule may repair one derivation and destroy another. Both results deserve attention, for a failure locates the point at which phonology ceases to suffice and philology must decide among analogy, borrowing, dialect mixture, uncertain attestation, or a faulty reconstruction.

No new principle is involved. The Neogrammarians already demanded exceptionless sound laws and an account of apparent exceptions (?). Their formula was the *Ausnahmslosigkeit der Lautgesetze*. Formalization merely enforces that demand mechanically: every eligible form bears the stated consequences. The computer cannot decide which reconstruction is historically defensible, but it prevents the investigator from forgetting what that reconstruction entails.

I apply this method to the development of Proto-Germanic and early West Germanic forms into Old English. Germanic makes a severe test. Its historical grammar rests on two centuries of philological labor, while Old English offers abundant but orthographically and dialectally varied testimony (?; ?; ?; ?). A computational account cannot plead scarcity of evidence. It must reproduce familiar developments, identify the evidence for their order, and explain why its input sometimes differs from the headword printed in an etymological dictionary.

1.2 The formal claim

A sound change defines a relation between strings. For each etymology I supply a reconstructed input to an ordered series of such relations and compare the result with an Old English form. Kaplan and Kay demonstrated that the familiar rewrite rules of phonology admit a finite-state interpretation (?); Foma gives that interpretation executable form (?). A toy rule deleting final **z* may be written:

```
define ToyFinalZLoss [{*z} -> 0 || _ .#.];
```

The notation states a target, an output, and an environment. Actual rules also refer to segment classes, stress, syllable weight, word boundaries, and the output of earlier changes. A loose prose formulation can conceal choices that code must resolve. Does the rule affect every vowel or only short vowels? Does a following consonant block it? Does it apply before or after the loss of a final syllable? A transducer forces each question into the open.

Order gives the cascade its historical content. An early change may create the environment for a later one and thus feed it; alternatively, it may remove that environment and bleed it. A word affected by both changes can therefore establish their relative chronology. Other changes commute across the available lexicon: reversing them alters no checked output. Such a result does not prove simultaneity or historical indifference. It shows only that the present evidence fails to order them. Throughout this book I distinguish chronology compelled by the tested forms from placement adopted on the authority of the historical grammars.

Backward reconstruction requires a further restriction. An unrestricted inverse transducer will propose formally possible strings that no Germanic language could have inherited. I therefore restrict backward reconstruction with an inventory and a statement of permissible ancestral forms. Reconstruction always combines correspondences with a theory of what could count as a word in the *Grundsprache*; this restriction states that theory rather than leaving it tacit.

1.3 Inputs, targets, and success

I compare a selected earlier Germanic form with a selected Old English target. Neither selection is innocent. Dictionaries cite lexemes, but sound change operates on word-forms. The ancestor of an Old English plural, preterite, or oblique case may differ from the reconstructed lemma in precisely the material on which a later sound law acts. Kroonen and Orel provide indispensable lexical reconstructions, while the grammars often supply the paradigm history needed to choose the actual input (?; ?; ?).

I therefore distinguish the citation reconstruction from the transducer input. The former identifies the etymon; the latter represents the paradigm cell or remodeled stem whose history is at issue. Where the two differ, the lexical entry states the difference and argues for it. This prevents a convenient input from masquerading as a received reconstruction. An unmotivated alteration made solely to secure the desired output would empty the exercise of historical meaning.

The distinction is concrete in Old English *sculdrum* ‘shoulders, dative plural’. Its inflectional history requires an input of the type **skuldramiz* ‘shoulders, reconstructed dative-plural input’; the singular headword represents a different paradigm cell.

The target also requires judgment. Old English spelling varies by date, dialect, manuscript, and editorial practice. The cascade may produce an internal phonological symbol that an orthographic transducer then maps to a normalized written form. A string match at this final stage cannot by itself establish an etymology; conversely, a superficial spelling mismatch need not disprove one. I treat phonological development and orthographic normalization as separate operations so that exact computation does not confuse notation with history.

Within these limits a successful derivation has three senses. It succeeds formally when the output string matches the target. It succeeds philologically when the chosen input and comparator are the proper forms to compare. It succeeds historically when the proposed path agrees with the wider Germanic evidence. The first kind of success is cheap. The argument of the book concerns the conjunction of all three.

1.4 The evidence of failure

Regular sound change makes irregularity legible. If an inherited form refuses to pass through an otherwise successful cascade, the mismatch demands a name. Analogy may have replaced the expected reflex with a form drawn from another paradigm cell. Borrowing may have introduced the word after the relevant changes. A dialectal form may lie outside the modeled West Saxon

path. The target may be late, corrupt, or normalized beyond what the manuscript evidence warrants. Finally, either the reconstruction or the rule may be wrong.

These possibilities should not be suppressed by narrow, lexeme-specific “sound laws.” A system that derives every target by multiplying exceptions has only encoded its answers. I instead distinguish several classes of non-regular result: attested variants, early and late analogy, reconstructed Old English comparators, known but unmodeled remodeling, and unexplained exceptions. The categories are claims, not housekeeping labels. They identify where regular phonology ends and what additional history the evidence requires.

This treatment follows the original Burmish CAPR work, in which resistant forms often disclosed loans or mistaken cognate assignments. Old English shifts the balance toward morphology and analogy, but the methodological advantage remains the same. Failure concentrates inquiry. It tells us which assumption—input, target, environment, order, or lexical history—must bear the explanation.

1.5 Evidence for relative chronology

The chronology chapters combine three kinds of evidence. First come the statements of the standard historical grammars. These establish the received description and often the broad order of developments (?; ?; ?; ?). Second come individual witness words. A derivation that succeeds under one order and fails under the reverse order supplies direct lexical evidence for that relation. Third come exhaustive order tests across the active dataset. These reveal whether an apparently decisive relation is local, whether other words contradict it, and how far a rule can move without disturbing any output.

Suppose that a consonant change creates the environment for a later vowel change. Under the received order both apply and the Old English target emerges. Reverse them and the vowel change misses its environment. The word then supports the priority of the consonant change. The converse case is equally informative: if an early rule creates a segment that a later rule would wrongly alter, the creating rule must follow the other. When every checked form remains unchanged under reversal, the lexical evidence leaves the order open even if philological considerations still favor one placement.

A note on notation: a superscript dagger placed immediately before an italicized form marks a concrete predicted output generated by a counterfactual manipulation of the rule order (for example, moving a change earlier or later than argued). This dagger indicates that the marked form is a model prediction, not an independently attested historical form; reconstructions remain starred in the usual way.

Sims-Williams argues for mechanizing precisely this kind of reasoning (?). Computation does not replace the historical argument; it makes the extent of that argument measurable. A traditional chronology may prove correct but less tightly constrained than its customary presentation suggests. A relation described as local may in fact rest only on a broad terminus. Such negative results are salutary. They separate what the data demonstrate from what a convenient exposition merely presupposes.

1.6 Rules and words

The book accordingly moves twice through the same history. Part I begins with the ordered rules. For each development it states the historical problem, gives the rule in formal notation, and examines the evidence for placement. The code appears because it is part of the claim, but the code-name is never an explanation. OEIUmlaut, for example, is only a rule name; its linguistic content lies in the stated environment, the historical discussion, and the words whose derivations depend upon it.

Part II begins with the words. Each entry identifies the reconstruction, the selected input, the Old English comparator, and the derivational class. A trace shows which rules altered the form; the accompanying prose explains the philological decisions that the trace cannot make. Short entries

record ordinary derivations. Longer ones treat doubtful reconstructions, variant attestations, paradigm-cell selection, or analogy. The asymmetry is intentional: equal database rows do not warrant equal historical discussion.

The reader can thus move in either direction. A chronology chapter names the lexical witnesses that constrain a rule; their entries display the complete derivations. A lexical entry invokes a change; the corresponding chapter explains its formulation and place in the cascade. The index verborum provides a third route through the material, gathering reconstructed and attested forms by language.

1.7 Reproducibility and disagreement

An executable derivation identifies the exact point of disagreement. One reader may accept a sound change but reject its environment; another may accept the rule and dispute its order; a third may object to the selected paradigm cell or to the normalization of the Old English target. Each objection addresses a recorded decision. Given the same inputs, rules, and order, the stated outputs follow. Reproducibility here concerns those consequences, not the surrender of philological judgment to an algorithm.

Prose can sound settled while concealing several incompatible derivations. Code is less accommodating. A rule that is too broad damages words it was never designed to explain; one that is too narrow leaves legitimate reflexes untouched. A reconstruction plausible in isolation may fail after three later changes. The resulting embarrassment is a virtue of the method. Applying every rule to the whole lexicon subjects local analyses to a global test.

The converse danger is false precision. A deterministic cascade may tempt the reader to mistake exact strings for exact history. Every result still depends on choices about segmentation, symbol inventory, reconstruction, morphology, dialect, chronology, and orthography. I therefore cite the philological sources, display the selected forms, preserve unresolved exceptions, and distinguish tested order from source-based order. Those choices remain accountable to philology.

1.8 The argument

A book must advance an argument. Mine is that traditional rule-based reconstruction becomes clearer when every proposed derivation can be executed, every rule must face the whole lexicon, and every failure is reported. The Germanic-to-Old-English case shows both the power and the boundary of that claim. Much of the history admits a coherent ordered account. The residue does not disappear: it resolves into morphology, analogy, variation, borrowing, imperfect attestation, and a small number of genuine problems.

The sound-change chapters ask how tightly the lexical evidence constrains the cascade. The lexical chapters ask whether individual etymologies survive that cascade without philological sleight of hand. Neither inquiry suffices alone. A formal system that reaches the targets by hiding doubtful inputs is bad historical linguistics; elegant prose that never reproduces its derivations is an incomplete formal account. Executable derivations make those two standards answer to one another.

Part I

Sound changes, formalization, and relative chronology

Chapter 2

The ordered sound-change sequence

2.1 Scope and orientation

The sequence begins with early West Germanic consonant and vowel changes and ends with Old English r-metathesis.

Rhotacism, brightening, breaking, umlaut, and apocope alternate with narrowly conditioned changes whose relative order rests on particular witness words.

The evidence ranges from broadly attested sound laws to lexical constraints that establish only one chronological boundary.

Numbering note

The sequence follows the established rule numbering.

SC038, SC062, and SC084 mark technical or prosodic stages rather than sound changes; SC077 is unused.

2.2 West Germanic rhotacism

Historical discussion

Hogg states that Germanic *z yielded *r in intervocalic position in Old English, while final *z was generally lost (?). Ringe and Taylor argue that this merger of *z with *r was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (?). Crist likewise places rhotacism after earlier West Germanic *z-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (?; ?).

The label SC003 PGmcRhotacism is historically misleading: the change is a later West Germanic rhotacism, not a Proto-Germanic one. It is also distinct from SC020 PGmcFinalZDeletion, which removes final *z before the surviving medial consonant becomes *r.

SC003. West Germanic rhotacism (PGmcRhotacism)

```
define PGmcRhotacism [  
  {*z} -> {*r} || EnglishStarVocalic _ ?  
];
```


Breaking supplies the decisive upper boundary. If rhotacism is delayed until after SC044 OEBreaking, PGmc **líznojanq* ‘learn’ yields [†]*lirnian* rather than expected OE *liornian* ‘learn’, PGmc **líznoθi* ‘learns’ yields [†]*lirnaþ* rather than expected *liornaþ* ‘learns’, PGmc **lízno* ‘learn’ yields [†]*lirna* rather than expected *liorna* ‘learn’, and PGmc **mízdai* ‘meed’ yields [†]*merde* rather than expected OE *meorde* ‘meed’. Moving rhotacism earlier within the tested range changes none of the checked forms.

The lexical evidence thus supplies a terminus ante quem but no terminus post quem. Its placement after the earlier loss of final **z* rests on the historical analyses cited above.

2.3 Proto-West-Germanic ai-monophthongization

Historical discussion

Ringe and Taylor treat the reduction of unstressed **ai* as one of the major early vowel shifts shared across the Northwest Germanic area (?).

The historical support is strongest for unstressed **ai*, especially word-finally. The rule extends the change to nonfinal **ai* > **ā*, a generalization stated more sharply than in the current handbook discussion. Both developments have inherited **ai* as their input.

SC004. Proto-West-Germanic ai-monophthongization (PWGmcAiMonophthongization)

```
define PWGmcAiMonophthongization [
  [{*ai} -> {*ē} || _ .#. ]
  .O.
  [{*ai} -> {*ā}]
  .O.
  [{*ái} -> {*ā}]
];
```

The soul form fixes the relation to interstress raising. If monophthongization is delayed until after that change, PGmc **sáiwalo* ‘soul’ yields [†]*sāwel* rather than expected OE *sāwol* ‘soul’. No earlier placement changes a checked output.

This witness proves that monophthongization preceded interstress raising; it says nothing about the date of the wider nonfinal **ai* > **ā* generalization. The word-final merger with long mid **ē* belongs among the early Northwest Germanic vowel shifts; the broader chronology remains less certain.

2.4 Unstressed **a*-raising before final **m*

Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (?, §373). Fulk likewise includes the development of early unstressed **o* to *u* before *m* among the similarities shared by North and West Germanic (?, §5.2).

I restrict the change to unstressed vowels in inflectional material because the strongest evidence concerns noninitial unstressed material before final **m*. Final **m* conditions the raising.

SC005. Unstressed **a*-raising before final **m* (NWGmcAToUBeforeM)

```
define NWGmcAToUBeforeM [
  {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
  ↪ ({*z})? .#.
];
```

Here the witness word and the comparative evidence serve different purposes. If raising is delayed until after SC017 NWGmcULowering, PGmc **skúldramiz* ‘shoulders’ yields *†scoldrum* rather than expected OE *sculdrum* ‘shoulders’; earlier placements converge on the expected output. The shoulder family therefore tests the chronology, while the inflectional endings justify restricting the change to noninitial unstressed material before **m*.

The evidence is confined to inflectional material.

2.5 Early i-apocope

Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geogub* ‘youth’ (?; ?). Campbell’s *dugup* ‘troop’ and *geogup* ‘youth’ examples belong to the same pattern (?).

The ending vowel disappears in a weak suffixal environment early enough to block later umlaut. This anti-umlaut chronology distinguishes the change from later final-vowel losses.

SC006. Early i-apocope (PWGmcEarlyIApocope)

```
define PWGmcEarlyIApocope [
  {*i} -> Ø || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
  ↪ PGmcStarConsonant+ _ .#.,
  {*i} -> Ø || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
  ↪ PGmcStarConsonant+ _ {*z} .#.
];
```

The absence of umlaut in *geogub* ‘youth’ provides the historical argument for early deletion. The ordered derivation supplies a different test: if apocope is delayed until after SC034 OEAwLong-Diphthong, PGmc **skáwōθi* ‘shows’ yields *†scēawep* rather than expected OE *scēawap* ‘shows’.

Early i-apocope must therefore precede the long-diphthong development. Moving it earlier within the tested range leaves every checked output unchanged; its early date rests on the anti-umlaut evidence, not on a lower boundary supplied by the witness words.

2.6 Final *ō-lowering before *r

Historical discussion

Ringe and Taylor treat the West Germanic lowering of final bimoric *ō before word-final *r as a specific inherited development and illustrate it above all with the families behind *fēower* ‘four’ and *wæter* ‘water’ (?).

The rule is historically secure but narrow: final or pre-final *ō before word-final *r. The clearest evidence remains concentrated in the *four* and *water* material. No broader environment for *ō is attested.

SC007. Lowering of final bimoric *ō before *r (PWGmcFinalOrLowering)

```
define PWGmcFinalOrLowering [
  { *ō } -> { *a } || _ { *r } .#.
];
```

OE *wæter* ‘water’ reveals why lowering must precede SC043 AngloFrisianBrightening. If SC007 PWGmcFinalOrLowering is delayed until afterwards, PGmc **wátōr* ‘water’ yields †*water* rather than expected OE *wæter* ‘water’: brightening can affect the vowel only after lowering has created its input. Moving the change earlier within the tested range alters no checked output.

The witness thus supplies a terminus ante quem at brightening but no earlier boundary. The *fēower* ‘four’ and *wæter* ‘water’ families support the narrow environment before word-final *r; no broader lowering of *ō is attested.

2.7 Coronal-w assimilation

Historical discussion

Ringe and Taylor treat the assimilation of **dw* and **zw* to **ww* as a shared Proto-West-Germanic innovation and support it through the four family and plural-pronominal forms such as *you* and *your* (?).

The historical support rests on a small witness set. Both coronal inputs assimilate before **w*, but only the numeral and the pronominal forms directly support the generalization.

SC008. Assimilation of coronal consonants before **w* (PWGmcCoronalWAssimilation)

```
define PWGmcCoronalWAssimilation [
  {*d} -> {*w} || _ {*w},
  {*z} -> {*w} || _ {*w}
];
```

OE *fēower* ‘four’ exposes a feeding relation: coronal assimilation must create **ww* while simplification can still reduce it. If SC008 PWGmcCoronalWAssimilation is delayed until after SC031 OEWWsimplification, PGmc **fédwōr* ‘four’ yields *†fēowwer* rather than expected OE *fēower* ‘four’. Earlier placements alter no checked output.

The numeral fixes that relative order. The pronouns extend both input clusters beyond four; the earlier boundary remains undetermined.

2.8 ij-contraction in *friend*

Historical discussion

Ringe and Taylor describe a change of **ijo* to **iu* in the friend family, with the pathway PGmc **frijond-* > PWGmc **friund* ‘friend’ > OE *frēond* ‘friend’ (?). The same source immediately warns that the **ijo* sequence is unique enough that wider generalization is inadvisable (?).

The change concerns a rare sequence confined to the friend family and cannot safely be generalized into a broadly productive rule.

SC009. *ij-contraction in friend* (PWGmcIjContraction)

```
define PWGmcIjContraction [
  {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
  {*ī} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
];
```

Only the friend family tests this contraction. If the rare **ijō* sequence survives until after SC032 OEDiphthongLeveling, PGmc **frījōndz* ‘friend’ yields *†friund* rather than expected OE *frēond* ‘friend’; moving contraction earlier within the tested range changes no checked output.

That single contrast places SC009 PWGmcIjContraction before diphthong leveling but gives no lower boundary. It cannot establish a productive sound law beyond this family, precisely the reservation made by Ringe and Taylor.

2.9 West Germanic j-gemination

Historical discussion

Fulk treats West Germanic consonant gemination before *j after a short vowel as a regular development and illustrates it with forms such as OE *settan* ‘set’ and *lecgan* ‘lay’ (? , §6.15).

The change applies specifically after a short vowel before *j, not to geminate consonants generally.

SC010. West Germanic j-gemination (PWGmcJGemination)

```
define PWGmcJGemination [
  {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
  {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
  {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
  {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
  {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
  {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
  {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
  {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
  {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
  {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
  {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
  {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
  {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
];
```

OE *nett* ‘net’ fixes the order because the syllabic-*j* development would remove the glide that conditions gemination. If SC011 PWGmcSyllabicJ precedes SC010 PWGmcJGemination, PGmc **nátjɑ* ‘net’ yields [†]*nete* rather than expected OE *nett* ‘net’. Earlier movement of gemination changes no checked output.

The chronology is phonologically transparent: the consonant must geminate before *j ceases to be consonantal. The witness establishes no earlier boundary.

2.10 Syllabic *j* after final-vowel loss

Historical discussion

Ringe and Taylor state directly that after final unstressed **a* and **ą* were lost, postconsonantal **j* became syllabic **i*, with outcomes behind OE *here* ‘army’ and *rice* ‘kingdom’ (?).

The sources establish the development, although the checked lexicon supplies little independent evidence for its position. Its scope is postconsonantal *j*, not high-vowel vocalization generally.

SC011. Syllabic **j* after final-vowel loss (PWGmcSyllabicJ)

```
define PWGmcSyllabicJ [
  {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#. ,
  {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
];
```

The same PGmc **nátją* ‘net’ witness supplies the only firm boundary. Placing SC011 PWGmcSyllabicJ before SC010 PWGmcJGeminatio yields [†]*nete* rather than expected OE *nett* ‘net’; moving it later changes no checked output.

Comparative evidence establishes postconsonantal **j* to syllabic **i* after final unstressed **a* or **ą* loss, with *here* ‘army’ and *rice* ‘kingdom’ as outcomes. The lexicon adds only that vocalization followed gemination, not where it falls among subsequent changes.

2.11 *lp*-voicing

Historical discussion

Ringe and Taylor treat word-internal **lp* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan* ‘fold’, *beald* ‘bold’, *wuldor* ‘glory’, and *gylden* ‘golden’ (?). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold* ‘gold’, and *feld* ‘field’ (? , §414).

The comparative evidence supports *lp* > *ld* most clearly in northern West Germanic, not as an unqualified pan-PWGmc development.

SC012. *lp*-voicing (PWGmcLThVoicing)

```
define PWGmcLThVoicing [
  {*θ} -> {*d} || {*l} _
];
```

The *field*, *fold*, *gold*, and *wold* families preserve **lp* to **ld*, but none dates the change against a neighboring rule. Every checked output remains unchanged when the voicing is moved in either direction.

Comparative reconstruction therefore establishes northern West Germanic *lp* > *ld*, but the witness forms fix no date. Neither a pan-PWGmc attribution nor an exact local placement follows from the evidence presented here.

2.12 Dental hardening

Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative **ð* became stop **d* in all positions (?).

The change is systemic across early West Germanic and extends beyond any one lexical family.

SC013. Dental hardening (PWGmcDentalHardening)

```
define PWGmcDentalHardening [  
    {*ð} -> {*d}  
];
```

Dental hardening has systemic scope: voiced fricative **ð* became stop **d* throughout early West Germanic. Moving SC013 PWGmcDentalHardening earlier or later changes no checked output.

Comparative evidence establishes the sound law; the present lexicon leaves its exact position approximate.

2.13 Early unstressed vowel changes

Historical discussion of the earliest unstressed vowel changes

The first change removes the remaining diphthongal quality of unstressed **ai*; the second carries early unstressed front-vowel leveling farther in forms such as *weorold* ‘world’. Their chronological evidence differs: monophthongization is historically clear but not closely dated by the witness forms, whereas **i*-lowering has a diagnostic later boundary.

Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the broad Northwest Germanic reduction of unstressed **ai* to a long mid vowel that merges with unstressed **e* (?). The historical change is thus established, although the order test determines no closer relative position.

SC014. Monophthongization of unstressed **ai* (NWGmcUnstressedAiMonophthongization)

```
define NWGmcUnstressedAiMonophthongization [
  {*āi} -> {*ē}
];
```

Moving SC014 NWGmcUnstressedAiMonophthongization earlier or later changes no checked form. The lexicon therefore cannot refine its source-based placement among the earliest Northwest Germanic simplifications of unstressed vowels.

Ringe and Taylor’s merger of unstressed **ai* with **e* establishes the historical development; the current witnesses do not distinguish its position relative to neighboring changes.

Historical discussion of early unstressed front-vowel leveling

Campbell treats the merger of unstressed front vowels directly and also records the variation of *weorold* ‘world’ and *weoruld* ‘world’ (?). These forms supply SC015 NWGmcILowering with a firmer lexical basis than the preceding change.

SC015. Leveling of early unstressed front vowels (NWGmcILowering)

```
define NWGmcILowering [
  {*i} -> {*e}
  || .#. EnglishStarNonVelarConsonant* _
  EnglishStarCoronal+ EnglishStarNonHighVowel,
  {*í} -> {*é}
  || .#. EnglishStarNonVelarConsonant* _
  EnglishStarCoronal+ EnglishStarNonHighVowel
];
```

The *weorold* ‘world’ and *weoruld* ‘world’ variants turn the general source claim into an ordering test. If SC015 NWGmcILowering is delayed until after SC036 OEInterStressRaising, PGmc **wír-āldu* ‘world’ yields [†]*wuruld* rather than expected OE *weorold* ‘world’; earlier movement changes no checked output.

The derivation thus fixes front-vowel leveling before interstress raising while leaving its earlier boundary open.

SC016 OEWsPalatalGlide and SC017 NWGmcULowering follow with a more tightly constrained local chronology.

2.14 West Saxon palatal glide and u-lowering

Historical discussion of West Saxon palatal glide and u-lowering

The derivation of *ġeoc* ‘yoke’ passes through both rules. Campbell treats the West Saxon rising-diphthong spellings before back vowels, while the same handbook tradition describes the lowering of *u* before a following non-high vowel separately (?, §44; ?, §115; ?, §4.3).

The first change creates the West Saxon *ġeoc* type; the second carries the same material into the subsequent vowel history.

Historical discussion of West Saxon palatal glide

West Saxon spellings such as *ġeoc* ‘yoke’, *ġeong* ‘young’, and *ġeogub* ‘youth’ reflect an early development before back vowels. Campbell gives the most direct handbook statement of the phenomenon (?, §44).

The sources establish SC016 OEWsPalatalGlide, although the checked forms provide only a later boundary.

SC016. West Saxon palatal glide before back vowels (OEWsPalatalGlide)

```
define OEWsPalatalGlide [
  {*j} {*u} -> {*j} {*e} {*u} || .#. _ ,
  {*j} {*ú} -> {*j} {*é} {*u} || .#. _
] .o. [
  {*ǵ} {*u} -> {*ǵ} {*e} {*u} || .#. _ ,
  {*ǵ} {*ú} -> {*ǵ} {*é} {*u} || .#. _
] .o. [
  {*tʃ} {*u} -> {*tʃ} {*e} {*u} || .#. _ ,
  {*tʃ} {*ú} -> {*tʃ} {*é} {*u} || .#. _
] .o. [
  {*f} {*u} -> {*f} {*e} {*u} || .#. _ ,
  {*f} {*ú} -> {*f} {*é} {*u} || .#. _
];
```

OE *ġeoc* ‘yoke’ fixes the close relation between glide insertion before back-vocalic *u* and the following change.

If glide insertion follows SC017 NWGmcULowering, PGmc **júkq* ‘yoke’ yields *†ġoc* rather than expected OE *ġeoc* ‘yoke’; earlier placement changes no checked output. The witness therefore dates SC016 OEWsPalatalGlide before u-lowering without supplying an earlier boundary. The *ġeoc* ‘yoke’, *ġeong* ‘young’, and *ġeogub* ‘youth’ material establishes the lexical scope of the West Saxon development.

Historical discussion of u-lowering

After the glide-conditioned West Saxon spellings are in place, the broader Northwest Germanic lowering of *u* to *o* before a following non-high vowel provides the clearest standard sound change in this small region. Campbell and Fulk both describe that change directly (?, §115; ?, §4.3).

SC017 NWGmcULowering thus rests on a broader source base than the preceding West Saxon rule.

SC017. Lowering of **u* before following non-high vowels (NWGmcULowering)

```
define NWGmcULowering [
```

```

{*u} -> {*o}
  || .#. EnglishStarConsonant* _
    [EnglishStarConsonantNoJ - EnglishStarNasal]
    EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
{*ú} -> {*ó}
  || .#. EnglishStarConsonant* _
    [EnglishStarConsonantNoJ - EnglishStarNasal]
    EnglishStarConsonantNoJ* EnglishStarNonHighVowel
];

```

Lowering of *u* to *o* is fixed on both sides by *ġeoc* ‘yoke’, *nosu* ‘nose’, *šcofl* ‘shovel’, and *sorg* ‘sorrow’.

Before SC016 OEWSPalatalGlide, PGmc **júkq* ‘yoke’ yields *†ġoc* rather than expected OE *ġeoc* ‘yoke’. After SC019 NWGmcFinalLongORaising, PGmc **núsō* ‘nose’ yields *†nusu* rather than expected *nosu* ‘nose’, PGmc **skúflō* ‘shovel’ yields *†šcufl* rather than expected *šcofl* ‘shovel’, and PGmc **súrgō* ‘sorrow’ yields *†surg* rather than expected *sorg* ‘sorrow’. The two witness sets place SC017 NWGmcULowering after glide formation and before final long-*o* raising.

2.15 Stressed monosyllable *ō-raising

Historical discussion

Campbell treats the development of final accented *ō* to *ū* in stressed monosyllables directly, with the familiar outcomes behind *cū* ‘cow’, *hū* ‘how’, *tū* ‘two’, and *bū* ‘both’ (? §122).

The change is historically secure, but the tested forms determine no close relative position for it. Its input is final **ō* in a stressed monosyllable.

SC018. Raising of final stressed monosyllabic *ō (NWGmcStressedMonosyllableORaising)

```
define NWGmcStressedMonosyllableORaising [
  { *ō } -> { *ū } || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
];
```

Campbell’s *cū* ‘cow’, *hū* ‘how’, and *tū* ‘two’ establish final stressed monosyllabic **ō* > **ū*.

Reversing SC018 NWGmcStressedMonosyllableORaising with neighboring changes leaves every checked output unchanged. The sound change is secure, but its exact position in the early history of long vowels rests on the handbooks.

2.16 Final long-*o* raising and final z-deletion

Historical discussion of final long-*o* raising and final z-deletion

The same final-syllable structure undergoes both changes. Ringe and Taylor describe the change of unstressed final non-nasalized long **ō* to short **u*, while Hogg and Crist treat word-final **z* loss as a separate later step in West Germanic (?; ?; ?).

The derivation of *ræste* ‘rest’ fixes their order: SC019 NWGmcFinalLongORaising must still see final **ō*, and SC020 PGmcFinalZDeletion removes final **z* only afterward.

Historical discussion of final long-*o* raising

The first change in the pair is the Northwest Germanic raising of unstressed final long **ō* to **u*. Ringe and Taylor state that development directly in comparative terms (?).

The change supplies the final vowel of forms such as *nosu* ‘nose’, *šcofl* ‘shovel’, and *sorg* ‘sorrow’.

SC019. Raising of final unstressed long **ō* (NWGmcFinalLongORaising)

```
define NWGmcFinalLongORaising [
  { *ō } -> { *u }
  || EnglishStarVocalic
  [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
];
```

Two groups of witnesses confine final unstressed long **ō* > **u*. The forms *nosu* ‘nose’, *šcofl* ‘shovel’, and *sorg* ‘sorrow’ fix its lower boundary.

Before SC017 NWGmcULowering, PGmc **nūsō* ‘nose’ yields [†]*nusu* rather than expected OE *nosu* ‘nose’, PGmc **skūflō* ‘shovel’ yields [†]*šcufl* rather than expected *šcofl* ‘shovel’, and PGmc **surgō* ‘sorrow’ yields [†]*surg* rather than expected *sorg* ‘sorrow’. After SC020 PGmcFinalZDeletion, PGmc **rāstōz* ‘rest’ yields [†]*rast* rather than expected *ræste* ‘rest’. These failures place SC019 NWGmcFinalLongORaising after u-lowering and before final z-loss.

Historical discussion of final z-deletion

The second change is the loss of word-final **z*. Standard handbook tradition and Crist’s West Germanic discussion establish the development within broader accounts of inflectional morphology (?; ?).

Final z-loss follows long-*o* raising and precedes the later changes in weak syllables.

SC020. Deletion of word-final **z* (PGmcFinalZDeletion)

```
define PGmcFinalZDeletion [{ *z } -> 0 || _ .#.];
```

The chronology of word-final **z*-loss is unusually well delimited: *ræste* ‘rest’ supplies its early boundary, while later weak syllables supply its late boundary.

Before SC019 NWGmcFinalLongORaising, PGmc **rāstōz* ‘rest’ yields [†]*rast* rather than expected OE *ræste* ‘rest’. After SC040 OEMedUnstressedULowering, PGmc **bébruz* ‘beaver’ yields [†]*befro* rather than expected *befer* ‘beaver’, PGmc **kwéðuz* ‘cud’ yields [†]*cwedo* rather than expected *cwedu* ‘cud’, and PGmc **félθuz* ‘field’ yields [†]*feldo* rather than expected *feld* ‘field’, alongside eight other newly failing rows. Final z-loss therefore follows SC019 NWGmcFinalLongORaising and precedes SC040 OEMedUnstressedULowering.

The **rástōz* ‘rest’ derivation fixes the local relation to SC019 NWGmcFinalLongORaising. The distant boundary at SC040 OEMedUnstressedULowering shows only that word-final **z*-loss precedes the later weak-syllable sequence; its placement within that wider interval follows the handbook chronology after final **ō*-raising.

2.17 Unstressed **o*-raising

Historical discussion

The older history of *heofon* ‘heaven’ requires an unstressed-vowel adjustment before the later reshaping of medial vowels in Old English. Campbell derives the -*o*- from an earlier unstressed environment, and Ringe and Taylor place the same family within the wider West Germanic record (? , §373; ?).

The change is historically recognizable, but the checked forms provide only a later boundary.

SC021. Raising of unstressed **o* before later **u* (NWGmcUnstressedORaising)

```
define NWGmcUnstressedORaising [
  {*o} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _
  ↪ EnglishStarConsonant* {*u}
];
```

After SC040 OEmedUnstressedULowering, PGmc **xémonu* ‘heaven’ yields †*heofun* rather than expected OE *heofon* ‘heaven’; earlier placement changes no checked output. The witness therefore places SC021 NWGmcUnstressedORaising before medial unstressed-*u* lowering.

Nothing in the present lexicon supplies the corresponding earlier boundary.

2.18 *mn*-dissimilation

Historical discussion

The handbooks describe the history of *mn* sequences as a limited descriptive pattern. Campbell discusses both loss of unstressed material and later assimilation in forms of this type, including the special status of *mōnaþ* ‘month’-type evidence (?, §§470, 484).

The pattern is historically established, but the checked forms do not constrain its position.

SC022. Dissimilation of *mn* sequences (NWGmcMnDissimilation)

```
define NWGmcMnDissimilation [
  {*m} -> {*β}
  || EnglishStarVocalic _
  EnglishStarVocalic EnglishStarConsonant* EnglishStarNasal
];
```

Campbell’s *heofon* ‘heaven’ and *mōnaþ* ‘month’ material supports early *m* > *β* before a later nasal, but supplies no ordering witness.

Moving SC022 NWGmcMnDissimilation earlier or later leaves every checked output unchanged. Its place among the early consonantal changes rests on the handbook account of *mn*-dissimilation.

2.19 N-stem *n*-loss

Historical discussion

The broader history is the reduction and leveling of older *n*-stem endings in West Germanic. Ringe and Taylor describe the resulting syncretism in the *n*-stems, which is the wider morphological setting for the narrower step isolated here (?).

The path to *dōn* ‘do’ provides the clearest witness, but the change remains narrow in scope.

SC023. Loss of *n*-stem **n* in final position (NWGmcNStemNLoss)

```
define NWGmcNStemNLoss [
  {*ō} {*n} -> {*ō} || _ .#.
];
```

After SC047 OEHeavySyllableNasalApocope, PGmc **dōnq* ‘do’ fails entirely (+?) instead of yielding expected OE *dōn* ‘do’; earlier placement changes no checked output. Thus SC023 NWGmcNStemNLoss must feed the later apocope.

This failed derivation supplies a terminus ante quem, while the lower boundary remains unattested.

2.20 Long ē-lowering

Historical discussion

The later West Saxon forms *scēap* ‘sheep’ and *ġēar* ‘year’ imply an earlier lowering of long *ē* before the palatal diphthongal outcomes described more fully later in the sequence. Campbell and Ringe and Taylor discuss those later West Saxon outputs directly (? §185; ? §6.5.1).

The change is historically recognizable, but the checked forms provide only a later boundary.

SC024. Lowering of long *ē* before non-nasal consonants (NWGmcLongELowering)

```
define NWGmcLongELowering [
  { *ē } -> { *æ } || _ [EnglishStarConsonant - EnglishStarNasal],
  { *é } -> { *æ } || _ [EnglishStarConsonant - EnglishStarNasal]
];
```

After SC056 OEWSPalatalDiphthongization, long *ē* > *æ* can no longer produce the expected West Saxon forms: PGmc **sképą* ‘sheep’ yields *†scīep* rather than OE *scēap* ‘sheep’, and PGmc **jérą* ‘year’ yields *†ġier* rather than *ġēar* ‘year’. Earlier placement changes no checked output, so SC024 NWGmcLongELowering has a secure upper boundary.

Its lower boundary remains a matter of handbook chronology.

2.21 Long $\bar{e}$ nasal-rounding

Historical discussion

Before nasals, older long $\bar{e}$ can round toward the $\bar{o}$ -vocalism seen later in *mōnaþ* ‘month’ and *mōna*/*mōn* ‘moon’-type material. Campbell treats this split directly in his discussion of Germanic long $\bar{e}$ before nasal consonants (?, §129).

The change is historically recognizable, but the tested forms supply no close relative chronology.

SC025. Rounding of long $\bar{e}$ before nasals (NWGmcLongENasalRounding)

```
define NWGmcLongENasalRounding [
  {*\bar{e}} -> {*\bar{o}} || _ EnglishStarNasal,
  {*\bar{e}} -> {*\bar{o}} || _ EnglishStarNasal
];
```

Reversing SC025 NWGmcLongENasalRounding with neighboring changes leaves every checked output unchanged. Its position beside the other $\bar{e}$ -developments therefore follows the handbooks.

2.22 Nasal spirant changes

Historical discussion of nasal loss before spirants and compensatory lengthening

The two rules state successive phases of a single development. Campbell describes nasal loss before voiceless spirants with compensatory lengthening and nasalization of the preceding vowel. Ringe and Taylor assign the same outcomes to inherited northern West Germanic, before late Old English (? , §121; ?).

SC026 NWGmcNasalSpirantLengthening adjusts the vowel while the nasal-plus-spirant sequence remains present; SC027 NWGmcNasalSpirantLoss then removes the nasal. The first rule must therefore precede the second.

SC026. Lengthening before nasal plus spirant (NWGmcNasalSpirantLengthening)

```
define NWGmcNasalSpirantLengthening [
  {*a} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*e} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*i} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*o} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*u} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*æ} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*á} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*é} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*í} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*ó} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
  {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative
];
```

All three witnesses require the vowel adjustment while the nasal is still present. If SC026 NWGmcNasalSpirantLengthening follows SC027 NWGmcNasalSpirantLoss, PGmc **fúnxstiz* ‘fist’ yields *†fyst* rather than expected OE *fȳst* ‘fist’, PGmc **gánsz* ‘goose’ yields *†geas* rather than expected *gōs* ‘goose’, and PGmc **júgunθ* ‘youth’ yields *†geogop* rather than expected *geogup* ‘youth’. Earlier placement changes no checked output. The evidence requires lengthening to precede nasal loss without supplying a lower boundary, in agreement with the handbook treatment of the two as successive phases.

SC027. Loss of the nasal before spirants (NWGmcNasalSpirantLoss)

```
define NWGmcNasalSpirantLoss [
  EnglishStarNasal -> 0 || _ EnglishStarVoicelessFricative
];
```

The converse test fixes the same boundary: placing SC027 NWGmcNasalSpirantLoss before SC026 NWGmcNasalSpirantLengthening produces the same errors in *fȳst* ‘fist’, *gōs* ‘goose’, and *geogup* ‘youth’. Later placement changes no checked output. These forms prove that the vowel was adjusted before the nasal disappeared; they provide no upper boundary for the loss.

2.23 Preconsonantal *x-loss

Historical discussion

Campbell explicitly treats loss of *x* and gives forms such as *fléam* ‘flight’ and *hēla* ‘heel’ as examples of the same broad development (?, §461).

The historical evidence is firmer than the chronology: the checked forms do not constrain the rule’s position.

SC028. Loss of preconsonantal *x (NWGmcPreconsonantalXLoss)

```
define NWGmcPreconsonantalXLoss [
  {*x} -> 0 || _ {*s} EnglishStarConsonant
];
```

No witness word dates preconsonantal *x-loss before *s plus another consonant: moving SC028 NWGmcPreconsonantalXLoss in either direction leaves every checked output unchanged. Its position within this stretch therefore rests on the handbook chronology for *x*-loss.

2.24 Awj glide formation and au-fronting

Historical discussion of awj glide formation and au-fronting

The *hīeg* ‘hay’ and *strīegan* ‘strew’ material undergoes both changes. Glide formation reshapes the older *awj* sequence, and fronting then affects the resulting *au*. Campbell’s discussion of these outcomes and Ringe and Taylor’s derivations of *hīeg* and *strīegan* describe the same sequence (? , §120; ?).

Glide formation creates the input to fronting; diphthong leveling follows both.

Historical discussion of awj glide formation

Older *awj* sequences are the source of forms such as *hīeg* ‘hay’ and *strīegan* ‘strew’. Campbell treats the relevant developments directly, and Ringe and Taylor likewise trace the same material through intermediate *auj*-type stages (? , §120; ?).

The sources establish glide formation, while the witness forms supply only a later boundary.

SC029. Glide formation in *awj (OEAwjGlideFormation)

```
define OEAwjGlideFormation [
  { *á } { *w } { *w } { *j } -> { *áu } { *j },
  { *a } { *w } { *w } { *j } -> { *au } { *j },
  { *á } { *w } { *j } -> { *áu } { *j },
  { *a } { *w } { *j } -> { *au } { *j }
];
```

The *hīeg* ‘hay’ and *strīegan* ‘strew’ derivations show that *awj* reshaping prepared the input to fronting. If fronting is applied first, PGmc **xáwwjɑ* ‘hay’ yields *†haug* rather than expected OE *hīeg* ‘hay’, and PGmc **stráwwjanɑ* ‘strew’ yields *†strauian* rather than expected *strīegan* ‘strew’. Earlier placement of glide formation changes no checked output, so these forms supply an upper boundary without a corresponding lower one.

Historical discussion of au-fronting

Once the glide sequence is in place, *au*-fronting produces the fronted diphthongal outcomes of the broader West Saxon vowel history. Campbell describes *au* > *ēa* (? , §135).

Fronting must follow glide formation and precede diphthong leveling, which applies to a wider set of derivations.

SC030. Fronting of *au (OE AuFronting)

```
define OE AuFronting [
  { *au } -> { *aeu },
  { *áu } -> { *áeu }
];
```

Two distinct failure sets confine fronting. Placed before glide formation, it produces the wrong forms: PGmc **xáwwjɑ* ‘hay’ yields *†haug* rather than expected OE *hīeg* ‘hay’, and PGmc **stráwwjanɑ* ‘strew’ yields *†strauian* rather than expected *strīegan* ‘strew’. Placed after diphthong leveling, PGmc **galáubijanɑ* ‘believe’, **bráudɑ* ‘bread’, and **dráugmaz* ‘dream’, together with sixteen other derivations, fail to produce output at all (+?) instead of yielding expected OE *gēliefan* ‘believe’, *brēad* ‘bread’, and *drēam* ‘dream’. The lexical errors require fronting to follow glide formation, while the failed derivations require it to precede diphthong leveling.

The later failure set consists of failed derivations, not competing Old English surface forms.

2.25 West Saxon diphthong sequence

Historical discussion of the West Saxon diphthong sequence

Four distinct developments shape the West Saxon diphthongal field. Campbell discusses inherited *aw/ew* outcomes, palatal-triggered diphthongization, and later Anglian smoothing in connected but separate parts of the vowel history; Hogg likewise distinguishes the palatal-diphthongal developments (?, §§120, 135–136, 170–176, 185, 223–227; ?).

The closest interaction joins *ww*-simplification and long-*aw* diphthongization, which together shape *dēaw* ‘dew’ and *hēawan* ‘hew’. Diphthong leveling regularizes a wider field, while long-*ew* diphthongization carries *ēow* into the later environment of breaking.

Historical discussion of WW simplification

West Germanic *ww* sequences lie behind forms such as *dēaw* ‘dew’ and *hēawan* ‘hew’, and Campbell treats them as part of the early West Germanic diphthong history (?, §120).

SC031 OEWWsimplification precedes the later long-diphthong outcomes.

SC031. Simplification of *ww sequences (OEWWsimplification)

```
define OEWWsimplification [
  {*w} {*w} -> {*w}
];
```

The *dēaw* ‘dew’ and *hēawan* ‘hew’ derivations establish that doubled *w* was simplified before the long *ēaw* development. If SC031 OEWWsimplification follows SC034 OEAwLongDiphthong, PGmc **dāwwō* ‘dew’ yields *†dawu* rather than expected OE *dēaw* ‘dew’, and PGmc **xāwwanq* ‘hew’ yields *†hawan* rather than expected *hēawan* ‘hew’. Earlier placement changes no checked output. The witnesses require simplification before the long-diphthong change and leave the lower boundary to the broader West Saxon chronology.

Historical discussion of diphthong leveling

Forms such as *hēafod* ‘head’ reflect the redistribution of diphthongal outcomes across a wider set of words. Campbell describes smoothing and related later monophthongization, although the rule below is more narrowly conditioned than any single textbook label (?, §§223–227).

The evidence for SC032 OEDiphthongLeveling is less self-contained than that for the *dēaw* ‘dew’ / *hēawan* ‘hew’ developments.

SC032. Leveling of diphthongal outputs (OEDiphthongLeveling)

```
define OEDiphthongLeveling [
  {*aeu} -> {*ēa},
  {*áeu} -> {*ēa},
  {*eu} -> {*ēo},
  {*éu} -> {*ēo},
  {*iu} -> {*ēo},
  {*íu} -> {*ēo},
  {*e} {*u} -> {*eo},
  {*é} {*u} -> {*éo},
  {*i} {*u} -> {*eo}
];
```

The two edges of this interval fail differently. Before SC030 OEAuFronting, PGmc **galáubijanq* ‘believe’, **bāug* ‘bow’, and **bráudq* ‘bread’ produce no output (+?) instead of expected OE *gēliefan*

‘believe’, *bēag* ‘bow’, and *brēad* ‘bread’, alongside fifteen other failed derivations. After SC040 OEmedUnstressedULowering, PGmc **xáubudq* ‘head’ yields [†]*hēafud* rather than expected *hēafod* ‘head’. Absence at the lower edge places diphthong leveling after fronting; the wrong surface form at the upper edge places it before medial unstressed-*u* lowering.

Historical discussion of long *ēow*

The long *ēow* forms of *cēowan* ‘chew’, *fēower* ‘four’, and *cnēow* ‘knee’ form part of the West Saxon vowel history, although their clearest ordering relation points forward. Campbell describes early *eu* in Old English, and Ringe and Taylor give the corresponding examples from *chew*, *four*, and *knee* (? §136; ?).

The only checked boundary for SC033 OEEwLongDiphthong lies ahead at SC044 OEBreaking.

SC033. Long *ēow* before following vowels and weak endings (OEEwLongDiphthong)

```
define OEEwLongDiphthong [
  {*e} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
  {*i} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
  {*é} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
  {*í} {*w} -> {*ēo} {*w} || _ OEEwLongContext
];
```

The long *ēow* of *cēowan* ‘chew’, *fēower* ‘four’, and *cnēow* ‘knee’ supplies only a terminus ante quem. If SC033 OEEwLongDiphthong follows SC044 OEBreaking, PGmc **kéwwanq* ‘chew’ yields [†]*cēowan* rather than expected OE *cēowan* ‘chew’, PGmc **fédwōr* ‘four’ yields [†]*feower* rather than expected *fēower* ‘four’, and PGmc **knéwq* ‘knee’ yields [†]*cneow* rather than expected *cnēow* ‘knee’. Earlier placement changes no checked output. The sources associate *ew* and *iw* with the same diphthongal history but furnish no lower boundary.

Historical discussion of long *ēaw*

After SC031 OEWWsimplification has reduced *ww* to single *w*, the remaining *aw* sequence can develop into the long *ēaw* seen in *dēaw* ‘dew’ and *hēawan* ‘hew’. Campbell treats these outputs in the early diphthong history of West Germanic and Old English (? §§120, 135–136). The resulting long diphthong is *ēaw*.

SC034 OEAwLongDiphthong follows SC031 OEWWsimplification locally and must also precede SC043 AngloFrisianBrightening.

SC034. Long *ēaw* before following vowels (OEAwLongDiphthong)

```
define OEAwLongDiphthong [
  {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
  {*á} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}]
];
```

A local feeding relation and a later vowel change confine *aw* > *ēaw*. Before SC031 OEWWsimplification, PGmc **dáwwō* ‘dew’ yields [†]*dawu* rather than expected OE *dēaw* ‘dew’, and PGmc **xáwwanq* ‘hew’ yields [†]*hawan* rather than expected *hēawan* ‘hew’. After SC043 AngloFrisianBrightening, PGmc **skáwōjanq* ‘show’ yields [†]*scawian* rather than expected OE *scēawian* ‘show’, PGmc **skáwōθi* ‘shows’ yields [†]*scawap* rather than expected *scēawap* ‘shows’, and PGmc **stráwq* ‘straw’ yields [†]*stræw* rather than expected *strēaw* ‘straw’. The *dēaw* and *hēawan* forms require long-diphthong formation after simplification, while *scēawian* requires it before brightening; the handbooks assign the same interval to the West Saxon development.

2.26 Prefix and compound adjustments

Historical discussion of prefixal **a*-reduction

Weakly stressed prefixes can lose their older low vowel early in Old English, and that is the historical setting for SC035 OEPrefixAReduction. Campbell treats the small class of pretonic losses directly, while Ringe and Taylor's derivation of **galaubijana* 'believe' supplies the comparative witness for the same development (?, §354; ?, ?).

The rule has a narrow historical range and gives prefixed forms the weak vowel inherited by later vocalic changes.

SC035. Reduction of prefixal **a* (OEPrefixAReduction)

```
define OEPrefixAReduction [
  {*a} -> {*ë}
  || .#. {*g} _
  [EnglishStarConsonant | EnglishPalatalConsonant]
  EnglishStarVocalic
];
```

The prefix of *ġeliefan* 'believe' supplies the upper boundary for **ga-* > **ge-*. If SC035 OEPrefixAReduction follows SC043 AngloFrisianBrightening, PGmc **galáubijanā* 'believe' yields *†ġealiefan* rather than expected OE *ġeliefan* 'believe'. Earlier placement changes no checked output, so the witness dates prefix reduction before brightening without locating its beginning.

Historical discussion of inter-stress raising

SC036 OEInterStressRaising has the strongest evidence of the three. Campbell's discussion of *weorold* 'world' / *weoruld* 'world' and Ringe and Taylor's derivation of **weraldu* 'world' > **weruldu* 'world' > OE *weorold* place the rule squarely in the history of low-stress medial vowels (?, §§338–339; ?, §6.3.3).

The rule changes the vowel between stronger stress peaks, and its witnesses consequently constrain the relative chronology.

SC036. Raising of medial **a* between stress peaks (OEInterStressRaising)

```
define OEInterStressRaising [
  {*a} -> {*u}
  || PGmcStarVowel EnglishStarConsonant* _
  [EnglishStarConsonant - {*j}]+ [{*u}|{*ū}],
  {*à} -> {*u}
];
```

The two boundaries have unequal force. Before SC019 NWGmcFinalLongORaising, PGmc **sáiwalō* 'soul' yields *†sāwel* rather than expected OE *sāwol* 'soul'; after SC040 OEmedUnstressedU-Lowering, it yields *†sāwul* rather than *sāwol*, while PGmc **wír-àldu* 'world' yields *†weoruld* rather than *weorold* 'world'. The distant lower boundary places inter-stress raising after final long-*o* raising, and the local upper boundary places it before medial unstressed-*u* lowering. In hand-book terms, medial **a* > **u* belongs to the *world*- and *soul*-type low-stress vocalism that followed the earlier final-vowel changes.

Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and ob-

scured compounds (?, §§356–357; ?, §367; ?, §§386–387).

SC037 OECompoundLinkingSyncope captures this pattern in compounds such as *reǵnboga* ‘rainbow’. Its only checked boundary is the immediately following technical stress-stripping stage, which is not a sound change.

SC037. Syncope of compound linking vowels (OECompoundLinkingSyncope)

```
define OECompoundLinkingSyncope [
  [{*a}|{*i}|{*u}] -> 0
  || PGmcStarAcuteVowel OEAnyConsonant+ _
  OEAnyConsonant+ PGmcStarGraveVowel
];
```

The *reǵnboga* ‘rainbow’ test exposes a bookkeeping dependency rather than a historical sound-change boundary. After SC038 OEStripSecondaryStress, PGmc **régna-bùgô* ‘rainbow’ yields [†]*reǵnefoga* rather than expected OE *reǵnboga* ‘rainbow’, because the technical stage has erased the stress information that licenses syncope. The handbooks instead place weakened compound junctures with the behavior described under SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

2.27 Medial unstressed vowel changes

Historical discussion of medial unstressed vowel changes

The history of *wuduwe* ‘widow’ orders these two changes within the same low-stress vocalic development. Campbell discusses both the *w*-conditioned *u* forms and the later *weorold* ‘world’ / *weoruld* ‘world’ alternation, while Ringe and Taylor give the same connection comparatively in **widuwon-*, **weraldu* ‘world’, and **jugunþi* ‘youth’ (? §218; ?, §332; ?, §§338–339; ?, §6.3.3).

SC039 OEWICombinativeUUmlaut feeds the vowel sequence subsequently reshaped by SC040 OEMedUnstressedULowering. Initial *w* conditions the first change.

SC039. Combinative **u*-umlaut in *wi*-forms (OEWICombinativeUUmlaut)

```
define OEWICombinativeUUmlaut [
  {*i} -> {*ú}
  || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
];
```

The *wuduwe* ‘widow’ derivation answers one narrow question about *wi*-forms. If SC039 OEWICombinativeUUmlaut follows SC040 OEMedUnstressedULowering, PGmc **wíduwōn* ‘widow’ yields [†]*wudowe* rather than expected OE *wuduwe*; earlier placement changes no checked output. The witness requires combinative u-umlaut to precede medial lowering and supplies no lower boundary.

SC040. Lowering of medial unstressed **u* (OEMedUnstressedULowering)

```
define OEMedUnstressedULowering [
  {*u} -> {*o}
  || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]]
      [EnglishStarConsonant | EnglishPalatalConsonant]+ _
      [[EnglishStarConsonant | EnglishPalatalConsonant] - {*m}]
];
```

The two witnesses date medial unstressed **u* > **o* at very different scales. Before SC039 OEWICombinativeUUmlaut, PGmc **wíduwōn* ‘widow’ yields [†]*wudowe* rather than expected OE *wuduwe* ‘widow’; after SC072 OEUnstressedLongVowelShortening, PGmc **júgunθ* ‘youth’ yields [†]*geogop* rather than expected *geogup* ‘youth’. The local *weorold* ‘world’ and widow evidence places lowering after combinative u-umlaut, while the youth form supplies only the distant requirement that lowering precede unstressed long-vowel shortening.

2.28 Final bare-*a* loss

Historical discussion

I isolate the loss of final short low vowels within the broader erosion of final syllables described by the handbooks (?, §341; ?).

Final bare-a loss follows the medial unstressed vowel changes and precedes restoration, which depends on the environment left by the loss.

SC041. Loss of final bare **a* (PWGmcFinalBareALoss)

```
define PWGmcFinalBareALoss [
  {*a} -> 0 || _ .#.
];
```

The two sides of final bare-*a* loss rest on different evidence. Applied before final z-deletion, the change gives the wrong outputs: PGmc **bárdaz* ‘beard’ yields [†]*bearda* rather than expected OE *beard* ‘beard’, and PGmc **kámbaz* ‘comb’ yields [†]*camba* rather than expected *camb* ‘comb’. Applied after restoration, PGmc **kráftaz* ‘craft’ yields [†]*craft* rather than expected OE *cræft* ‘craft’, and PGmc **dágaz* ‘day’ yields [†]*dag* rather than expected *dæg* ‘day’. The distant lower limit follows final z-loss; the local feeding relation precedes restoration, which requires the environment created by the vowel loss.

2.29 Surviving bimoric **ō* unrounding

Historical discussion

The handbooks do not isolate a large independent sound change under this label. The surviving bimoric **ō* in the pathway to *ræste* ‘rest’ nevertheless undergoes unrounding before SC043 AngloFrisianBrightening. Campbell, Hogg, and Ringe and Taylor describe the surrounding fronting and restoration history without naming this feeder separately (? , §§131, 157–158; ?; ?).

The sole witness establishes a local relation to brightening but supports no broader generalization.

SC042. Unrounding of the surviving bimoric **ō* (PWGmcSurvivingBimoricOUnrounding)

```
define PWGmcSurvivingBimoricOUnrounding [
  { *ō } -> { *ā } || EnglishStarVocalic [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant]+ _ .#.
];
```

The single *ræste* ‘rest’ derivation carries the chronology of bimoric **ō* > **ā*. Before SC020 PGmcFinalZDeletion or after SC043 AngloFrisianBrightening, PGmc **rástōz* ‘rest’ yields †*rasta* rather than expected OE *ræste*. Unrounding must therefore follow final z-loss and precede brightening, although only the relation to brightening is local.

2.30 Anglo-Frisian brightening

Historical discussion

Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments. Later Old English developments presuppose this fronted stage even where they partly conceal it. Campbell gives the classical statement of the change, Hogg supplies the standard modern labels, and Ringe and Taylor establish its local chronology with breaking and restoration (?, §131; ?; ?; ?, §§4.12–4.13).

Brightening creates the input to SC044 OEBreaking, while SC046 OEARestoration later partly reverses its outcome before back vowels.

SC043. Fronting of low **a* outside nasal environments (AngloFrisianBrightening)

```
define AngloFrisianBrightening [
  AngloFrisianBrighteningUnstressed .o.
  AngloFrisianBrighteningStressed .o.
  AngloFrisianBrighteningLongFinal
];
```

Two derivations place low **a* > **æ* between unrounding and breaking. Before SC042 PWGmc-SurvivingBimoricOUnrounding, PGmc **rástōz* ‘rest’ yields [†]*rasta* rather than expected OE *ræste* ‘rest’. After SC044 OEBreaking, PGmc **sláxanq* ‘slay’ yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. The first witness requires brightening to receive the outcome of the surviving-bimoric **ō* development; the second requires breaking to receive the fronted vowel.

2.31 Breaking and velar-fricative palatalization

Historical discussion of breaking and velar-fricative palatalization

Breaking creates *eo*-type outputs before *h*, *rC*, and *lC*; velar-fricative palatalization then operates in that reshaped environment. Campbell, Ringe and Taylor, and Fulk place breaking after brightening. The following fricative palatalization is more narrowly conditioned (?, §§139, 405–406; ?, §§6.2.1–6.2.3, 6.4.1–6.4.2; ?, §4.13).

Breaking has the fuller handbook treatment, while velar-fricative palatalization follows it locally in the *feoh* ‘cattle’ and *feohtan* ‘fight’ type derivations.

SC044. Breaking before *h*, *rC*, and *lC* (OEBreaking)

```
define OEBreaking OEBreakingA
  .o. OEBreakingE
  .o. OEBreakingI;
```

Breaking must encounter the vowel created by brightening and must precede the fricative change seen in *feoh* ‘fee’ and *feohtan* ‘fight’. Before SC043 AngloFrisianBrightening, PGmc **sláxanq* ‘slay’ yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. After SC045 OEVelarFricativePalatalization, PGmc **féxu* ‘cattle’ yields *†fehu* rather than expected OE *feoh*, and PGmc **féxtanq* ‘fight’ yields *†fehtan* rather than expected *feohtan*. The two feeding relations place breaking between brightening and velar-fricative palatalization.

SC045. Palatalization of velar fricatives beside front vowels (OEVelarFricativePalatalization)

```
define OEVelarFricativePalatalization [
  {*x} -> {*ç} || _ EnglishStarFrontVowel,
  {*y} -> {*j} || _ EnglishStarFrontVowel,
  {*x} -> {*ç} || EnglishStarFrontVowel _,
  {*y} -> {*j} || EnglishStarFrontVowel _,
  {*x} -> {*ç} || _ {*j},
  {*y} -> {*j} || _ {*j}
]
.o. EnglishStarAlphabet*;
```

The local chronology comes from *feoh* ‘cattle’ and *feohtan* ‘fight’. Before SC044 OEBreaking, palatalization of **x* and **y* beside front vowels or **j* makes PGmc **féxu* ‘cattle’ yield *†fehu* rather than expected OE *feoh*, and PGmc **féxtanq* ‘fight’ yield *†fehtan* rather than expected *feohtan*. The distant upper limit comes from *six* ‘six’: after SC060 OEWsPalatalUmlaut, PGmc **séxs* ‘six’ yields *†sihs* rather than expected OE *six*. Breaking therefore feeds velar-fricative palatalization directly, while palatal umlaut supplies only the broader upper limit.

2.32 A-restoration and nasal changes

Historical discussion of A-restoration

Campbell's restoration of *a* before following back vowels and Ringe and Taylor's later retraction describe the same post-brightening development (?, §§157–159; ?, §6.3.1; ?, §4.13). Some outcomes of Anglo-Frisian fronting survive only in environments where restoration does not return them to back *a*.

SC046 OEARestoration has firmer handbook support than the two following nasal rules.

SC046. Restoration of **a* before following back vowels (OEARestoration)

```
define OEARestoration (
  { *æ } -> { *a } || _
    OEARestorationIntervening OEARestorationTriggerVowel
    - OEARestorationIntervening OEARestorationWeakTailVowel
);
```

Restoration must receive fronted **æ* and return **a* before the nasal-tail changes. Before SC043 AngloFrisianBrightening, PGmc **bákanq* 'bake' yields *†bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáranq* 'fare' yields *†færan* rather than expected *faran* 'fare'. After SC048 OESecondaryNasalization, **bákanq* again yields *†bæcan* instead of *bacan*, while PGmc **wádanq* 'wade' yields *†wædan* instead of *wadan* 'wade'. These independent witness pairs place restoration after brightening and before secondary nasalization.

Historical discussion of heavy-syllable nasal loss and secondary nasalization

Heavy-syllable nasal apocope removes the final nasalized vowel; secondary nasalization then marks the preceding *a* before final *n*. The handbooks do not isolate both developments under equally prominent labels. Campbell describes later nasal loss and the back-mutation environment; Ringe and Taylor provide the later relation to back mutation (?, §§205–206, 403; ?, §6.9.4).

The reciprocal failure set fixes the order: apocope removes the ending before secondary nasalization acts on the remaining structure. Restoration receives the fuller historical treatment in the handbooks.

SC047. Heavy-syllable nasal apocope of final **q* (OEHeavySyllableNasalApocope)

```
define OEHeavySyllableNasalApocope [
  { *q } -> Ø || OEAnyConsonant _ .#.
];
```

The evidence for final nasalized **q* loss is sharply asymmetric. Before SC034 OEAwLongDiphthong, the single PGmc witness **stráwq* 'straw' yields *†stræw* rather than expected OE *strēaw* 'straw'. After SC048 OESecondaryNasalization, PGmc **bákanq* 'bake' yields *†bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndanq* 'bind' yields *†binden* rather than expected *bindan* 'bind', alongside a broad *-en* failure set. One lower witness places apocope after long-diphthong formation; many reciprocal upper failures place it before secondary nasalization.

SC048. Secondary nasalization before final **n* (OESecondaryNasalization)

```
define OESecondaryNasalization [
  {*a} -> {*ǣ} || _ {*n} .#.
];
```

The broad *-an/-en* split fixes the lower boundary of final **a* nasalization before *n*. Before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* ‘bake’ yields *†bacen* rather than expected OE *bacan* ‘bake’, and PGmc **bīdaną* ‘bind’ yields *†binden* rather than expected *bindan* ‘bind’. The upper boundary comes from back mutation. After SC059 OEBackMutation, PGmc **stélaną* ‘steal’ yields *†steolan* rather than expected OE *stelan* ‘steal’, and PGmc **wébaną* ‘weave’ yields *†weofan* rather than expected *wefan* ‘weave’. Reciprocal nasal-tail failures place secondary nasalization after apocope, and the later mutation witnesses place it before back mutation; SC046 OEARestoration retains the clearest independent historical support.

2.33 B allophony and Sievers-law syncope

Historical discussion of B allophony

The first change is the positional alternation of Germanic **b*. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (?). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (?), and Luick's spelling evidence shows the same labial fricative pattern in Old English (?).

The distribution is narrow, but later changes presuppose the stop-fricative alternation.

SC049. Distribution of **b* after vowels and liquids (PGmcBAllophony)

```
define PGmcBAllophony [
  { *b } -> { *β } || PGmcStarVocalic _ ,
  { *b } -> { *β } || [ { *l } | { *r } ] _
] .o. [
  { *β } -> { *b } || _ { *b }
];
```

The handbooks describe **b*/**bb* as a positional alternation within the consonant system, and one compound supplies its chronological consequence. Before SC037 OECompoundLinkingSyncope, *reǵnboga* ‘rainbow’ develops as [†]*reǵnfoga* rather than expected OE *reǵnboga*; later placement creates no comparable failure. The witness places b-allophony after compound-linking syncope without turning the alternation into an independent sound law.

Historical discussion of Sievers-law syncope

Sievers’ Law concerns a different historical problem. It is a prosodic and morphological adjustment in heavy stems, not a distributional allophone of a stop consonant. Adamczyk treats the Old English reflexes of the law as historical evidence from weak verbs and related formations (?). Fulk gives the compact comparative summary through familiar forms such as *biddan* ‘ask’, *sellan* ‘give’, and *nerian* ‘save’ (? , §6.15).

Sievers-law syncope is narrow in scope, but its relation to the following palatalization is lexically secure. Its earlier limit is less sharply defined than that of the preceding allophony rule.

SC050. Sievers-law syncope (SieversLawSyncope)

```
define SieversLawSyncope [
  { *i } -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ { *j }
];
```

The Sievers-law reduction **-CijV-** > **-CjV-**, including loss of **i* before **j*, must precede palatalization. If SC050 SieversLawSyncope follows SC052 OEVelarPalatalization, PGmc **strákkijaną* ‘stretch’ yields [†]*strecćan* rather than expected OE *strecćan* ‘stretch’; earlier placement creates no comparably precise error. The single cluster witness therefore places syncope before velar palatalization.

2.34 Palatalization of **sk* to **sc*

Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (? , §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (?). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (? , §§6.4.1, 6.5.1).

Luick places the cluster change within a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (? , §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (?). The result is a historically clear rule, but not an identity between the cluster change and the later umlautal developments.

SC051. Palatalization of **sk* to **sc* (OESkPalatalization)

```
define OESkPalatalization [
  {*s} {*k} -> {*ʃ} || .#. _
] .o. [
  {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
  {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
  {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
  {*s} {*k} -> {*ʃ} || {*j} _
];
```

The non-fronted vowels of *flasce* ‘flask’ and *wascan* ‘wash’ fix the lower boundary of **sk* > **sc*. Before SC046 OEARestoration, the forms are fronted too soon, yielding *flæsce* ‘flask’ and *wæscan* ‘wash’ rather than expected OE *flasce* and *wascan*. This places SC051 OESkPalatalization after restoration.

Five witnesses establish the upper boundary collectively. The palatal cluster must already underlie *sceaft* ‘shaft’, *scēar* ‘shear’, *scēap* ‘sheath’, *scēap* ‘sheep’, and *sciold* ‘shield’ before SC056 OEWsPalatalDiphthongization. The **scea-** ‘sea’/**scie-** set therefore places cluster palatalization before the West Saxon vowel change. The cluster change occupies the same palatalization zone as SC052 OEVelarPalatalization while remaining distinct from plain-velar palatalization and the later vowel changes.

2.35 Velar palatalization before front vowels

Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading “Frühe Verschiebungen in palataler Richtung,” he treats English *k* and *g* before bright vowels together with the larger field of palatal effects (?, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological sequence. The examples associated with that sequence, such as *ceaster* ‘town’, *geaf* ‘gave’, *giefan* ‘give’, and *giest* ‘guest’, already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (?, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone *sk* cluster. His remark that “[*sk*] is more prone to palatalization and assibilation than [*k*]” is brief, but it makes clear that different members of the larger palatal field need not behave identically (?, §440). Elsewhere in the same part of the grammar he uses forms such as *cild* ‘child’, *dæg* ‘day’, *giefan* ‘give’, and *giest* ‘guest’, which show how palatalized velars, palatal influence, and later umlaut outcomes meet in the same region of the lexicon without collapsing into one process (?, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant *j* (?). This formulation replaces a broad list of palatal outcomes with a phonological environment defined by adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that “after initial velars and **sk* had been palatalized” West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (?, §6.5.1). Their own examples of the plain-velar rule, such as *weccan* ‘wake’, *licgan* ‘lie’, *leccan* ‘lay’, *secg* ‘retainer’, *ecg* ‘edge’, *wicg* ‘horse’, and *brycg* ‘bridge’, illustrate the same point in lexical detail: front vowels and *j* create the palatal environment in which plain *k* and *g* cease to behave as plain velars (?, §6.4.1).

Luick describes a broad early movement; Campbell distinguishes plain velars from the *sk* complex; Hogg specifies the adjacency and syllable conditions; and Ringe and Taylor order the plain-velar change before West-Saxon diphthongization. Plain-velar palatalization thus forms part of a wider palatalizing environment without being identical to its neighboring changes.

SC052. Palatalization of **k* before front vowels and **j* (OEVelarPalatalizationKFront)

```
define OEVelarPalatalizationKFront [
  {*k} -> {*tʃ} || .#. _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || _ [{*i} | {*ī}],
  {*k} -> {*tʃ} || _ {*í},
  {*k} -> {*tʃ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || {*í} _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || [{*i} | {*ī}] _ .#. ,
  {*k} -> {*tʃ} || {*í} _ .#.
] .o. [
  {*k} {*k} -> {*tʃ} {*tʃ} || _ {*j}
] .o. [
  {*k} -> {*tʃ} || _ {*j}
] ;
```

The *weccan* ‘wake’, *licgan* ‘lie’, and *leccan* ‘lay’ set identifies front vowels and *j* as the environment for palatalization of *k* (?, §6.4.1). These forms establish the conditioning; different witnesses establish the chronology.

Applied before Sievers-law syncope, PGmc **strákkijaną* ‘stretch’ yields [†]*streccan* rather than expected OE *streccan* ‘stretch’. Applied after i-umlaut fronting, PGmc **kūi* ‘cow’ and **lúnganjō*

‘lungs’ yield *cȳ* ‘cows’ and *lungēn* ‘lungs’ rather than expected OE *cȳ* ‘cows’ and *lungen* ‘lungs’. The front-vowel *k* change therefore follows Sievers-law syncope and precedes i-umlaut fronting.

SC052. Velar palatalization before front vowels (OEVelarPalatalization)

```
define OEVelarPalatalization [
  OEVelarPalatalizationKFront
] .o. [
  {*g} -> {*ǵ} || _ EnglishStarFrontVowel,
  {*g} -> {*ǵ} || EnglishStarFrontVowel _ .#.,
  {*g} -> {*ǵ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
  {*g} -> {*ǵ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
  {*g} {*g} -> {*ǵ} {*ǵ} || _ {*j}
] .o. [
  {*g} -> {*ǵ} || _ {*j}
];
```

Plain *k* and *g* palatalization in front-vocalic and *j*-adjacent environments follows *sk*-palatalization and occupies a sharply defined pre-umlaut interval. Applied before Sievers-law syncope, PGmc **strákkijanǵ* ‘stretch’ yields [†]*strecċan* rather than expected OE *streċċan* ‘stretch’. Applied after general i-umlaut, PGmc **kūi* ‘cow’ yields [†]*cȳ* rather than expected *cȳ* ‘cows’, and PGmc **lúnganjō* ‘lungs’ yields [†]*lungēn* rather than expected *lungen* ‘lungs’. These witnesses place velar palatalization after Sievers-law syncope and before umlaut.

Luick, Campbell, and Ringe and Taylor place *cild* ‘child’ and *dæg* ‘day’ in a consonantal palatalization that precedes later vowel fronting (?, §168; ?, §440; ?, §§6.4.1, 6.5.1). The umlautal developments therefore receive plain *k* and *g* already reshaped beside front vowels and *j*.

The *sk* change belongs to the same palatalizing region with a separate scope. The *streċċan* ‘stretch’ evidence establishes a specific dependency on earlier syncope; it does not merge the two changes into one process.

2.36 Post-velar *w-loss and loss of *w before final *i

Historical discussion of early *w-loss before umlaut

The first rule is a narrow loss of *w after velars in the *ngw sequence. Ringe and Taylor derive PGmc *singwan ‘sing’ to Old English *singan* ‘sing’ (?, §6.4.2). This comparative evidence establishes the change, although no checked form fixes its order relative to a neighboring rule.

The second rule is historically more legible. Campbell notes the recurring loss of *w before *i in unstressed position (?, §406). Ringe and Taylor trace the development of *sǣ* ‘sea’ from earlier *saiwi- / *sawi- (?, §6.7.1), and Luick gives the same trajectory in his own historical grammar (?, §187). The first rule is restricted to the *ngw sequence; the second has a specific lexical witness and defined earlier and later limits.

SC053. Loss of *w after velars (OEPPostVelarWLoss)

```
define OEPPostVelarWLoss [
  { *w } -> 0 || { *n } { *g } _
];
```

The comparative development *singwan > *singan* establishes narrow post-velar *w-loss in the *ngw sequence, yielding *singan* ‘sing’. Moving SC053 OEPPostVelarWLoss earlier or later leaves every checked output unchanged. Its pre-umlaut position therefore rests on comparative evidence, while the present lexicon supplies no neighboring boundary.

SC054. Loss of *w before final *i (OEWLossBeforeI)

```
define OEWLossBeforeI [
  { *w } -> 0 || EnglishStarVocalic _ { *i } .#.
];
```

The history of *sǣ* ‘sea’ explains why non-initial *w disappeared before final unstressed *i. Campbell describes the loss, Ringe and Taylor derive the form from *saiwi-/*sawi-, and Luick gives the parallel trajectory (?, §406; ?, §6.7.1; ?, §187). Loss of the glide allowed the preceding vowel to undergo the later fronting and lengthening.

The same witness supplies two distant limits. Before SC020 PGmcFinalZDeletion or after SC063 OEHighVowelApocope, SC054 OEWLossBeforeI yields *sǣw* ‘sea’ rather than expected OE *sǣ* ‘sea’. The loss must therefore follow final z-deletion and precede high-vowel apocope, while its exact position within that broad interval remains source-based.

2.37 The Old English i-umlaut and West Saxon palatal diphthongization

Historical discussion of i-umlaut and West Saxon palatal diphthongization

Luick gives the change its traditional scale:

Der wichtigste Fall von palataler Beeinflussung ... war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.

(?, §182)

Campbell gives the most compact classical formulation in English when he writes that “the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE” (?, §190). He immediately defines the core conditioning environment as a following i or j, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* ‘guest’, *giefan* ‘give’, *hierde* ‘shepherd’, and *ieldra* ‘older’ (?, §§190–197).

Hogg continues in the same vein: “we come now to a change which is almost as uncontroversial as it is important” (?). His examples, such as *bryd* ‘bride’, *trymman* ‘strengthen’, *bedd* ‘bed’, *ciest* ‘chest’, and *wiersa* ‘worse’, likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (?).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (?, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it “diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)” (?, §4.13). Ringe and Taylor’s examples such as *gielðan* ‘pay’, *sciold* ‘shield’, and *scieppan* ‘create’ show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (?, §6.5.1).

Luick, Campbell, and Hogg treat i-umlaut as a system-wide change. Ringe and Taylor and Fulk distinguish from it a narrower West-Saxon process affecting words after initial palatals. The two changes act in different environments and produce different lexical consequences.

SC055. Fronting under i-umlaut (OEIUmlautFronting)

```
define OEIUmlautFronting [
  {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*á} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

The breadth of i-umlaut appears in lexical classes that share only a following high front vocoid. The forms *fylgan* ‘follow’, *gylden* ‘golden’, *wyrm* ‘worm’, and *giest* ‘guest’ exemplify the same i- or j-conditioned fronting across different vowels (?, §6.6.1; ?, §§190–191).

The cow and lung forms establish the lower boundary. If fronting precedes velar palatalization, PGmc **kūi* ‘cow’ yields *cȳ* ‘cows’ rather than expected OE *cȳ* ‘cows’, and **lúnganjō* ‘lungs’ yields *lungen* ‘lungs’ rather than expected OE *lungen* ‘lungs’. The consonantal change must therefore precede fronting.

The gift and sheath forms establish the upper boundary. If West Saxon palatal diphthongization precedes fronting, PGmc **gēftiz* ‘gift’ yields *gieft* ‘gift’ rather than expected OE *gift* ‘gift’, and **skáiθiz* ‘sheath’ yields *scēþ* ‘sheath’ rather than expected *scēap* ‘sheath’. Fronting consequently follows velar palatalization and precedes the West Saxon change; the other components of i-umlaut share those bounds.

SC055. Raising under i-umlaut (OEIUmlautRaising)

```
define OEIUmlautRaising [
  { *æ } -> { *e } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Raising of umlauted æ to e continues the same assimilatory event as fronting and therefore shares the chronology of general i-umlaut.

The same four forms fix both boundaries. If raising precedes velar palatalization, **kūi* ‘cow’ yields *†cȳ* instead of expected *cȳ* ‘cows’ and **lúnganjō* ‘lungs’ yields *†lungen* instead of expected *lungen* ‘lungs’. If West Saxon palatal diphthongization precedes raising, **gēftiz* ‘gift’ yields *†gieft* rather than expected *gift* ‘gift’, and **skáiθiz* ‘sheath’ yields *†scēþ* rather than expected *scēap* ‘sheath’. These forms place raising after velar palatalization and before West Saxon palatal diphthongization.

The sources do not describe umlaut as simple fronting alone. Campbell notes that the low front vowel changes again before m and n in most dialects (? , §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (?).

SC055. Diphthongal outcomes under i-umlaut (OEIUmlautDiphthong)

```
define OEIUmlautDiphthong [
  { *ea } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *ēa } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *io } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *īo } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *eo } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *ēo } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *ēa } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *ēo } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  { *io } -> { *ie } || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Diphthongal outcomes belong to the same system-wide assimilation as simple-vowel fronting and raising. All three therefore belong to a single historical event.

The relevant examples are the recurring West-Saxon ie forms cited in the handbooks, including *giest* ‘guest’, *giefan* ‘give’, and *hierde* ‘shepherd’ in Campbell and *ciest* ‘chest’ in Hogg (? , §§190–191, 248–251; ?). These diphthongal outcomes form a distinct part of the general umlautal development alongside simple fronting.

The chronology comes from the cow/lung and gift/sheath contrasts. Placed before velar palatalization, diphthongal mutation over-palatalizes **kūi* ‘cow’ and **lúnganjō* ‘lungs’; placed after West Saxon palatal diphthongization, it yields *gieft* ‘gift’ and *scēþ* ‘sheath’ instead of expected *gift* ‘gift’ and *scēap* ‘sheath’. These failures place diphthongal mutation after velar palatalization and before West Saxon palatal diphthongization.

SC055. The composite i-umlaut rule (OEIUmlaut)

```
define OEIUmlaut OEIUmlautFronting
.o. OEIUmlautRaising
```

```
.o. OEIUmlautDiphthong;
```

The literature presents fronting, raising, and diphthongal mutation as effects of one historical development. They consequently occupy a single place in the Old English chronology.

The lower boundary is consonantal. If general umlaut precedes velar palatalization, PGmc **kui* ‘cow’ yields *cȳ* ‘cows’ rather than expected *cȳ* ‘cows’, and PGmc **lūnganjō* ‘lungs’ yields *lunġen* ‘lungs’ rather than expected *lungen* ‘lungs’. These over-palatalized forms place general umlaut after velar palatalization.

The upper boundary separates general umlaut from the narrower West Saxon process. If West Saxon palatal diphthongization precedes umlaut, PGmc **gēftiz* ‘gift’ yields *ġieft* ‘gift’ rather than expected OE *ġift* ‘gift’, and **skáiθiz* ‘sheath’ yields *scēþ* ‘sheath’ rather than expected *scēap* ‘sheath’. Together the two witness pairs place general umlaut after velar palatalization and before the West Saxon process.

SC056. West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

```
define OEWsPalatalDiphthongization [
  { *æ } -> { *ea } || .#. [ { *tj } | { *tj̥ } | { *f } | { *j } ] _ [ EnglishStarConsonant |
    ↪ EnglishPalatalConsonant | .#. ],
  { *ǣ } -> { *ēa } || .#. [ { *tj } | { *tj̥ } | { *f } | { *j } ] _ [ EnglishStarConsonant |
    ↪ EnglishPalatalConsonant | .#. ],
  { *e } -> { *ie } || .#. [ { *tj } | { *tj̥ } | { *f } | { *j } ] _ [ EnglishStarConsonant |
    ↪ EnglishPalatalConsonant | .#. ],
  { *ē } -> { *īe } || .#. [ { *tj } | { *tj̥ } | { *f } | { *j } ] _ [ EnglishStarConsonant |
    ↪ EnglishPalatalConsonant | .#. ],
  { *é } -> { *īe } || .#. [ { *tj } | { *tj̥ } | { *f } | { *j } ] _ [ EnglishStarConsonant |
    ↪ EnglishPalatalConsonant | .#. ],
  { *ē } -> { *īe } || .#. [ { *tj } | { *tj̥ } | { *f } | { *j } ] _ [ EnglishStarConsonant |
    ↪ EnglishPalatalConsonant | .#. ]
];
```

West Saxon *gielðan* ‘pay’, *sciold* ‘shield’, and *scieppan* ‘create’ show diphthongization after an already palatal consonant (? , §6.5.1). Their dialectal and phonological restriction separates this development from system-wide i-umlaut.

Hogg’s *giefan* ‘give’ and *sceap* ‘sheep’ belong to the same palatal-consonant environment (?). Fulk likewise assigns this diphthongization a place before front mutation and distinguishes the two processes (? , §4.13).

The forms *ġift* ‘gift’ and *scēap* ‘sheath’ fix the lower boundary. If West Saxon palatal diphthongization precedes general i-umlaut, PGmc **gēftiz* ‘gift’ yields *ġieft* ‘gift’ rather than expected *ġift*, and PGmc **skáiθiz* ‘sheath’ yields *scēþ* ‘sheath’ rather than expected *scēap*. These witnesses place West Saxon palatal diphthongization after general umlaut; no tested lexical item supplies a later terminus ante quem.

The one-sided chronology reflects the difference in scale. General umlaut reorganizes the vowel system, whereas West Saxon palatal diphthongization affects a narrower dialectal class after palatal consonants. Its exact later placement remains undemonstrated by the present lexicon.

2.38 J-cluster coalescence

Historical discussion

Only a small lexical group reveals the coalescence of velars with **j*. Plain-velar and **sk* palatalization must already have run before **gj* and **kj* acquire their later outcomes. Campbell, Ringe and Taylor, and Fulk discuss the palatalized and fronted outcomes in *bīēgan* ‘bend’ and *sēcān* ‘seek’ without assigning this later cluster adjustment the status of a major sound law (?, §§170, 248–251; ?, §§6.4.1, 6.5.1, 6.6.1–6.6.4; ?, §§4.7, 4.13).

SC057. Coalescence of velar + **j* clusters (OEJClusterCoalescence)

```
define OEJClusterCoalescence (
  [{*g} {*j} -> {*ǵ}]
  .o. [{*k} {*j} -> {*ǵ}]
);
```

The forms *bīēgan* ‘bend’ and *sēcān* ‘seek’ determine the earlier boundary. If coalescence precedes SC052 OEVelarPalatalization, the developments behind *bīēgan* ‘bend’ and *sēcān* ‘seek’ are lost. Related forms such as *fylġan* ‘follow’, *heċġ* ‘hedge’, and *sengan* ‘sing’ fail in the same broader palatalization zone. PGmc **bāugijaną* yields *bēaġan* ‘bend’ rather than expected OE *bīēgan*, and PGmc **sōkijaną* yields *sōcān* ‘seek’ rather than expected *sēcān*. This demonstrates that velar palatalization preceded coalescence. Nothing in the present lexicon supplies a terminus ante quem.

2.39 Nasal dissimilation

Historical discussion

Most accounts introduce nasal dissimilation to explain individual forms rather than as a regular sound law. Luick records *enetre* ‘yearling’ (spelled *enitre* ‘yearling’ in his text) (?); Campbell discusses *heofon* ‘heaven’ with suffixal variation (?); and Hogg encounters the same form while treating back mutation (?).

Fulk supplies the clearest general formulation: “In the cluster *mn*, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular” (? §6.11). Ringe and Taylor stay close to the lexical evidence and note that *enetre* ‘yearling’ reflects “loss of the second **n* by dissimilation” (?).

The disagreement concerns scope. Fulk’s formulation recognizes a recurrent but irregular development in *mn*; the remaining discussions stay with particular lexical outcomes. None warrants a sound law comparable in scope to the major Old English vowel changes.

SC058. Nasal dissimilation in short-vowel environments (OENasalDissimilation)

```
define OENasalDissimilation [
  {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n}
  ↪ [EnglishStarShortVowel | .#. ]
];
```

I adopt a narrower environment than the handbook observations might suggest. Fulk formulates the tendency at the level of *mn* clusters and illustrates it with *heofon* ‘heaven’ and *fæstenn* ‘fasting’ (? §6.11). Ringe and Taylor show the same kind of development in *enetre* ‘yearling’ (?). Campbell’s “*heofon* is for older *hefzen*” and Hogg’s sequence **hefon* > *heofon* preserve outcomes of the same kind (?; ?). The short-vowel environment adopted here covers a recurrent subset of these outcomes, not every dissimilatory development involving nasals.

No witness word fixes the position of nasal dissimilation within the Old English sequence. Reversing its order with any tested neighbor leaves every checked output unchanged. A more precise relative chronology would therefore require lexical evidence not represented here.

2.40 Back mutation

Historical discussion

West Saxon *giefan* ‘give’ and *wefan* ‘weave’ stand against non-West-Saxon *geofad* ‘gave’ and *weofan* ‘weave’. Ringe and Taylor use this contrast to define the dialectal profile of back mutation (?, §6.9.4). Campbell’s treatment of diphthongization before following back vowels includes *heofon* ‘heaven’ (?, §207), while Hogg draws the instructive comparison with breaking (?). Fulk accordingly separates back mutation from the earlier umlautal changes (?, §4.8).

SC059. Back mutation before labials and liquids (OEBackMutation)

```
define OEBackMutation [
  {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
  {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
    ↪ EnglishBackMutationTrigger,
  {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

Three witness forms bracket the chronology. If back mutation precedes SC048 OESecondaryNasalization, forms such as **gébanq* ‘give’ produce *geofan* ‘give’; the expected form is *giefan* ‘give’. **stélanq* ‘steal’ likewise produces *steolan* ‘steal’; the expected form is *stelan* ‘steal’. At the other edge, delaying back mutation until after SC078 OEWeakTailReduction makes **wébanq* ‘weave’ yield *weofan* ‘weave’; the expected form is *wefan* ‘weave’. Thus back mutation follows secondary nasalization but precedes the weak-tail reductions.

2.41 West Saxon palatal umlaut

Historical discussion

The reflexes *miht* ‘might’ and *niht* ‘night’ place West Saxon palatal umlaut after the principal umlautal developments. Campbell and Ringe and Taylor describe the forms themselves; Fulk supplies the broader chronology of palatal-vowel change (?, §§248–251; ?, §§6.5.1, 6.6.1–6.6.4; ?, §§4.7, 4.13).

SC060. West Saxon palatal umlaut before **h*-clusters (OEWsPalatalUmlaut)

```
define OEWsPalatalUmlaut [
  {*eo} -> {*i} || _ OEHCluster .#.,
  {*io} -> {*i} || _ OEHCluster .#.,
  {*ie} -> {*i} || _ OEHCluster .#.,
  {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
  {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
  {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
  {*éo} -> {*i} || _ OEHCluster .#.,
  {*ío} -> {*i} || _ OEHCluster .#.,
  {*íe} -> {*i} || _ OEHCluster .#.,
  {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
  {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
  {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

The change to **i* before **h*-clusters can be ordered only on its earlier side. If palatal umlaut precedes SC055 OEIUmlaut, the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage [†]*mieht* and [†]*nieht* rather than expected OE *miht* and *niht*. Consequently, i-umlaut precedes palatal umlaut. Reordering the latter against any tested later change leaves both witness forms unchanged.

2.42 Weak-tail nasal loss

Historical discussion

The pathway from **dōnq* ‘do’ to *dōn* ‘do’ supplies the sole lexical thread through this reduction. Campbell, Hogg, and Fulk place such weak-tail losses among apocope and related late reductions (? , §§345–349; ?; ? , §5.6). The witness, however, ties the change to a much older development. Its immediate neighbors remain untested.

SC061. Reduction of final nasal weak-tail endings (OEWeakTailNasalLoss)

```
define OEWeakTailNasalLoss [
  { *n } { *q } -> { *n } | | _ .# . ,
  { *m } { *q } -> { *m } | | _ .# .
];
```

Final weak-tail **-nq* and **-mq* accordingly yield plain **-n* and **-m*.

Only *dōn* ‘do’ constrains the relative order. Placing this loss before the older n-stem loss makes the derivation record no output instead of expected OE *dōn* ‘do’. The older loss must therefore precede weak-tail nasal loss. Nothing in the current lexicon distinguishes among its possible later positions, and one witness cannot establish a wider historical development.

2.43 High-vowel apocope

Historical discussion

Final high vowels must survive long enough to condition umlaut before apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk agree on this Old English development, though they differ over the extent of the surrounding syncope (? , §§345–349; ?; ? , §§6.8.1, 6.8.4; ? , §5.6).

SC063. High-vowel apocope after heavy syllables and in trisyllables (OEHighVowelApocope)

```
define OEHighVowelApocope [
  {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
    ↪ EnglishStarShortVowel OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
    ↪ EnglishStarShortVowel OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
    ↪ EnglishStarShortVowel OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
    ↪ OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
    ↪ EnglishStarShortVowel OEAnyConsonant+ _ .#.,
  {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
    ↪ EnglishStarShortVowel OEAnyConsonant+ _ .#.,
  {*y} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
    ↪ EnglishStarShortVowel OEAnyConsonant+ _ .#.,
  {*i} -> 0 || EnglishStarLongVowel _ .#.,
```

{*u}	-> 0		{*x}	_	.#.	,
{*u}	-> 0		{*x}	_	.#.	,
{*i}	-> 0		{*x}	_	.#.	

l;

Final **i*, **u*, and **u* cannot disappear before completing their umlautal work. Applied before SC055 OEIUmlaut, PGmc **kūi* ‘cow’ yields [†]*cū* rather than expected OE *cȳ* ‘cow’, and PGmc **brūdiz* ‘bride’ yields [†]*brūd* rather than expected OE *brȳd* ‘bride’. Conversely, if apocope waits until after SC072 OEUnstressedLongVowelShortening, PGmc **fūrxtīnaz* ‘fright’ yields [†]*fyrht* rather than expected OE *fyrhte* ‘fright’. The three witnesses establish the sequence i-umlaut, high-vowel apocope, unstressed long-vowel shortening.

2.44 Post-apocope **n*-loss and medial syncope

Historical discussion of post-apocope **n*-loss and medial syncope

Evidence for post-apocope reduction is strikingly uneven. The inherited **furht-* family makes the survival of one nasal diagnostic and fixes both sides of stem-final *n*-loss (?). No comparable witness orders the medial syncope that follows. Hogg, Ringe and Taylor, and Fulk describe both processes within the late history of weak syllables (?; ?, §§6.7.3–6.8.4; ?, §5.6).

SC064. Loss of stem-final **n* after long **ī* (NWGmcInStemNLoss)

```
define NWGmcInStemNLoss [{*n} -> 0 || {*ī} _ .#.];
```

Only final **n* after long **ī* is at issue, as in the inherited family behind *fyrhte* ‘fright’.

The same proto-form fixes both edges. Before SC041 PWGmcFinalBareALoss, PGmc **fūrxtīnaz* ‘fright’ yields *†fyrhten* rather than expected OE *fyrhte* ‘fright’. After SC072 OEUnstressedLongVowelShortening, PGmc **fūrxtīnaz* again yields *†fyrhten* rather than expected *fyrhte* ‘fright’. I therefore order final bare-a loss, stem-final *n*-loss, and unstressed long-vowel shortening in that sequence. Both boundaries are firm within the derivation, but depend upon one lexical family.

SC065. Medial syncope before dentals after heavy syllables (OEMedialSyncope)

Loss of medial **i* before dentals belongs to the late weak-tail history described by Hogg, Ringe and Taylor, and Fulk (?; ?, §§6.7.3–6.8.4; ?, §5.6).

```
define OEMedialSyncope [
  {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
  {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
  {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
    ↳ [{*θ}|{*ð}|{*d}|{*t}]
];
```

No diagnostic word establishes a local chronology. Moving medial syncope to either end of the tested range leaves every checked output unchanged. Its handbook placement after apocope and before later cluster simplification therefore remains preferable, but the present lexicon cannot demonstrate it.

2.45 Late syncope and degemination

Historical discussion of late syncope and degemination

Vowel loss creates the clusters upon which later assimilation and degemination operate. Hogg and Ringe and Taylor describe this dependence, while Brunner's *netle* 'nettle' beside later *nete* 'nettle' supplies a concrete lexical type (?, ?, §§6.7.3–6.8.2; ?, §§158–159). Fulk places this syncope after i-umlaut (?, §5.6).

The three relations are not equally secure. Lexical evidence orders syncope and degemination; the intervening dental assimilation has no independent ordering witness.

SC066. L-adjacent syncope in medial syllables (OELAdjacentSyncope)

```
define OELAdjacentSyncope [
  {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
  {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
  {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
];
```

The loss of medial **i* before **l* is late enough to preserve earlier umlaut, as *netle* 'nettle' and *spinl* 'spindle' demonstrate.

Placed before i-umlaut, PGmc **nátílōn* 'nettle' yields [†]*nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* 'spindle' yields [†]*spenl* rather than expected *spinl* 'spindle'. Placed after preconsonantal degemination, PGmc **spénnilō* yields [†]*spinnl* rather than expected *spinl*. The witnesses therefore establish the sequence i-umlaut, l-adjacent syncope, preconsonantal degemination. The first relation separates two historical phases; the second is a direct feeding relation, since syncope creates the cluster that degemination simplifies.

SC067. Dental assimilation in newly formed clusters (OEDentalAssimilation)

```
define OEDentalAssimilation [
  {*θ} -> 0 || {*t} _
];
```

Loss of **θ* after **t* resolves a dental cluster produced by syncope (?, ?, §§6.7.5, 6.8.2). No witness distinguishes its position: moving dental assimilation across every tested neighbor leaves the outputs unchanged. I nevertheless place it after syncope, which supplies its input, and before the more general cluster simplification described in the handbooks. This order is phonologically motivated, not established by a lexical contrast.

SC068. Preconsonantal degemination before sonorants (OEPreconsonantalDegemination)

```
define OEPreconsonantalDegemination OEPreconsonantalDegemTT .o.
  ⇨ OEPreconsonantalDegemNN;
```

Preconsonantal **tt* and **nn* simplify only after syncope has created a following sonorant cluster, as in *spinl* 'spindle' (?, §§6.7.5, 6.8.2).

Placed before l-adjacent syncope, PGmc **spénnilō* 'spindle' yields [†]*spinnl* rather than expected OE *spinl* 'spindle'. Syncope must therefore create the cluster before degemination simplifies it. Reordering degemination against any tested later change leaves the witness unchanged, so no terminus ante quem is known.

2.46 Early o-shortening

Historical discussion

After the principal palatal andumlautal changes, unstressed vowels undergo shortening, fronting, merger, and sometimes complete loss. Campbell describes the early shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk relate it to apocope, syncope, and the later reductions (? , §355; ?; ? , §§6.8.3–6.9.3; ? , §§5.6–5.7).

Early o-shortening has only a distant earlier boundary. The rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening, have more closely defined relations.

SC069. Early shortening of unstressed **ō* before nasals (OEEarlyOShortening)

```
define OEEarlyOShortening [
  { *ō } -> { *a } || EnglishStarVocalic [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant]+ _ EnglishStarNasal
];
```

The rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Moving the rule before SC023 NWGmcNStemNLoss, PGmc **nēdrōn* ‘adder’ yields †*nædran* rather than expected OE *nædre* ‘adder’, PGmc **érθōn* ‘earth’ yields †*eorþan* rather than expected *eorþe* ‘earth’, and PGmc **fláskōn* ‘flask’ yields †*flaskan* rather than expected *flasce* ‘flask’. The same earlier shift also disrupts forms such as *heorte* ‘heart’ and *līne* ‘line’. This broad set of failures requires SC069 OEEarlyOShortening to follow SC023 NWGmcNStemNLoss.

If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one. The checked forms therefore do not identify a corresponding later constraint. The sources place early **ō*-shortening before the later weak-tail changes without fixing a closer local order.

2.47 Early unstressed fronting and later o-shortening

Historical discussion of early unstressed fronting and later o-shortening

Campbell distinguishes the shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk place fronting and shortening within a later history of syncope and final-vowel adjustment (?, §355; ?, §6.8.3–6.9.3; ?, §§5.6–5.7). Earlier unstressed fronting precedes later o-shortening.

SC070 OEUnstressedFrontingEarly has both an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms their reciprocal order, but no checked form fixes its later boundary.

SC070. Early fronting of unstressed *a (OEUnstressedFrontingEarly)

```
define OEUnstressedFrontingEarly OEUnstressedAFronting;
```

The rule fronts unstressed *a to *æ after the earlier shortening has created a frontable vowel but before the later shortening of unstressed *ō. It produces endings such as OE *-en* in *lungen* ‘lungs’.

If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* ‘lungs’ yields †*lungen* rather than expected OE *lungen* ‘lungs’. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* ‘bears’ yields †*boreþ* rather than expected OE *borap* ‘bears’, and PGmc **ménōθz* ‘month’ yields †*mōneþ* rather than expected *mōnaþ* ‘month’. The witness forms require SC070 OEUnstressedFrontingEarly to follow SC052 OEVelarPalatalization and precede SC071 OELateOShortening.

The relation to SC071 OELateOShortening is local. The earlier boundary at SC052 OEVelarPalatalization places fronting after the older palatal developments.

SC071. Later shortening of unstressed *ō (OELateOShortening)

The following rule handles the later shortening stage.

```
define OELateOShortening [
  { *ō } -> { *a } || EnglishStarVocalic [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant]*
];
```

The rule shortens the remaining unstressed long *ō after fronting, producing the later “stable a” endings in OE *borap* ‘bears’ and *liornaþ* ‘learns’.

Moving the rule before SC070 OEUnstressedFrontingEarly makes PGmc **búrōθi* ‘bears’ yield †*boreþ* rather than expected OE *borap* ‘bears’, and PGmc **líznoθi* ‘learns’ yield †*liorneþ* rather than expected *liornaþ* ‘learns’. The contrast requires SC071 OELateOShortening to follow SC070 OEUnstressedFrontingEarly. Moving it later within the tested range creates no equally sharp failure.

2.48 Unstressed long-vowel shortening and ae-merger

Historical discussion of unstressed long-vowel shortening and ae-merger

Campbell describes the shortening of unaccented long vowels, and Ringe and Taylor place it among the last prehistoric Old English changes before the merger of unstressed **æ* with **e* (§355; ?, §6.8.3–6.9.3; §5.6–5.7).

SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a reciprocal ordering relation. SC064 NWGmcInStemNLoss supplies the earlier boundary of shortening, and SC085 OEHLoss the later boundary of the merger.

SC072. Shortening of unstressed long vowels (OEUnstressedLongVowelShortening)

```
define OEUnstressedLongVowelShortening OEUnstressedLongVowelShortening1
  .o. OEUnstressedLongVowelShortening2
  .o. OEUnstressedLongVowelShortening3
  .o. OEUnstressedLongVowelShortening5
  .o. OEUnstressedLongVowelShortening6
  .o. OEUnstressedLongVowelShortening7
  .o. OEUnstressedLongVowelShortening8;
```

The rule shortens the remaining unstressed long vowels before weak final syllables reach their later forms. A small group of lexical witnesses fixes its chronology.

If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* ‘fright’ yields *†fyrhten* rather than expected OE *fyrhte* ‘fright’. If the rule is delayed until after SC073 OEUnstressedAEMerger, PGmc **nédrōn* ‘adder’ yields *†nædræ* rather than expected OE *nædre* ‘adder’, and PGmc **fādēr* ‘father’ yields *†fædær* rather than expected *fæder* ‘father’. These outputs require SC072 OEUnstressedLongVowelShortening to follow SC064 NWGmcInStemNLoss and precede SC073 OEUnstressedAEMerger.

Shortening therefore follows the earlier weak-tail preparation and immediately precedes the merger.

SC073. Merger of unstressed **æ* with **e* (OEUnstressedAEMerger)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger OEWeakTailReduction3;
```

The rule merges unstressed **æ* with **e* after shortening has produced the weak final vowels, yielding the ordinary OE *-e* spellings.

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **nédrōn* ‘adder’ yields *†nædræ* rather than expected OE *nædre* ‘adder’, and PGmc **fādēr* ‘father’ yields *†fædær* rather than expected *fæder* ‘father’. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* ‘toe’ yields *†tāæ* rather than expected OE *tā* ‘toe’. These failures show that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The checked forms fix the local order after SC072 OEUnstressedLongVowelShortening and place the merger before the later h-loss and contraction.

2.49 Medial unstressed-*i* lowering

Historical discussion of medial unstressed-*i* lowering and **ng* retention

Hogg and Ringe and Taylor treat the late weakening and merger of unstressed vowels as a continuing history (?; ?, §§6.9.5–6.9.6). SC074 OEMedUnstressedILowering1 lowers medial unstressed *i*; SC075 OEMedUnstressedILowering preserves *i* before **ng* in words of the *scilling* ‘shilling’ type.

General lowering precedes the restricted restoration before **ng*. The evidence is narrower than that for SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

SC074. First medial unstressed-*i* lowering (OEMedUnstressedILowering1)

```
define OEMedUnstressedILowering1 [
  {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant]+ _
];
```

The rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. The resulting *e*-outcome is reversed before **ng*.

If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* ‘fright’ yields [†]*fyrhti* rather than expected OE *fyrhte* ‘fright’. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skillingaz* ‘shilling’ yields [†]*scilleng* rather than expected *scilling* ‘shilling’. The derivations require SC074 OEMedUnstressedILowering1 to follow SC072 OEUnstressedLongVowelShortening and precede SC075 OEMedUnstressedILowering.

The evidence is narrow on each side. The rule follows unstressed long-vowel shortening and precedes the more specific **ng* preservation.

SC075. Preservation of medial unstressed **i* before **ng* (OEMedUnstressedILowering)

The following rule reverses the lowering before **ng*.

```
define OEMedUnstressedILowering [
  {*e} -> {*i} || _ {*n} {*g}
];
```

The rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *scilling* ‘shilling’.

Moving the rule before SC074 OEMedUnstressedILowering1 makes PGmc **skillingaz* ‘shilling’ yield [†]*scilleng* rather than expected OE *scilling* ‘shilling’. On this evidence, I take SC075 OEMedUnstressedILowering to follow SC074 OEMedUnstressedILowering1. Moving it later within the tested range creates no equally sharp failure.

2.50 Prefix i-reduction

Historical discussion

Late weak-tail reduction affects unstressed prefixes as well as inflectional endings and medial vowels. Fulk's discussion of prefix vowels accounts for OE **be-* and **ne-* (? , §5.7). Hogg and Ringe and Taylor place such weakening within the broader late history of unstressed vowels (? , §§6.8.3–6.9.6).

The tested forms do not determine the rule's position relative to a neighboring change.

SC076. Reduction of prefixal **i* in unstressed position (OEPrefixIReduction)

```
define OEPrefixIReduction [
  {*i} -> {*ë} || .#. [{*b} | {*n}] _ [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant] EnglishStarVocalic
];
```

The rule reduces unstressed prefixal **i* to a weaker vowel in the *bi-* and *ni-* type prefixes before a consonant plus a following vowel. The development accounts for later prefix spellings such as OE **be-* and **ne-*.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC076 OEPrefixIReduction before or after any specific neighboring change.

The handbooks attest late prefix-vowel weakening, but the precise placement remains approximate. No lexical failure fixes it.

2.51 Weak-tail reduction

Historical discussion

Campbell, Hogg, Ringe and Taylor, and Fulk describe a late history in which apocope, shortening, contraction, and further weak-tail reductions reshape final syllables (? §355; ?; ? §§6.8.3–6.9.3; ? §5.6). Lexical failures place the remaining weak-tail reduction after unstressed fronting and before contraction.

SC078. Reduction of remaining weak-tail vowels (OEWeakTailReduction)

```
define OEWeakTailReduction OEWeakTailReduction1;
```

The rule reduces the remaining weak-tail vowels, preventing a broad class of *-en* and extra-vowel outcomes.

I place the change after SC070 OEUnstressedFrontingEarly and before SC086 OEContraction. Moving it before SC070 OEUnstressedFrontingEarly, PGmc **bákanq* ‘bake’ yields [†]*bacen* rather than expected OE *bacan* ‘bake’, and PGmc **bíndanq* ‘bind’ yields [†]*binden* rather than expected *bindan* ‘bind’, alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc **fléuxanq* ‘flee’ yields [†]*flēoan* rather than expected OE *flēon* ‘flee’, and PGmc **sláxanq* ‘slay’ yields [†]*sleaan* rather than expected *slēan* ‘slay’.

The earlier boundary spans a wide interval and does not establish a close neighboring relation. The later boundary is narrower: SC078 OEWeakTailReduction precedes SC086 OEContraction.

2.52 Final-j loss and final geminate simplification

Historical discussion of final-j loss and final geminate simplification

After SC079 OEJLossAfterHeavy removes **j* in heavy environments, forms such as *lungen* ‘lungs’ acquire a final geminate. SC080 OEFinalGeminateSimplification then removes the second nasal.

SC079 OEJLossAfterHeavy has a broad earlier boundary at SC055 OEIUmlaut. SC080 OEFinalGeminateSimplification is fixed only by the final *nn* outcome in the following derivation.

SC079. Loss of **j* after heavy syllables (OEJLossAfterHeavy)

```
define OEJLossAfterHeavy [
  {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
  ↳ [EnglishStarConsonantNoR | EnglishPalatalConsonant] _ ,
  {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
  ↳ EnglishPalatalConsonant] [EnglishStarConsonantNoR |
  ↳ EnglishPalatalConsonant] _
];
```

The rule removes **j* after the relevant heavy-syllable configurations, after the earlier umlaut-sensitive vocalism has developed. The affected glide is **j*.

If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijanq* ‘believe’ yields *†gelēafan* rather than expected OE *geliefan* ‘believe’, PGmc **báugijanq* ‘bow’ yields *†bēagan* rather than expected *bīegan* ‘bow’, and PGmc **fúlgijanq* ‘follow’ yields *†fulgan* rather than expected *fylgan* ‘follow’. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* ‘lungs’ yields *†lungenn* rather than expected OE *lungen* ‘lungs’. I accordingly take SC079 OEJLossAfterHeavy to follow SC055 OEIUmlaut and precede SC080 OEFinalGeminateSimplification.

The earlier boundary is broad, but the relation to final geminate simplification is local.

SC080. Simplification of final geminates (OEFinalGeminateSimplification)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
  {*n} -> 0 || {*n} _ .#.
];
```

The rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Moving the rule before SC079 OEJLossAfterHeavy makes PGmc **lúnganjō* ‘lungs’ yield *†lungenn* rather than expected OE *lungen* ‘lungs’. These failures require SC080 OEFinalGeminateSimplification to follow SC079 OEJLossAfterHeavy. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

2.53 J-strengthening, vocalization, and ei-contraction

Historical discussion of j-strengthening, vocalization, and ei-contraction

SC081 OEJStrengtheningAfterFrontDiphthong preserves a consonantal outcome after front diphthongs. SC082 OEIntervocalicJVocalization then vocalizes the remaining intervocalic **j*, and SC083 OEUnstressedEIContraction removes the resulting *ei*-like sequence in weak verbal endings.

The output of each rule conditions the next. SC082 OEIntervocalicJVocalization has local lexical evidence on both sides; SC081 OEJStrengtheningAfterFrontDiphthong has a distant earlier boundary, and SC083 OEUnstressedEIContraction has no tested later boundary.

SC081. Strengthening of **j* after front diphthongs (OEJStrengtheningAfterFrontDiphthong)

```
define OEJStrengtheningAfterFrontDiphthong [
  {*j} -> {*ʒ} || [{*ēa}|{*ēā}|{*īe}|{*īē}|{*éa}] _ EnglishStarVocalic
];
```

After the relevant front diphthongs, **j* first strengthened to a consonantal outcome; otherwise it would have vocalized too early.

If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjanq* ‘strew’ yields *†strēagan* rather than expected OE *strīegan* ‘strew’. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields *†strīeian* rather than *strīegan*. The order test requires SC081 OEJStrengtheningAfterFrontDiphthong to follow SC055 OEIUmlaut and precede SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval. The *strīegan* ‘strew’ derivation fixes the local relation to SC082 OEIntervocalicJVocalization.

SC082. Intervocalic vocalization of **j* (OEIntervocalicJVocalization)

```
define OEIntervocalicJVocalization [
  {*j} -> {*i} || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule vocalizes intervocalic **j* to **i*, creating the *ei*-like sequence later removed by SC083 OEUnstressedEIContraction in many weak verb forms.

Moving the rule before SC081 OEJStrengtheningAfterFrontDiphthong makes PGmc **stráwjanq* ‘strew’ yield *†strīeian* rather than expected OE *strīegan* ‘strew’. Delaying it until after SC083 OEUnstressedEIContraction makes PGmc **búrōjanq* ‘bore’ yield *†boreian* rather than expected OE *borian* ‘bore’, PGmc **xándlōjanq* ‘handle’ yield *†handleian* rather than expected *handlian* ‘handle’, and PGmc **mákōjanq* ‘make’ yield *†maceian* rather than expected *macian* ‘make’. The witness forms require SC082 OEIntervocalicJVocalization to follow SC081 OEJStrengtheningAfterFrontDiphthong and precede SC083 OEUnstressedEIContraction.

SC082 OEIntervocalicJVocalization is therefore ordered between strengthening and contraction.

SC083. Contraction of unstressed *ei* (OEUnstressedEIContraction)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
  {*e} -> ∅ || EnglishStarVocalic [EnglishStarConsonant |
    ↪ EnglishPalatalConsonant]+ _ {*i}
];
```

1;

The rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* ‘bore’ and *liccian* ‘lick’.

Moving the rule before SC082 OEIntervocalicJVocalization makes PGmc **būrōjaną* ‘bore’ yield [†]*boreian* rather than expected OE *borian* ‘bore’, PGmc **līznōjaną* ‘learn’ yield [†]*liorneian* rather than expected *liornian* ‘learn’, and PGmc **līkkōjaną* ‘lick’ yield [†]*licceian* rather than expected *liccian* ‘lick’. The contrast requires SC083 OEUnstressedEIContraction to follow SC082 OEIntervocalicJVocalization. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

2.54 H-loss and contraction

Historical discussion of h-loss and contraction

When SC085 OEHLoss removes intervocalic **h*, it creates hiatus. SC086 OEContraction immediately resolves the resulting vowel sequence.

Ringe and Taylor describe this late sequence of *h*-loss and contraction (?, §§6.9.1–6.9.3). Fulk places the contracted verbs in a broader Germanic context (?, §12.21), and Luick describes the corresponding West Germanic contractions (?).

SC085. Loss of intervocalic **h* (OEHLoss)

```
define OEHLoss [
  {*x} -> Ø || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* ‘toe’ yields [†]*tāæ* rather than expected OE *tā* ‘toe’. If it is delayed until after SC086 OEContraction, PGmc **fléuxanq* ‘flee’ yields [†]*flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxanq* ‘slay’ yields [†]*sleaan* rather than expected *slēan* ‘slay’, PGmc **téxun* ‘draw’ yields [†]*teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* yields [†]*tāæ* rather than expected *tā*. These outputs require SC085 OEHLoss to follow SC073 OEUnstressedAEMerger and precede SC086 OEContraction.

The earlier boundary rests on one witness; the four later witnesses establish the immediate relation to contraction.

SC086. Contraction of the resulting hiatus (OEContraction)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
  {*a} {*a} -> {*ā},
  {*e} {*e} -> {*ē},
  {*i} {*i} -> {*ī},
  {*o} {*o} -> {*ō},
  {*u} {*u} -> {*ū},
  {*ea} {*a} -> {*ēa},
  {*ēa} {*a} -> {*ēa},
  {*eo} {*a} -> {*ēo},
  {*ēo} {*a} -> {*ēo},
  {*eo} {*o} -> {*ēo},
  {*ēo} {*o} -> {*ēo},
  {*éo} {*o} -> {*éo},
  {*éo} {*o} -> {*éo},
  {*ā} {*a} -> {*ā},
  {*ā} {*e} -> {*ā},
  {*ē} {*a} -> {*ē},
  {*ē} {*e} -> {*ē},
  {*é} {*a} -> {*é},
  {*é} {*e} -> {*é},
  {*ī} {*a} -> {*ī},
  {*ī} {*e} -> {*ī},
  {*í} {*a} -> {*í},
  {*í} {*e} -> {*í},
  {*ō} {*a} -> {*ō},
  {*ō} {*e} -> {*ō},
  {*ū} {*a} -> {*ū},
  {*ū} {*e} -> {*ū}
];
```

1;

The rule contracts the vowel sequences created after *h*-loss, producing *flēon* ‘flee’, *slēan* ‘slay’, and *tēon* ‘draw’.

Moving contraction before SC085 OEHLoss makes PGmc **fléuxanq* ‘flee’ yield *†flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxanq* ‘slay’ yield *†sleaan* rather than expected *slēan* ‘slay’, PGmc **tēxun* ‘draw’ yield *†teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* ‘toe’ yield *†tāe* rather than expected *tā* ‘toe’. The derivations require SC086 OEContraction to follow SC085 OEHLoss. Moving it later within the tested range before SC087 OERMetathesis creates no new failure. The more distant SC078 OEWeakTailReduction relation establishes only that weak-tail reduction precedes contraction.

2.55 R-metathesis

Historical discussion

Sievers-Brunner describes r-metathesis in forms such as *berstan* ‘burst’, *forst* ‘frost’, and *cærse* ‘cress’ (? , §179). Luick likewise treats it as a later rearrangement whose interaction with breaking remains variable (?).

The evidence establishes that breaking precedes metathesis. It does not establish an ordering relation between SC086 OEContraction and SC087 OERMetathesis.

SC087. Metathesis of **r* with a following short vowel (OERMetathesis)

```
define OERMetathesis [
  {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
  {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

The rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* ‘burst’ where an earlier order would still show a broken vowel sequence.

Moving the rule before SC044 OEBreaking makes PGmc **bréstanq* ‘burst’ yield *†beorstan* rather than expected OE *berstan* ‘burst’. On this evidence, I take SC087 OERMetathesis to follow SC044 OEBreaking. Moving it later within the tested sequence alters none of the checked outputs.

The checked forms fix the earlier relation but do not identify a corresponding later constraint. The sources treat r-metathesis as a late rearrangement after breaking without placing it immediately beside contraction.

Part II

Lexical derivations

Chapter 3

Word-by-word derivations

3.1 Introduction

The catalogue groups words by the kind of historical explanation they require. Regular derivations establish the reach of the sound laws. The remaining classes mark, without concealment, the places where attestation, morphology, analogy, or an unresolved mismatch intervenes.

The catalogue proceeds from words; the preceding part proceeds from rules. Together they allow each proposed sound law to be tested against its lexical witnesses and each etymology against the complete ordered history.

3.2 Data and sources

This volume assembles the lexical corpus from the aligned Germanic dataset and the compact derivation traces that accompany each entry. Comparative dictionaries, Old English dictionaries, and historical grammars are cited in the prose where they bear on particular lexical arguments.

The result is a lexical catalogue rather than a separate report on citation method or trace machinery.

3.3 Transducer and derivation method

Four objects must be distinguished in every derivation: the citation reconstruction, the selected input, the transducer outcome, and the Old English target. The summary identifies them where they differ; the boxed trace then divides the changes into Earlier Germanic and Old English stages.

3.4 Derivation classes

The lexical catalogue is ordered by seven derivation classes in the current manifest. Counts in this alpha are:

- Regular derivations: 70
- Attested variants: 4
- Early analogy: 35
- Late analogy: 28
- Reconstructed Old English comparators: 3
- Known but unmodelled remodellings: 2
- Unexplained or deliberately unmodelled exceptions: 5

3.5 Regular derivations

In these entries the selected Germanic input yields the Old English reflex by regular sound change. They establish the standard against which the following analogical and exceptional histories must be judged.

adder — OE *nædre*

Derivation: **nédrōn* > *nædre* (regular).

Derivation trace

Proto input: **nédrōn*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel	<i>*nædræ</i>
[no change]		Shortening	
Northwest Germanic		OE Unstressed AE Merger	<i>*nædre</i>
NWGmc N Stem N Loss	<i>*nédrō</i>		
NWGmc Long E Lowering	<i>*nædrō</i>		

Old English form: *nædre*

Reconstruction and comparative evidence

Kroonen distinguishes the masculine snake word **nadra-* from a feminine ablauting formation **nēdrōn-*, and gives Old English *nædre*, *næddre* under the latter (?). Orel likewise points from the masculine entry to a feminine **nēdrōn* ‘adder’ ~ **nadrōn* ‘adder’ type (?).

The derivational input therefore is not a reshaped convenience form. It is the comparative reconstruction that specifically underlies the Old English noun.

Old English evidence

The Old English word is securely represented by *nædre*, with *næddre* as a secondary variant. Clark Hall cross-references *næddre* to *nædre*, and Fulk treats *næddre* as the later geminated form beside the older base (?; ?).

Development to Old English

From **nédrōn* ‘adder’, the stressed long mid vowel develops to Old English *nædre*, and the weak feminine ending remains as final *-e*, giving *nædre*. The doubled consonant of *næddre* is secondary and does not alter the inherited base form.

bake — OE *bacan*

Derivation: **bákaną* > *bacan* (regular).

Derivation trace

Proto input: **bákaną*

Earlier Germanic changes		Old English changes
West Germanic		Anglo Frisian Brightening
[no change]		OE A Restoration
Northwest Germanic		OE Heavy Syllable Nasal Apocope
[no change]		OE Secondary Nasalization
		OE Weak Tail Reduction

Old English form: *bacan*

Reconstruction and comparative evidence

Orel reconstructs the verb as **bakanan* ‘bake’ and cites Old English *bacan* beside Old High German *backan*, *bahhan* (?). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from **bakan* ‘bake’ to Old English *bacan* (?; ?).

Old English evidence

Bosworth-Toller and Clark Hall both record *bacan* as the ordinary Old English verb ‘to bake’ (?; ?). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

Development to Old English

From **bákanq* ‘bake’, Anglo-Frisian brightening first gives **bækanq* ‘bake’. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* (?; ?). The development is therefore straightforward: **bákanq* > *bacan*.

beech — OE *bōc*

Derivation: **bōkō* > *bōc* (regular).

Derivation trace

Proto input: **bōkō*

Earlier Germanic changes		Old English changes
West Germanic		OE High Vowel Apocope
[no change]		<i>*bōk</i>
Northwest Germanic		
NWGmc Final Long O Raising	<i>*bōku</i>	

Old English form: *bōc*

Reconstruction and comparative evidence

Kroonen gives the beech noun as **bōk(j)ō-* and cites Old English *boc*, *bēce* among its reflexes (?). The derivational input **bōkō* ‘beech’ is the nominative-singular shape of that family, which is the relevant comparison form here.

Old English evidence

Kroonen’s Old English evidence already separates the paradigm material: *boc* as the nominative form and *bēce* as an oblique form (?). The relevant comparator is therefore *bōc*; *bēce* remains related paradigm evidence rather than the form chosen for this comparison.

Development to Old English

With nominative input **bōkō* ‘beech’, the development is compact. Northwest Germanic final long *ō* raises to *u*, and later high-vowel apocope leaves *bōc*. The regular comparison is therefore **bōkō* > *bōc*.

begin — OE *beġinnan*

Derivation: **bigínnanq* > *beġinnan* (regular).

Derivation trace

Proto input: **bigínnanq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	<i>*bigínnan</i>
[no change]	OE Secondary Nasalization	<i>*bigínnan̥</i>
Northwest Germanic	OE Velar Palatalization	<i>*biǵínnan̥</i>
[no change]	OE Prefix I Reduction	<i>*bēǵínnan̥</i>
	OE Weak Tail Reduction	<i>*bēǵínnan</i>

Old English form: *beġinnan*

Reconstruction and comparative evidence

The verb is modeled here as inherited **bigínnanq* ‘begin’. Ringe and Taylor state that intervocalic **g* is palatalized between front vowels in Old English (?), and Campbell lists *ginnan* among familiar examples of palatal *g* in this verb family (?).

Old English evidence

Bosworth-Toller and Clark Hall lemmatize the verb as *be-ginnan* / *beġinnan* (?; ?). Those plain-*g* dictionary spellings support the same verb that appears here in normalized form as *beġinnan*.

Development note

The prefix deserves separate notice. Ringe and Taylor explicitly cite *bi-* > *be-* as an Old English unstressed-prefix development (?).

Development to Old English

From **bigínnanq* ‘begin’, heavy-syllable nasal apocope yields **bigínnan* ‘begin’. Intervocalic **g* between front vowels then palatalizes to *ǵ*, and the unstressed prefix reduces *bi-* to *be-*, giving *beġinnan*.

bier — OE *bæ̃r*

Derivation: **bérō* > *bæ̃r* (regular).

Derivation trace

Proto input: **bérō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	<i>*bǣr</i>
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	<i>*bēru</i>		
NWGmc Long E Lowering	<i>*bǣru</i>		

Old English form: *bǣr*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **bērō*- f. ‘bier’ and cites Old English *bar*, *bær* among the reflexes (?). The derivational input **bērō* ‘bier’ is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller lemmatize the noun as *bær*, and Kroonen also records *bar* beside it (?; ?; ?). The target *bǣr* is therefore a normalized long-vowel spelling of the same noun.

Source note

Lexicographic spellings vary between *bær* and *bar*. The normalized target *bǣr* represents the same long vowel (?; ?; ?).

Development to Old English

From **bērō* ‘bier’, Northwest Germanic final long *ō* raises to *u*, long *ē* lowers to *æ*, and high-vowel apocope yields *bǣr*. The resulting noun matches the normalized Old English target.

birth — OE *byrd*

Derivation: **búrdiz* > *byrd* (regular).

Derivation trace

Proto input: **búrdiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	<i>*byrdi</i>
[no change]		OE High Vowel Apocope	<i>*byrd</i>
Northwest Germanic			
PGmc Final Z Deletion	<i>*búrdi</i>		

Old English form: *byrd*

Reconstruction and comparative evidence

Kroonen cites the noun under stem-level **burdi-* and gives Old English (*ge-*)*byrd* among the reflexes (?). The derivational input **búrdiz* ‘birth’ is the nominative-style form that stands behind that stem label.

Old English evidence

Clark Hall and Bosworth-Toller both attest simplex *byrd* as an Old English noun meaning ‘birth’ (?; ?). The prefixed form *gebyrd* is also well established in the tradition: Kroonen lists *(ge-)byrd*, Bosworth-Toller has a separate *ge-byrd* entry, and Campbell cites *gebyrd* and *gebyrdu* in his grammatical discussion (?; ?; ?).

Form note

The relevant comparator here is the simplex noun *byrd*. The prefixed forms remain related attested material within the same lexical family, and Hogg’s discussion of deverbal feminines provides the broader derivational setting (?).

Development to Old English

From **búrdiz* ‘birth’, loss of final *z* gives **búrdi* ‘birth’. I-umlaut fronts *u* to *y*, and high-vowel apocope then yields *byrd*. The result is the ordinary simplex Old English noun.

bone — OE *bān*

Derivation: **báiną* > *bān* (regular).

Derivation trace

Proto input: **báiną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	<i>*bān</i>
PWGmc Ai	<i>*bāną</i>		
Monophthongization			
Northwest Germanic			
[no change]			

Old English form: *bān*

Reconstruction and comparative evidence

Kroonen cites the noun as **baina-*, and Orel gives the same lexeme under **bainan* ‘bone’ (?; ?). Both are comparative headword conventions for the same neuter noun whose Old English reflex is *bān*.

Old English evidence

Clark Hall and Bosworth-Toller record *bān* as the ordinary Old English noun (?; ?). Bright’s glossary also distinguishes citation-form *bān* from oblique *bāne* (?).

Source note

The comparative headwords **baina-* and **bainan* ‘bone’ provide lexeme background. The relevant comparison form here is the nominative-accusative singular **báiną* ‘bone’.

Development to Old English

West Germanic monophthongization turns stressed **ai* into *ā*, giving **bāną* ‘bone’; heavy-syllable nasal apocope then yields *bān*. The resulting form matches the attested Old English citation noun.

both — OE *bū*

Derivation: **bō* > *bū* (regular).

Derivation trace

Proto input: **bō*

Earlier Germanic changes	Old English changes
West Germanic	[no change]
[no change]	
Northwest Germanic	
NWGmc Stressed Monosyllable O Raising <i>*bū</i>	

Old English form: *bū*

Reconstruction and comparative evidence

Kroonen treats the Germanic numeral under **ba-* and gives the inherited paradigm **bai*, *bans*, *bōz/bōns*, *bō* ‘both’, with Old English *bēgen*, *bā*, and neuter *bū* (?). For the present entry, the relevant inherited form is the unextended neuter dual **bō*.

The older explanation of *bēgen* derives it from **bō-jen-*, and Orel still gives OE *bezen* (< **bō-jenō*) beside ON *báðir*, OFris *bēthe*, OS *be-thia*, and OHG *bēde* (?). Fulk reports that explanation cautiously and notes Seebold’s preference for a **bō-p-* analysis instead (?). That debate concerns *bēgen* and the extended forms behind Modern English *both*, German *beide*, and Dutch *beide*; it does not displace the inherited neuter **bō* > *bū* treated here.

Old English evidence

The Old English dual paradigm is well established. Brunner gives masculine *bēgen*, feminine *bā*, and neuter *bū* beside *bā*, with compounds such as *bā twā*, *bū tū*, and *bām twām* (? Anm. 2). Campbell and Fulk present the same basic pattern: masculine *bēgen*, feminine *bā*, neuter *bā*, *bū*, genitive *bēgra*, *bēg(e)a*, and dative *bæm* (?; ?).

bū is therefore an attested neuter dual form, not a reconstruction. It is the cleanest target for this entry because *bēgen* belongs to the historically more contested **bō-jen-* / analogical zone, while *bā* remains a partner form within the dual paradigm rather than the most straightforward monosyllabic comparison.

Development to Old English

**bō* is a stressed monosyllabic form. Campbell cites *cū*, *hū*, *tū*, and *bū* as examples of final accented *ō* > *ū* in the West Germanic stage leading to Old English (?). Brunner states the same development more directly: Auslautendes *ō* erscheint als *û* in *bū* ... *cu* ... *hū*, *tū* (?).

The development is therefore straightforward: **bō* > *bū*.

Form comparison

The comparison below sets the relevant forms side by side. It separates the inherited OE target from the other forms that belong to the same broader lexical history.

Form	Source / stage	Status	Relevance to this entry
<i>*bō</i> > <i>bū</i>	PGmc neuter dual > OE neuter dual	selected regular comparison	main line of the entry

Form	Source / stage	Status	Relevance to this entry
<i>bēgen</i>	OE masculine dual	attested, but historically contested and at least partly analogical in Kroonen	real OE evidence, not the Old English form here
<i>bā</i>	OE feminine dual; also neuter variant	attested partner form	part of the OE paradigm, but not the chosen monosyllabic comparator
<i>báðir</i> , German <i>beide</i> , Dutch <i>beide</i> , Modern English <i>both</i>	Norse, continental West Germanic, Modern English extended forms	related but different formation	useful background, not the direct continuation of OE <i>bū</i>

bow — OE *bīegan*

Derivation: **bāugijanq* > *bīegan* (regular).

Derivation trace

Proto input: **bāugijanq*

Earlier Germanic changes	Old English changes
West Germanic	OE Au Fronting
[no change]	OE Diphthong Leveling
	OE Heavy Syllable Nasal Apocope
Northwest Germanic	OE Secondary Nasalization
[no change]	Sievers Law Syncope
	OE Velar Palatalization
	OE I Umlaut
	OE Weak Tail Reduction
	OE J Loss After Heavy

Old English form: *bīegan*

Reconstruction and comparative evidence

Kroonen reconstructs the weak verb as **baugjan-* ‘to (make) bend’ and cites Old English *biegan* among its reflexes (?). Ringe and Taylor give the northwest Germanic preform **baugijana* ‘bow’, with later **béagian* ‘bow’ behind West Saxon Old English *biegan* (?). The entry therefore concerns the weak causative member of the bend-family, alongside the related strong verb and noun.

Old English evidence

Clark Hall lemmatizes *biegan*, and Bosworth-Toller records *bigan* with examples such as Ic *bēge mīne cneowa* and Se ord *bīgde upp tō pām hiltum* (?; ?). The form *bīegan* used here is a normalized spelling of that attested Old English weak verb.

Development to Old English

From **bāugijanq* ‘bow’, the stem reaches pre-Old-English **bēagian* ‘bow’, after which palatalization of **gj* and i-umlaut yield West Saxon *biegan*; Campbell lists *biegan* among the regular *ie* outcomes of **éa* under i-umlaut (?; ?). The development is therefore straightforward: **bāugijanq* > *bīegan*.

breeches — OE *brēc*

Derivation: **brōkiz* > *brēc* (regular).

Derivation trace

Proto input: **brōkiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Palatalization	<i>*brōtʃi</i>
[no change]		OE I Umlaut	<i>*brētʃi</i>
Northwest Germanic		OE High Vowel Apocope	<i>*brēʃ</i>
PGmc Final Z Deletion	<i>*brōki</i>		

Old English form: *brēc*

Reconstruction and comparative evidence

Kroonen cites the noun under **brōk-*, with Old English *brōc* and plural ‘breeches’ among its reflexes (?). Ringe and Taylor give the plural development directly as PNWGMc **brokiz* > **breeci* > OE *brēc* (?). The deeper verbal base belongs to the noun’s etymological background, while the derivational input here is the plural noun form **brōkiz* ‘breeches’.

Old English evidence

Bright notes *brōc* with plural *brēc*, and Clark Hall gives *brēc* fp. breeches while also listing *broc* as a feminine noun probably represented chiefly in the plural (?; ?). I write *brēc* for the long vowel and palatal consonant; the attested plural is *brēc*.

Development to Old English

After loss of final -z, the stem ends in -ki, so the velar palatalizes and *ō* undergoes i-umlaut to *ē*; final high-vowel apocope then yields *brēc* (?). The development is therefore regular: **brōkiz* > *brēc*.

calf — OE *ċealf*

Derivation: **kálbaz* > *ċealf* (regular).

Derivation trace

Proto input: **kálbaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGMc Final Bare A Loss	<i>*kálb</i>
[no change]		Anglo Frisian Brightening	<i>*kælb</i>
Northwest Germanic		OE Breaking	<i>*kealb</i>
PGmc Final Z Deletion	<i>*kálba</i>	PGmc B Allophony	<i>*kealβ</i>
		OE Velar Palatalization	<i>*tʃealβ</i>

Old English form: *ċealf*

Reconstruction and comparative evidence

Kroonen treats the noun under **kalbiz-* and notes an older s-stem **kalbaz*, pl. **kalbizō* ‘calf’, while Orel cites **kalbaz* ‘calf’ as the citation form and Ringe and Taylor derive West Saxon *Cealf* from **kalbaz*, **kalbiz-* (?; ?; ?). The derivational input here is the singular **kálbaz* ‘calf’, since the entry concerns the citation-form noun.

Old English evidence

Clark Hall gives *cealf* I. (æ, e) nm. (nap. *cealfu*), and Bosworth-Toller likewise records *Caelf* / *Cealf* beside plural forms such as *calfur* and *cealfu* (?; ?). Campbell and Fulk show the same singular-plus-*-r*- plural pattern (?; ?). I write *cealf* for the palatalized initial; the attested dictionary headword is *cealf*.

Development to Old English

After loss of final -z and bare -a, Anglo-Frisian brightening gives **kælb* ‘calf’, and breaking before *l* plus consonant yields **kealb* ‘calf’. Ringe and Taylor’s account of the lexeme and their rule for initial *k* in front-vocalic environments support the West Saxon palatalized onset represented here as *č*-, so **kálbaz* ‘calf’ develops regularly to *cealf* (?).

corn — OE *corn*

Derivation: **kurnq* > *corn* (regular).

Derivation trace

Proto input: **kurnq*

Earlier Germanic changes	Old English changes
West Germanic	OE Heavy Syllable Nasal Apocope <i>*kórn</i>
[no change]	
Northwest Germanic	
NWGmc U Lowering <i>*kórnq</i>	

Old English form: *corn*

Reconstruction and comparative evidence

Kroonen cites the noun as **kurna-*, and Orel gives the citation form **kurnan* ‘corn’, both with Old English *corn* among the reflexes (?; ?). The singular form **kurnq* ‘corn’ is the nominative-accusative singular appropriate to the citation noun.

Old English evidence

Clark Hall gives *corn* *n.* ‘*corn*,’ *grain*, Bright’s glossary lists *corn*, *n.* with genitive singular *corne*s, and Bosworth-Toller treats *corn* as an ordinary noun headword (?; ?; ?). The target is therefore an attested citation form, while forms such as *corne*s simply provide paradigm background.

Development to Old English

With northwest Germanic lowering, **kurnq* ‘corn’ becomes **kórnq* ‘corn’, and later loss of final nasal after a heavy syllable yields **kórn* ‘corn’, whence *corn*. The oblique form **kurnǣn* ‘corn’ belongs to comparative background rather than to the derivational input of this entry.

deed — OE *dǣd*

Derivation: **dédiz* > *dǣd* (regular).

Derivation trace

Proto input: **dédiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	<i>*dǣd</i>
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	<i>*dédi</i>		
NWGmc Long E Lowering	<i>*dǣdi</i>		

Old English form: *dǣd*

Reconstruction and comparative evidence

Orel reconstructs the noun as **dēdiz* ‘deed’, and Ringe and Taylor derive the same inherited i-stem from Proto-Germanic **dédiz* ‘deed’ through northwest Germanic **dadiz* ‘deed’ (?; ?). The acute accent in **dēdiz* ‘deed’ marks stress on the same reconstructed long vowel.

Old English evidence

Campbell states that Primitive Germanic *ē* appears as West Saxon *ǣ* but in other Old English dialects mostly as *ē*, and Brunner gives the contrast explicitly as West Saxon *dǣd* beside non-West-Saxon *dēd* (?; ?). Clark Hall likewise lists *dǣd* and cross-refers Anglian *dēd* to it (?). West Saxon *dǣd* is therefore the relevant Old English form here, with Anglian *dēd* as a dialectal doublet.

Development to Old English

From inherited **dēdiz* ‘deed’, loss of final -z and the West Saxon lowering of stressed long *ē* yield *dǣd*; Anglian *dēd* preserves the non-West-Saxon outcome (?; ?). The development treated here is therefore the regular West Saxon line.

door — OE *dor*

Derivation: **dúrą* > *dor* (regular).

Derivation trace

Proto input: **dúrą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	<i>*dór</i>
[no change]			
Northwest Germanic			
NWGmc U Lowering	<i>*dórą</i>		

Old English form: *dor*

Reconstruction and comparative evidence

Kroonen reconstructs a neuter **dura-* ‘gate, (single) door’ and cites Old English *dor* among its reflexes. In the same entry he separates Old English *duru*, Old Frisian *dore*, and Old High German *tura* as reflexes of **durō-* instead (?).

Old English evidence

Clark Hall records *dor* as a neuter noun and separately records feminine *duru* with its own inflection (?). Ringe and Taylor likewise treat *duru* as an early Old English u-stem, originally a root noun shifted into that class (?). The Old English form here is therefore the attested neuter *dor*, while *duru* remains a parallel Old English reflex from another stem history.

Development to Old English

From **dúrą* ‘door’, Northwest Germanic u-lowering gives **dórą* ‘door’, and heavy-syllable nasal apocope then yields *dor*. The regular development treated in this entry is therefore **dúrą* > *dor*; the feminine *duru* belongs to the separate line identified by Kroonen and Ringe-Taylor (?; ?).

fare — OE *faran*

Derivation: **fáraną* > *faran* (regular).

Derivation trace

Proto input: **fáraną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	<i>*færaną</i>
[no change]	OE A Restoration	<i>*faraną</i>
Northwest Germanic	OE Heavy Syllable Nasal Apocope	<i>*faran</i>
[no change]	OE Secondary Nasalization	<i>*farən</i>
	OE Weak Tail Reduction	<i>*faran</i>

Old English form: *faran*

Reconstruction and comparative evidence

Kroonen gives the inherited strong verb as **faran-*, and Orel gives the same lexeme as **faranan* ‘fare’, both with Old English *faran* among the reflexes (?; ?). Campbell also uses *faran* as a standard example of Old English A-restoration (?).

Old English evidence

Clark Hall lemmatizes the strong verb as *faran* and separately records weak *færan* ‘to frighten’; *fære*, *færst*, and *færð* belong to present-tense forms of *faran* rather than to the infinitive itself (?). Bosworth-Toller preserves the same distinction (?). The Old English form here is therefore the attested citation infinitive *faran*.

Development to Old English

From **fáraną* ‘fare’, Anglo-Frisian brightening first gives **færaną* ‘fare’, but A-restoration before single *r* returns **faraną* ‘fare’; later apocope and weak-tail reduction yield *faran* (?). Fulk’s contrast with participial *faren-* < **faræn-* < **faran-* shows why fronting elsewhere in the paradigm does not alter the infinitive headword (?).

fell — OE *fell*

Derivation: **féllą* > *fell* (regular).

Derivation trace

Proto input: **féllą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	<i>*féll</i>
[no change]		
Northwest Germanic		
[no change]		

Old English form: *fell*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **fella-* ‘membrane, skin, hide’ and cites Old English *fell* beside Dutch *vel* and German *Fell* (?). The derivational input **féllq* ‘fell’ is the derivable singular form of that same inherited noun.

Old English evidence

Clark Hall records *fell* as the noun ‘fell, skin, hide’, and Bright’s glossary likewise gives *fell* with inflected forms such as accusative singular *fel* and dative plural *fellum* (?; ?). The target is therefore the attested noun *fell*, not the verb *fellan* or the preterite *feoll*.

Development to Old English

With **féllq* ‘fell’, no special earlier reshaping is needed: heavy-syllable nasal apocope yields **féll* ‘fell’, surfacing as *fell*. The regular development treated here is therefore **féllq* > *fell*.

fern — OE *fearn*

Derivation: **fárnaz* > *fearn* (regular).

Derivation trace

Proto input: **fárnaz*

Earlier Germanic changes	Old English changes	
West Germanic	PWGmc Final Bare A Loss	<i>*fárn</i>
[no change]	Anglo Frisian Brightening	<i>*færn</i>
Northwest Germanic	OE Breaking	<i>*fearn</i>
PGmc Final Z Deletion <i>*fárna</i>		

Old English form: *fearn*

Reconstruction and comparative evidence

Kroonen cites the noun as masculine **farna-* and gives Old English *fearn*, *fern*, while Orel gives the same lexeme as neuter **farnan* ‘fern’ with Old English *fearn* (?; ?). Those are comparative headword conventions rather than competing Old English outcomes; the modeled input here is the nominative-style **fárnaz* ‘fern’.

Old English evidence

Clark Hall gives *fearn* as an Old English noun, and Bosworth-Toller records *fearn* with inflected forms such as *fearnes*, *fearna*, and *fearne* (?; ?). Kroonen also records *fern*, but the local lexical sources give stronger support to *fearn* as the citation target (?).

Development to Old English

From **fárnaz* ‘fern’, loss of final -z and final -a gives **fárn* ‘fern’; Anglo-Frisian brightening then yields **færn* ‘fern’, and breaking before *r* plus consonant gives *fearn* (?; ?). The development treated here is therefore the regular *rC*-breaking line.

field — OE *feld*

Derivation: **félθuz* > *feld* (regular).

Derivation traceProto input: **félθuz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	
PWGmc L Th Voicing	<i>*félðuz</i>		<i>*féld</i>
Northwest Germanic			
PGmc Final Z Deletion	<i>*féldu</i>		

Old English form: *feld***Reconstruction and comparative evidence**

Ringe and Taylor treat Old English *feld* as one of the cases where earlier **felþu-* ~ **feldu-* may reflect either inherited **þ* **d* ‘field’ alternation or the regular West Germanic development **þ* > *ld* (?). The broader proto label **félθuz* ‘field’ can remain as comparative background, while that narrower historical ambiguity does not affect the regular classification.

Old English evidence

Clark Hall records *feld* with oblique forms such as *felda* and *felde*, and Campbell notes early place-name spellings in *-felth* beside the later standard form (?; ?). The Old English form here is therefore the attested citation noun *feld*, with the older *-felth* spellings as historical support rather than as rival targets.

Development to Old English

In the modeled pathway, medial **þ* becomes *ld*, final *-z* is lost, and high-vowel apocope then yields *feld*. Whether the voiced dental ultimately reflects inherited alternation or the regular **þ* > *ld* development, both accounts converge on the same Old English form (?).

fly — OE *flēogan*Derivation: **flēuganaþ* > *flēogan* (regular).**Derivation trace**Proto input: **flēuganaþ*

Earlier Germanic changes		Old English changes	
West Germanic		OE Diphthong Leveling	<i>*flēoganƿ</i>
[no change]		OE Heavy Syllable Nasal Apocope	<i>*flēogan</i>
		OE Secondary Nasalization	<i>*flēogaŋ</i>
Northwest Germanic		OE Weak Tail Reduction	<i>*flēogan</i>
[no change]			

Old English form: *flēogan***Reconstruction and comparative evidence**

Ringe and Taylor derive the verb as **fleugana* > OE *flēogan* and elsewhere contrast West Saxon *flēogan* with Anglian *flégan*, alongside related forms *fléoge* / *flége* (?). The derivational input **flēuganaþ* ‘fly’ represents that inherited strong verb in the notation used here.

Old English evidence

Clark Hall and Bosworth-Toller record *flēogan* as the ordinary Old English strong verb, and Bright gives the familiar paradigm *flēag*, *flugon*, *flogen* with present *fleogeð* (?; ?; ?). The target in this entry is therefore the attested verbal infinitive itself.

Form note

Ringe and Taylor also list related *fléoge* / *flége* and Anglian *flégan*, which belong to the same family but do not replace the infinitive *flēogan* treated here (?).

Development to Old English

From **fléuganq* ‘fly’, Old English diphthong leveling gives **flēoganq* ‘fly’; heavy-syllable nasal apocope and weak-tail reduction then yield *flēogan* (?). The development is therefore regular: **fléuganq* > *flēogan*.

forlorn — OE *lēosan*

Derivation: **léusanq* > *lēosan* (regular).

Derivation trace

Proto input: **léusanq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	<i>*lēosanq</i>
[no change]	OE Heavy Syllable Nasal Apocope	<i>*lēosan</i>
Northwest Germanic	OE Secondary Nasalization	<i>*lēosq̃n</i>
[no change]	OE Weak Tail Reduction	<i>*lēosan</i>

Old English form: *lēosan*

Reconstruction and comparative evidence

Kroonen reconstructs the verb under **leusan-* and cites prefixed daughters such as Gothic *fraliusan* and Old English *for-lēosan*; Orel likewise gives Old English *for-leósan* (?; ?). The inherited verbal base is therefore clear, though the daughter set often appears with the prefix.

Old English evidence

The direct Old English evidence behind English *forlorn* lies in the prefixed verb *forlēosan* and especially in the participle *forloren*, recorded by Ringe and Taylor and in the dictionaries (?; ?; ?). The simplex infinitive *lēosan* represents the verbal base itself.

Form note

As a base-form comparison, the simplex infinitive is *lēosan*, while the English adjective continues the prefixed Old English family *forlēosan* / *forloren* (?).

Development to Old English

From **léusanq* ‘forlorn’, Old English diphthong leveling gives **lēosanq* ‘forlorn’, and later nasal apocope and weak-tail reduction yield *lēosan* (?). The prefixed forms follow the same verbal base with added *for-*.

gang — OE *gang*

Derivation: **gángaz* > *gang* (regular).

Derivation trace

Proto input: **gángaz*

Earlier Germanic changes		Old English changes
West Germanic		PWGmc Final Bare A * <i>gáng</i>
[no change]		Loss
Northwest Germanic		
PGmc Final Z Deletion	* <i>gánga</i>	

Old English form: *gang*

Reconstruction and comparative evidence

Orel reconstructs the noun as **gangaz* ‘gang’ and cites Old English *gang* beside Old Norse *gangr*, Old Frisian *gang* / *gong*, Old Saxon *gang*, and Old High German *gang* (?). The derivational input **gángaz* ‘gang’ is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller both record *gang* as the noun ‘going, journey, way’, and Bright’s glossary gives *gong* (*gang*), *m.*, *path*, *course* (?; ?; ?). The target is therefore the attested noun headword itself.

Form note

This entry concerns the noun *gang*, not the separate verb *gangan* (?; ?).

Development to Old English

From **gángaz* ‘gang’, loss of final -z gives **gánga* ‘gang’, and later loss of final bare -a yields *gang*. The development is therefore regular: **gángaz* > *gang*.

give — OE *giefan*

Derivation: **gébanq* > *giefan* (regular).

Derivation trace

Proto input: **gébanq*

Earlier Germanic changes		Old English changes
West Germanic		OE Heavy Syllable Nasal Apocope
[no change]		OE Secondary Nasalization
Northwest Germanic		PGmc B Allophony
[no change]		OE Velar Palatalization
		OE Ws Palatal Diphthongization
		OE Weak Tail Reduction

Old English form: *giefan*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **geban-* and cites Old English *giefan* among its reflexes (?). Ringe and Taylor contrast West Saxon *giefan* with Mercian *for-geofan* and Northumbrian *geafa*, showing that the inherited verb takes different later dialectal shapes (?).

Old English evidence

Campbell gives *gefan* (W-S *giefan*) among examples of initial palatalization, and Clark Hall records the verb under plain *giefan* with forms such as *geaf* and *giefen* (?; ?). I write *giefan* for the palatal initial.

Dialect note

West Saxon *ie* here reflects palatal diphthongization after initial palatalization; non-West-Saxon forms such as *geafa* or *for-geofan* continue the same verb without the West Saxon vocalism (?).

Development to Old English

From **gēbanq* ‘give’, initial *g* palatalizes before *e*; West Saxon palatal diphthongization then yields *ie*, and later tail reduction gives *giefan* (?; ?). The result is therefore the regular West Saxon infinitive.

gold — OE *gold*

Derivation: **gúlθq* > *gold* (regular).

Derivation trace

Proto input: **gúlθq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	
PWGmc L Th Voicing	<i>*gúldq</i>		<i>*góld</i>
Northwest Germanic			
NWGmc U Lowering	<i>*góldq</i>		

Old English form: *gold*

Reconstruction and comparative evidence

Ringe and Taylor cite the noun as **gulþa-* / **gulda-*, and Kroonen gives the same pair (?; ?). The derivational input **gúlθq* ‘gold’ preserves the older consonantal form while leaving open whether the medial stop reflects inherited alternation or regular West Germanic development.

Old English evidence

Bosworth-Toller and Clark Hall both record *gold* as the ordinary Old English neuter noun (?; ?). The target is therefore the attested citation form itself.

Development note

Ringe and Taylor note that the medial stop can be understood either as alternation **gulþa-* / **gulda-* or as the ordinary West Germanic change **lþ* > *ld*; both routes lead to the same Old English consonantism (?).

Development to Old English

From **gúlθq* ‘gold’, the regular consonant development gives **gúldq* ‘gold’; Northwest Germanic / Old English lowering then yields **góldq* ‘gold’, and apocope gives *gold* (?; ?).

grave — OE *grafan*

Derivation: **grábanq* > *grafan* (regular).

Derivation trace

Proto input: **grábanq*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	<i>*græbanq</i>
[no change]	OE A Restoration	<i>*grabana</i>
Northwest Germanic	OE Heavy Syllable Nasal Apocope	<i>*graban</i>
[no change]	OE Secondary Nasalization	<i>*grabān</i>
	PGmc B Allophony	<i>*graβān</i>
	OE Weak Tail Reduction	<i>*graβan</i>

Old English form: *grafan*

Reconstruction and comparative evidence

Campbell gives *grafan* among the standard examples of Old English a-restoration before a single consonant and a following back vowel, and Ringe and Taylor describe the same development for Class VI infinitives (?; ?).

Old English evidence

Clark Hall records *grafan* as the verb ‘to dig, grave’ and separately records noun *græf* ‘grave, trench’ (?). The target here is the attested infinitive headword of the verb.

Development to Old English

From **grábanq* ‘grave’, Anglo-Frisian brightening first gives a fronted stem vowel. A-restoration then returns *a* before single *b* plus the back-vocalic infinitive ending, and later apocope and weak-tail reduction yield *grafan* (?; ?).

Form note

Noun *græf* and verbal forms such as *græfð* or past participial *græfen* belong to other lexical or paradigm positions and do not replace the infinitive *grafan* as the target here (?).

guest — OE *ġiest*

Derivation: **gástiz* > *ġiest* (regular).

Derivation trace

Proto input: **gástiz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	* <i>gæsti</i>
[no change]	Northwest Germanic	OE Velar Palatalization	* <i>ǵæsti</i>
		OE I Umlaut	* <i>ǵesti</i>
PGmc Final Z Deletion		* <i>gásti</i>	OE Ws Palatal Diphthongization
		OE High Vowel Apocope	* <i>ǵiest</i>

Old English form: *giest*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the noun as an ordinary i-stem whose West Saxon development shows palatal diphthongization, while non-West-Saxon evidence preserves forms of the *gest* type (?; ?).

Old English evidence

Bosworth-Toller and Clark Hall record the word under forms such as *gist*, *gest*, *giest*, and *gyst* (?; ?). The Old English form here, *giest*, is the normalized West Saxon form within that attested family.

Development to Old English

From **gástiz* ‘guest’, Anglo-Frisian brightening gives a *gæst*- stage, and i-mutation affects the front vowel before the lost high-vocalic ending. In West Saxon the initial palatal environment then produces *ie*, so the regular outcome is *giest* (?; ?).

Dialect note

West Saxon *giest* is the Old English form here. Anglian *gest* and related spellings remain real Old English comparators rather than corrections to that choice (?; ?).

hair — OE *hær*

Derivation: **xérq* > *hær* (regular).

Derivation trace

Proto input: **xérq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	*ǵæra
[no change]		OE Heavy Syllable Nasal Apocope	*ǵær
Northwest Germanic			
NWGmc Long E Lowering	*xæra		

Old English form: *hær*

Reconstruction and comparative evidence

Kroonen cites the ordinary Proto-Germanic hair word as **hēra*- (?). The derivational input **xérq* ‘hair’ represents that same long-ē stem in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hær* / *hær* as the ordinary Old English noun ‘hair’ (?; ?). The target is therefore the attested headword itself.

Development to Old English

From **xérq* ‘hair’, Northwest Germanic lowering gives a long front vowel, and later loss of the final nasal leaves the Old English form *hær*. The development treated here is straightforward and does not require any special paradigm choice.

Form note

Older references to **xazwǣz* ‘hair’ belong to a different lexeme, and the separate *haddr* / *heordan* / *hād-* material does not displace the ordinary simplex *hær* treated here (?; ?).

harvest — OE *hierfest*

Derivation: **xárbistuz* > *hierfest* (regular).

Derivation trace

Proto input: **xárbistuz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	<i>*xærbistu</i>
[no change]	Northwest Germanic	OE Breaking	<i>*xearbistu</i>
		OE Velar Fricative Palatalization	<i>*çearbistu</i>
PGmc Final Z Deletion	<i>*xárbistu</i>	PGmc B Allophony	<i>*çearβistu</i>
		OE I Umlaut	<i>*çierβistu</i>
		OE High Vowel Apocope	<i>*çierβist</i>
		OE Med Unstressed I Lowering ¹	<i>*çierβest</i>

Old English form: *hierfest*

Reconstruction and comparative evidence

Bammesberger and Ringe-Taylor treat **harbist-* as the inherited base and explain that the regular native West Saxon development would be of the *hierfest* / *hyrfest* type (?; ?).

Old English evidence

Bosworth-Toller and Clark Hall record *hærfest*, with *herfest* as a variant in the lexical tradition (?; ?). Those attested forms remain the main dictionary evidence for the noun.

Development to Old English

From **xárbistuz* ‘harvest’, Anglo-Frisian brightening, breaking, and i-mutation produce a *hierbist-* stage, and later lowering of unstressed medial *i* to *e* gives *hierfest*. That is the regular West Saxon development treated here (?; ?).

Source note

The Old English form here, *hierfest*, represents the regular native West Saxon outcome discussed by Bammesberger and Ringe-Taylor. The attested Old English lexical tradition, however, is chiefly *hærfest* / *herfest*, commonly treated as non-West-Saxon or Anglian material in West Saxon transmission (?; ?).

hedge — OE *heġġ*

Derivation: **xágjaz* > *heġġ* (regular).

Derivation trace

Proto input: **xágjaz*

Earlier Germanic changes	Old English changes
West Germanic	PWGmc Final Bare A Loss * <i>xággj</i>
PWGmc J Gemination * <i>xággjaz</i>	Anglo Frisian Brightening * <i>xæggj</i>
Northwest Germanic	OE Velar Fricative Palatalization * <i>çæggj</i>
PGmc Final Z Deletion * <i>xággja</i>	OE Velar Palatalization * <i>çæḡḡj</i>
	OE I Umlaut * <i>çeḡḡj</i>
	OE J Loss After Heavy * <i>çeḡḡ</i>

Old English form: *heḡḡ*

Reconstruction and comparative evidence

The derivational input models a palatal *-*gj*-* ‘hedge’ noun whose Old English development includes gemination, palatalization, and i-mutation. The current derivation therefore reaches a palatal-geminate outcome of the *heḡḡ* type (?).

Old English evidence

Bosworth-Toller and Clark Hall record the noun under standard spellings *hecg* / *heçḡ* (?; ?). The lexical item itself is therefore well attested even though the form compared here is normalized.

Development to Old English

From **xágjaz* ‘hedge’, West Germanic j-gemination first yields a geminate stop, and later Old English palatalization and loss of final *j* produce *heḡḡ*. The development is treated as regular rather than exceptional.

Form note

Standard dictionary spelling is *heçḡ* or *hecg*. Normalized *heḡḡ* is the Old English form here, while the ordinary lexicographic forms remain the main Old English citation evidence (?; ?).

helm — OE *helm*

Derivation: **xélmaz* > *helm* (regular).

Derivation trace

Proto input: **xélmaz*

Earlier Germanic changes	Old English changes
West Germanic	PWGmc Final Bare A Loss * <i>xélm</i>
[no change]	OE Velar Fricative Palatalization * <i>çélm</i>
Northwest Germanic	
PGmc Final Z Deletion * <i>xélma</i>	

Old English form: *helm*

Reconstruction and comparative evidence

Kroonen cites the helmet noun as **helma*- and separately distinguishes a different lexeme **helman*- ‘rudder’ (?). The derivational input **xélmaz* ‘helm’ is the nominative-style form used for the helmet noun itself.

Old English evidence

Clark Hall and Bosworth-Toller record *helm* as the ordinary Old English noun for ‘helmet’, while *helma* belongs to a separate rudder lexeme (?; ?).

Development to Old English

From **xélmaz* ‘helm’, loss of final *z* and later loss of the short final vowel yield *helm*. The development is therefore a straightforward citation-form match.

Form note

Comparative **helma-* is headword notation for the helmet cognate set. It should not be confused with Old English *helma*, which is a different noun meaning ‘helm, rudder’ (?; ?).

help — OE *helpan*

Derivation: **xélpānq* > *helpan* (regular).

Derivation trace

Proto input: **xélpānq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Velar Fricative Palatalization	* <i>çélpānq</i>
[no change]	OE Heavy Syllable Nasal Apocope	* <i>çélpān</i>
Northwest Germanic	OE Secondary Nasalization	* <i>çélpān</i>
[no change]	OE Weak Tail Reduction	* <i>çélpān</i>

Old English form: *helpan*

Reconstruction and comparative evidence

The entry treats the strong verb itself rather than the separate noun *help*. Bright’s principal parts *helpan*, *healp*, *hulpon*, *holpen* show the ordinary Old English strong-verb family continued by this input (?).

Old English evidence

Clark Hall and Bosworth-Toller record *helpan* as the verbal headword ‘to help’ (?; ?). The target is therefore the attested infinitive citation form.

Development to Old English

From **xélpānq* ‘help’, no special repair is needed beyond the ordinary reduction of the infinitive ending. The derivation therefore reaches *helpan* directly.

Form note

Noun *help* belongs to a separate lexical line and should not replace verbal *helpan* as the target here (?; ?).

hind — OE *hind*

Derivation: **xéndjō* > *hind* (regular).

Derivation trace

Proto input: *xéndjō

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	*çéndju
[no change]		OE I Umlaut	*çindju
Northwest Germanic		OE High Vowel Apocope	*çindj
NWGmc Final Long O Raising	*xéndju	OE J Loss After Heavy	*çind

Old English form: *hind*

Reconstruction and comparative evidence

Kroonen cites the animal name as *hindō- f. ‘hind’ (?). The derivational input *xéndjō ‘hind’ represents that same noun in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hind* as the noun ‘hind, female deer’ (?; ?). The target is therefore the attested lexical item itself.

Development to Old English

From *xéndjō ‘hind’, i-mutation produces the front-vocalic Old English stem, and later apocope plus loss of final *j* yield *hind*. The outcome is therefore a regular citation-form derivation.

Form note

hindan ‘from behind, behind’ is a different Old English lexeme and does not belong to the noun history of *hind* (?; ?).

hold — OE *healdan*

Derivation: *xáldanq > *healdan* (regular).

Derivation trace

Proto input: *xáldanq

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	*xældanq
[no change]		OE Breaking	*xealdanq
Northwest Germanic		OE Velar Fricative Palatalization	*çealdanq
[no change]		OE Heavy Syllable Nasal Apocope	*çealdan
		OE Secondary Nasalization	*çealdqn
		OE Weak Tail Reduction	*çealdan

Old English form: *healdan*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the verb as a regular *a + *lC* ‘hold’ breaking case, with West Saxon *healdan* opposed to Anglian and Mercian *haldan* (?; ?).

Old English evidence

Bright gives the ordinary strong-verb citation form and principal parts *healdan*, *heold*, *heoldon*, *healden* (?). The target is therefore the attested infinitive headword itself.

Development to Old English

From **xáldanq* ‘hold’, Anglo-Frisian brightening first yields a fronted vowel, and West Saxon breaking then produces *ea* before *ld*. Later reduction of the infinitive ending gives *healdan* (?; ?).

Dialect note

West Saxon *healdan* is the Old English form here. Anglian and Mercian *haldan* are genuine non-West-Saxon doublets rather than corrections to that choice (?; ?).

horn — OE *horn*

Derivation: **xúrnq* > *horn* (regular).

Derivation trace

Proto input: **xúrnq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	
[no change]		<i>*xórn</i>	
Northwest Germanic			
NWGmc U Lowering		<i>*xórnq</i>	

Old English form: *horn*

Reconstruction and comparative evidence

Kroonen and Orel cite lemma-style Proto-Germanic headwords of the **hurna-* / **xurnan* ‘horn’ type for this noun (?; ?). The derivational input **xúrnq* ‘horn’ is the nominative-style form used in the derivation here.

Old English evidence

Clark Hall, Bosworth-Toller, and Bright all record *horn* as the ordinary Old English noun (?; ?; ?). The target is therefore the attested citation form.

Development to Old English

From **xúrnq* ‘horn’, Northwest Germanic u-lowering gives **xórnq* ‘horn’, and later loss of the final nasal leaves *horn*. The development treated here is fully regular.

Form note

The note’s oblique **xurnān* ‘horn’ belongs to comparative stem background only. It does not replace the derivational input **xúrnq* ‘horn’ as the derivational form used here (?; ?).

lead — OE *lædan*

Derivation: **láidijanq* > *lædan* (regular).

Derivation trace

Proto input: **láidijanq*

Earlier Germanic changes	Old English changes
West Germanic	OE Heavy Syllable Nasal Apocope <i>*lādijan</i>
PWGmc Ai <i>*lādijanq</i>	OE Secondary Nasalization <i>*lādijqn</i>
Monophthongization	Sievers Law Syncope <i>*ladjqn</i>
Northwest Germanic	OE I Umlaut <i>*lædjqn</i>
[no change]	OE Weak Tail Reduction <i>*lædjan</i>
	OE J Loss After Heavy <i>*lædan</i>

Old English form: *lædan*

Reconstruction and comparative evidence

Ringe and Taylor derive Old English *lædan* from Proto-Germanic **laidijanq* ‘lead’, and Kroonen likewise cites a weak verb of the **laidjan-* type for ‘lead’ (?; ?).

Old English evidence

Clark Hall and Bosworth-Toller both record *lædan* / *lædan* as the ordinary Old English verb ‘to lead, guide, conduct’ (?; ?).

Development to Old English

From **láidijanq* ‘lead’, monophthongization of **ai* first gives a **lād-* stage. Later syncope, i-mutation, weak-tail reduction, and loss of *j* after a heavy stem yield *lædan*, so the development represented here is fully regular (?).

learn — OE *liornian*

Derivation: **líznojanq* > *liornian* (regular).

Derivation trace

Proto input: **líznojanq*

Earlier Germanic changes	Old English changes
West Germanic	OE Breaking <i>*líornōjanq</i>
[no change]	OE Heavy Syllable Nasal Apocope <i>*líornōjan</i>
Northwest Germanic	OE Secondary Nasalization <i>*líornōjān</i>
[no change]	OE I Umlaut <i>*líornejān</i>
	OE Unstressed Long Vowel Shortening <i>*líornejān</i>
	OE Weak Tail Reduction <i>*líornejan</i>
	OE Intervocalic J Vocalization <i>*líorneian</i>
	OE Unstressed EI Contraction <i>*líornian</i>

Old English form: *liornian*

Reconstruction and comparative evidence

Ringe and Taylor place the verb in a class-II weak family of the **lízno-* type (?), and Kroonen keeps the same comparative base for the Germanic lexeme (?). The derivational input therefore requires no change of stem class or paradigm cell.

The non-obvious issue is dialectal. Campbell records Northumbrian *liornian* beside *leornian*, and he states that the *eo* of *leornian* is secondary, while Northumbrian preserves forms with *io*, where original *eo* and *io* remain distinct (? n. 2).

Old English evidence

**liornian* ‘learn’ is the Northumbrian member of the Old English family. Dictionary practice more often privileges *leornian* as the ordinary headword (?; ?), but Campbell’s dialect evidence shows that *liornian* is a genuine Old English form (? n. 2).

This entry therefore remains compact. The point is to state clearly that the Old English form here belongs to the Northumbrian side of the OE evidence rather than to the leveled *leornian* headword tradition.

Development to Old English

From **līznōjanq* ‘learn’, the expected Old English developments include rhotacism of *z*, followed by breaking before *r* plus consonant, and the ordinary reduction of the weak verbal ending. The result is *liornian*, preserving the *io* spelling that Campbell associates with Northumbrian where original *eo* and *io* remain distinct (? n. 2).

Leornian reflects the later *eo* development that Campbell treats as secondary [Campbell (?), §123 n. 2; §296]. The selected Northumbrian target is therefore the regular comparison form for the *i*-grade member of the family.

Form comparison

The comparison below sets the relevant forms side by side. It distinguishes the regular Northumbrian form modeled here from the better-known West Saxon headword.

Form	Status	Relevance to this entry
* <i>līznōjanq</i> > <i>liornian</i>	computed regular output; attested Northumbrian comparison form	selected comparison
<i>leornian</i>	attested later <i>eo</i> form and dictionary headword	useful control, but not the target of this entry

lid — OE *hlid*

Derivation: **xlīdq* > *hlid* (regular).

Derivation trace

Proto input: **xlīdq*

Earlier Germanic changes	Old English changes
West Germanic	OE Heavy Syllable Nasal Apocope * <i>xlīd</i>
[no change]	
Northwest Germanic	
[no change]	

Old English form: *hlid*

Reconstruction and comparative evidence

Orel cites a neuter lexeme of the **xlīd-* type with Old English *hlid*, and Lloyd includes OE *hlid* beside ON *hlipó* and OHG (*h*)*lit* among forms that retain *i* (?; ?).

Old English evidence

Clark Hall and Bosworth-Toller record *hlid* as the noun ‘lid, cover, door, gate’ (?; ?).

Development to Old English

The derivational input already represents the later Germanic *hlid-* stage used for the derivation here. From **xlídą* ‘lid’, heavy-syllable apocope yields *hlid*, and the form belongs to the retained-*i* set noted by Lloyd rather than to the lowered *e* type (?).

Form note

An earlier etymological stage **lībuz* ‘lid’ belongs to comparative background only. The form represented here is the later **xlídą* > *hlid* line that matches the attested Old English noun (?; ?).

light — OE *liehtan*

Derivation: **léuxtijaną* > *liehtan* (regular).

Derivation trace

Proto input: **léuxtijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	<i>*lēuxtijaną</i>
[no change]	OE Heavy Syllable Nasal Apocope	<i>*lēuxtijan</i>
Northwest Germanic	OE Secondary Nasalization	<i>*lēuxtijən</i>
[no change]	Sievers Law Syncope	<i>*lēuxtjən</i>
	OE I Umlaut	<i>*liextjən</i>
	OE Weak Tail Reduction	<i>*liextjan</i>
	OE J Loss After Heavy	<i>*liextan</i>

Old English form: *liehtan*

Reconstruction and comparative evidence

Fulk gives Proto-Germanic **liuxtijanan* ‘light’ with Old English *liehtan* ‘illuminate’, and Ringe and Taylor likewise derive West Saxon *liehtan* from the same weak-verb formation (?; ?).

Old English evidence

Clark Hall and Bosworth-Toller preserve the verb family under spellings such as *liehtan*, *lihtan*, and *lihtan*, distinct from the related noun *lēoht* and adjective *leoht/liht* (?; ?).

Development to Old English

From **léuxtijaną* ‘light’, the regular verbal line preserves **xt*, passes through a West Saxon *liehtan* stage, and is represented here by normalized *liehtan*. The word treated in this entry is therefore the verb ‘to light, illuminate’, not the related noun from **leuxtą* ‘light’ (?; ?).

Dialect note

Ringe and Taylor and Campbell distinguish West Saxon *liehtan* from Anglian *lihtan*, while later West Saxon also shows *lyhtan* (?; ?).

linden — OE *lind*

Derivation: **lindō* > *lind* (regular).

Derivation trace

Proto input: **lindō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	<i>*lind</i>
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	<i>*lindu</i>		

Old English form: *lind*

Reconstruction and comparative evidence

Kroonen cites **lindō*- ‘lime tree’ and gives Old English *lind* as the relevant reflex (?).

Old English evidence

Clark Hall and Bosworth-Toller both record *lind* as the noun ‘lime-tree, linden’ (?; ?).

Development to Old English

From **lindō* ‘linden’, Northwest Germanic final **ō* raising first gives a **lindu* ‘linden’ stage, and later high-vowel apocope yields *lind*. The development is therefore straightforward and regular.

Form note

The Old English noun represented here is *lind*. Clark Hall also has a separate adjectival *linden* ‘made of linden-wood’, but that is not the noun counterpart for this entry (?).

milk — OE *meoloc*

Derivation: **mélukz* > *meoloc* (regular).

Derivation trace

Proto input: **mélukz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Med Unstressed U Lowering	<i>*méluk</i>
[no change]		OE Back Mutation	<i>*méoluk</i>
Northwest Germanic			
PGmc Final Z Deletion	<i>*méluk</i>		

Old English form: *meoloc*

Reconstruction and comparative evidence

Kroonen and Orel reconstruct the noun as **meluk-* / **melukz* ‘milk’, and the nominative-style input used here is **mélukz* ‘milk’ (?; ?).

Old English evidence

Old English preserves a mixed dossier for this noun. Ringe and Taylor describe West Saxon *meolc* < *meoluc* < **meluk*, Campbell likewise discusses *meoluc* and *meoloc*, and Anglian shows *milc* (?; ?).

Development to Old English

The unsyncopated line from **mélukz* ‘milk’ loses final **z*, lowers unstressed *u* to *o*, and with back mutation yields *meoloc*. That fuller unsyncopated outcome is the form represented here (?).

Form comparison

Syncopated *meolc* and Anglian *milc* belong to the competing leveled tradition associated with oblique forms, whereas *meoloc* / *meoluc* preserves the fuller nominal shape (?; ?).

mother — OE *mōder*

Derivation: **mōdēr* > *mōder* (regular).

Derivation trace

Proto input: **mōdēr*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel	<i>*mōdær</i>
[no change]		Shortening	
Northwest Germanic		OE Unstressed AE Merger	<i>*mōder</i>
NWGmc Long E Lowering	<i>*mōdēr</i>		

Old English form: *mōder*

Reconstruction and comparative evidence

Kroonen and Orel cite the Proto-Germanic r-stem kinship noun as **mōder-* / **mōdēr* ‘mother’ (?; ?).

Old English evidence

The transmitted Old English headword tradition is *mōdor* / *modor*, with oblique *mēder* in the paradigm. Clark Hall, Campbell, and Ringe and Taylor all preserve that contrast (?; ?; ?).

Development to Old English

From **mōdēr* ‘mother’, the regular suffixal development yields *mōder*. That regular nominative reflex is the form represented here, while the more familiar citation form *mōdor* reflects later levelling within the r-stem paradigm (?; ?).

Form comparison

The note therefore concerns inherited vocalism rather than a different lexeme: *mōder* is the regularized nominative represented here, but dictionaries usually print *mōdor* / *modor*, and the oblique evidence survives in *mēder* (?; ?).

net — OE *nett*

Derivation: **nátjq* > *nett* (regular).

Derivation trace

Proto input: **nátjɑ*

Earlier Germanic changes	Old English changes
West Germanic	Anglo Frisian Brightening <i>*nættjɑ</i>
PWGmc J <i>*nátjɑ</i>	OE Heavy Syllable Nasal Apocope <i>*nættj</i>
Gemination	OE I Umlaut <i>*nettj</i>
Northwest Germanic	OE J Loss After Heavy <i>*nett</i>
[no change]	

Old English form: *nett*

Reconstruction and comparative evidence

Orel gives **natjan* ‘net’ with Old English *nett*, and Fulk’s account of West Germanic gemination before *j* explains the geminate outcome after a short vowel (?; ?).

Old English evidence

Clark Hall and Bosworth-Toller record *nett* as the noun, and Campbell notes that final geminates are often graphically simplified in Old English spelling (?; ?; ?).

Development to Old English

From **nátjɑ* ‘net’, West Germanic j-gemination first gives **nátjɑ* ‘net’. Later brightening, loss of the weak ending, and loss of final *j* after a heavy stem yield *nett*, so the development represented here is regular (?).

Form note

Spellings in *net* can therefore be graphic simplifications, but the lexical target supported by the dictionary evidence is *nett* (?; ?).

nightmare — OE *mare*

Derivation: **mārōn* > *mare* (regular).

Derivation trace

Proto input: **mārōn*

Earlier Germanic changes	Old English changes
West Germanic	Anglo Frisian Brightening <i>*mæ̃rō</i>
[no change]	OE A Restoration <i>*marō</i>
Northwest Germanic	OE Unstressed Long Vowel Shortening <i>*maræ</i>
NWGmc N Stem N <i>*mārō</i>	OE Unstressed AE Merger <i>*mare</i>
Loss	

Old English form: *mare*

Reconstruction and comparative evidence

Ringe and Taylor treat the lexeme as Proto-Germanic / Proto-Northwest-Germanic **mārōn*-, with Old English *mare*, *maran*, and variant *mere*; Orel preserves the same comparative lemma though with a different Old English headword tradition (?; ?).

Old English evidence

Clark Hall records *mare* ‘nightmare, monster’ and also preserves related variant forms *mera* / *mere* (?).

Development to Old English

The selected simplex input **mārōn* ‘nightmare’ regularly gives *mare* after brightening, A-restoration before the n-stem ending, and later reduction of the final vowel. The word represented here is the attested simplex noun, not an attested compound (?).

Form note

The concept corresponds to an unattested compound **nihtmare* ‘nightmare’, but the Old English lexical evidence is for simplex *mare*, with oblique *maran* and variant *mere* / *mera* (?; ?).

coat — OE *rocc*

Derivation: **rúkkaz* > *rocc* (regular).

Derivation trace

Proto input: **rúkkaz*

Earlier Germanic changes		Old English changes
West Germanic		PWGmc Final Bare A <i>*rókk</i>
[no change]		Loss
Northwest Germanic		
NWGmc U Lowering	<i>*rókkaz</i>	
PGmc Final Z Deletion	<i>*rókka</i>	

Old English form: *rocc*

Reconstruction and comparative evidence

Orel cites a masculine **rukaz* ‘coat’ for the garment word, while Kroonen gives **hrukka-*. Both treat this as the garment lexeme and not as the separate stone word (?; ?).

Old English evidence

Clark Hall and Bosworth-Toller record *rocc* as an over-garment or tunic and preserve compounds such as *bisceoprocc* and *breóstrocc* (?; ?).

Development to Old English

With **rúkkaz* ‘coat’ as the derivational input, Northwest Germanic u-lowering and later loss of final *-a* yield *rocc* as a regular outcome.

Source note

This entry concerns the garment noun only. The stone word seen in *stānrocc* belongs to a different lexical history (?; ?).

sheep — OE *scēap*

Derivation: **sképą* > *scēap* (regular).

Derivation trace

Proto input: **skēpq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	<i>*skāep</i>
[no change]		OE Sk Palatalization	<i>*fæp</i>
Northwest Germanic		OE Ws Palatal Diphthongization	<i>*jeap</i>
NWGmc Long E Lowering	<i>*skāepq</i>		

Old English form: *scēap*

Reconstruction and comparative evidence

Ringe and Taylor cite a later West Germanic **skap* > WS *scēap*, while Orel preserves a Proto-Germanic noun of the **skēp-* type for the same lexeme (?; ?).

Old English evidence

Clark Hall records *scēap* with spelling variation, and Campbell likewise lists West Saxon *scēap* among the palatal-diphthongized forms (?; ?).

Development to Old English

From **skēpq* ‘sheep’, Northwest Germanic lowering gives **skāpq* ‘sheep’; after apocope and palatalization the West Saxon branch diphthongizes to *scēap*. The development represented here is therefore fully regular.

Dialect note

Ringe and Taylor contrast West Saxon *scēap* with Mercian and Kentish *scēp*, and Campbell also notes Northumbrian *scip*. The form represented here is the West Saxon headword (?; ?).

shilling — OE *scilling*

Derivation: **skillingaz* > *scilling* (regular).

Derivation trace

Proto input: **skillingaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	<i>*skilling</i>
[no change]		OE Sk Palatalization	<i>*filling</i>
Northwest Germanic		OE Med Unstressed I Lowering ¹	<i>*fīlleng</i>
PGmc Final Z	<i>*skillinga</i>	OE Med Unstressed I Lowering	<i>*fīlling</i>
Deletion			

Old English form: *scilling*

Reconstruction and comparative evidence

Kroonen treats the cognate set under **skellinga-* ~ **skillinga-* and connects it with **skeld-linga-*, while Orel likewise gives the coin word with OE *scilling* among the reflexes (?; ?). The derivational input **skillingaz* ‘shilling’ is the nominative-style form used here to represent that inherited *-ing-* derivative.

Old English evidence

Clark Hall records *scilling*, and Campbell cites it among nouns with unstressed *i* in derivational *-ing* (?; ?). The target is the ordinary OE citation form, normalized as *scilling*.

Development to Old English

From **skillingaz* ‘shilling’, loss of final *-az* yields **skilling* ‘shilling’. Old English palatalization of initial *sk* before front vocalism then gives *scilling*. The *i* of derivational *-ing-* remains, so the regular outcome is *scilling*, not **scilleng* ‘shilling’ (?; ?).

Form note

Kroonen’s **skellinga-* ~ **skillinga-* and his internal analysis **skeld-linga-* belong to the etymological background of the cognate set. The derivational input **skillingaz* ‘shilling’ is the specific form used for the derivation represented here (?).

show — OE *scēawian*

Derivation: **skáwōjaną* > *scēawian* (regular).

Derivation trace

Proto input: **skáwōjaną*

Earlier Germanic changes	Old English changes
West Germanic	OE Aw Long Diphthong <i>*skéawōjaną</i>
[no change]	OE Heavy Syllable Nasal Apocope <i>*skéawōjan</i>
Northwest Germanic	OE Secondary Nasalization <i>*skéawōjən</i>
[no change]	OE Sk Palatalization <i>*jéawōjən</i>
	OE I Umlaut <i>*jéawējən</i>
	OE Unstressed Long Vowel Shortening <i>*jéawejan</i>
	OE Weak Tail Reduction <i>*jéawejan</i>
	OE Intervocalic J Vocalization <i>*jéaweian</i>
	OE Unstressed EI Contraction <i>*jéawian</i>

Old English form: *scēawian*

Reconstruction and comparative evidence

Orel and Kroonen cite a Class II verb of the type **skawōjan-*, with OE *scēawian* among the reflexes (?; ?). Brunner likewise records the Old English family as *scēawian*, *scāwian*, confirming that it belongs to the ordinary show-verb set rather than to a special finite-cell formation (?).

Old English evidence

Bright lists *scēawian* (W. II.) and also the related form *scēawa* (?). The sources therefore use *scēawian*, while *scēawian* supplies a normalized spelling.

Development to Old English

From **skáwōjaną* ‘show’, Old English *aw* before a following vowel yields *ēaw*, and **ō* survives between **w* and **j* in the Class II suffix. The development therefore runs regularly to *scēawian*, without the direct **aw+j* ‘show’ problem seen in other verb types (?; ?).

Form note

The difference between *scēawian* and *scēawian* is orthographic normalization of initial <sc>, not a difference of lexeme or paradigm cell (?; ?).

sleep — OE *slēpan*

Derivation: **slēpanq* > *slēpan* (regular).

Derivation trace

Proto input: **slēpanq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	* <i>slēpan</i>
[no change]		OE Secondary Nasalization	* <i>slēpan</i>
Northwest Germanic		OE Weak Tail Reduction	* <i>slēpan</i>
NWGmc Long E Lowering	* <i>slēpanq</i>		

Old English form: *slēpan*

Reconstruction and comparative evidence

Kroonen preserves the comparative verb as **slēpan*-, and Fulk cites the same family under root **slēb*- (?; ?). The derivational input **slēpanq* ‘sleep’ is the infinitive-style form used here for that inherited sleep-verb.

Old English evidence

Clark Hall gives *slæpan* with preterite *slēp*, *slēap*, and Bright likewise lists *slæpan* (*slāpan*), *slēp* *slēpon* *slēpen* (?; ?). The target represented here is therefore the normalized infinitive *slēpan*, not the preterite forms and not the separate noun *slēp*.

Development to Old English

From **slēpanq* ‘sleep’, Northwest Germanic lowering gives **slēpanq* ‘sleep’. The later OE tail developments then yield *slēpan* regularly. Brunner and Bülbring show that the OE tradition also has variant spellings such as West Saxon *slāpan*/*slæpan* and Anglian or Kentish *slēpan*, but those do not displace the infinitive chosen here (?; ?).

Form note

The note concerns lemma type rather than a special derivational problem: this row represents the verb *slēpan*, whereas *slēp* belongs to noun or lookup background and *slēp*/*slēap* are preterite forms (?; ?).

smear — OE *smierwan*

Derivation: **smérwijanq* > *smierwan* (regular).

Derivation trace

Proto input: **smérwijanq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Breaking	* <i>sméorwijanq</i>
[no change]		OE Heavy Syllable Nasal Apocope	* <i>sméorwijan</i>
Northwest Germanic		OE Secondary Nasalization	* <i>sméorwija</i>
[no change]		Sievers Law Syncope	* <i>sméorwja</i>
		OE I Umlaut	* <i>smierwja</i>
		OE Weak Tail Reduction	* <i>smierwan</i>
		OE J Loss After Heavy	* <i>smierwan</i>

Old English form: *smierwan*

Reconstruction and comparative evidence

Kroonen gives the comparative headword as **smerwjan-* (?). Ringe and Taylor instead cite a later-stage **smirwijana* ‘smear’, from which they derive West Saxon *smierwan*, Mercian *smirwan*, and Northumbrian *smiriga* (?). The derivational input **smérwijaną* ‘smear’ therefore represents the Kroonen-aligned PGmc layer, while the later-stage dialect split belongs to a different chronological level.

Old English evidence

The target represented here is the West Saxon citation form *smierwan*. Campbell’s Anglian discussion explains the contrasting *smirwan*, and Clark Hall, Brunner, and Bright show that the same lexical family later also includes forms such as *smirian*, *smyrian*, and preterite *smyrode* (?; ?; ?).

Development to Old English

From **smérwijaną* ‘smear’, breaking before *r* + *consonant* yields *eo*, and later i-umlaut produces *ie*. The result is West Saxon *smierwan*. Anglian *smirwan* reflects the well-known failure of breaking in this environment, not a different lexeme (?; ?).

Dialect note

The entry therefore represents the West Saxon member of a broader OE family: *smierwan* in West Saxon, *smirwan* in Anglian or Mercian, and related later class-II forms such as *smirian* or *smyrian* in the same lexical field (?; ?).

span — OE *spannan*

Derivation: **spánnaną* > *spannan* (regular).

Derivation trace

Proto input: **spánnaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	<i>*spánnan</i>
[no change]	OE Secondary Nasalization	<i>*spánnan</i>
Northwest Germanic	OE Weak Tail Reduction	<i>*spánnan</i>
[no change]		

Old English form: *spannan*

Reconstruction and comparative evidence

Kroonen cites the inherited verb as **spannan-*, with OE *spannan* among the reflexes (?). The derivational input **spánnaną* ‘span’ is the infinitive-style form used here for that same verbal lexeme.

Old English evidence

Clark Hall lists noun *spann* and verb *spannan* as separate headwords, and Brunner likewise records *sponnan*, *spannan stv.* (?; ?). The target is the strong-verb infinitive, not the noun.

Development to Old English

From **spánnanq* ‘span’, the final nasal ending is lost and the regular OE weak-tail steps surface *spannan*. No paradigm-cell substitution is needed: the current derivation already lands on the infinitive directly.

Form note

English *span* can also reach the noun *spann* in local lookup material. The form represented here is the verb *spannan*, with the noun treated elsewhere (?).

spar — OE *spearra*

Derivation: **spárrô* > *spearra* (regular).

Derivation trace

Proto input: **spárrô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	<i>*spærrô</i>
[no change]	OE Breaking	<i>*spearrô</i>
Northwest Germanic	OE Unstressed Long Vowel Shortening	<i>*spearra</i>
[no change]		

Old English form: *spearra*

Reconstruction and comparative evidence

Kroonen and Orel place this noun in the beam or rafter set **spar(r)an-*, with cognates such as Old Saxon and Old High German *sparro* (?; ?). The derivational input **spárrô* ‘spar’ is the OE-facing nominal form used here for that same lexeme.

Old English evidence

The noun is *spearra*. English gloss overlap also reaches the unrelated verb *sperran* ‘to bar’, which does not belong to this row.

Development to Old English

From **spárrô* ‘spar’, Anglo-Frisian brightening gives **spærrô* ‘spar’, and OE breaking before geminate *rr* yields **spearrô* ‘spar’, later *spearra*. The development is therefore regular for a breaking-conditioned noun of this type (?).

Form note

This entry concerns the noun *spearra* only. It should be kept separate from verb *sperran*, even though the Modern English glosses overlap (?; ?).

still — OE *stillan*

Derivation: **stéllijanq* > *stillan* (regular).

Derivation traceProto input: **stéllijanq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	<i>*stéllijan</i>
[no change]	OE Secondary Nasalization	<i>*stéllijan</i>
Northwest Germanic	Sievers Law Syncope	<i>*stélljan</i>
[no change]	OE I Umlaut	<i>*stilljan</i>
	OE Weak Tail Reduction	<i>*stilljan</i>
	OE J Loss After Heavy	<i>*stillan</i>

Old English form: *stillan***Reconstruction and comparative evidence**

The wider West Germanic family includes adjective *still* and verb *stillen* (?). The derivational input **stéllijanq* ‘still’ represents the verbal j-formation used for the OE row.

Old English evidence

Clark Hall gives *stillan* as the verb and separately *stille* as the adjective (?). Bosworth-Toller likewise preserves a substantial prefixed verbal family under *ge-stillan* and related forms (?). The Old English form here is the verb *stillan*, not the adjective.

Development to Old English

As a heavy-stem Class I weak verb, **stéllijanq* ‘still’ undergoes the expected syncope and i-umlaut, and later loss of *j* after a heavy stem yields *stillan*. The development represented here is regular.

Form note

The note concerns lexical framing rather than sound law: *stillan* is the verb represented here, while *stille* belongs to the related adjectival branch of the family (?; ?).

summer — OE *sumer*Derivation: **súmaraz* > *sumer* (regular).**Derivation trace**Proto input: **súmaraz*

Earlier Germanic changes	Old English changes	
West Germanic	PWGmc Final Bare A Loss	<i>*súmar</i>
[no change]	Anglo Frisian Brightening	<i>*súmæ̃r</i>
Northwest Germanic	OE Unstressed AE Merger	<i>*súmer</i>
PGmc Final Z Deletion		<i>*súmarā</i>

Old English form: *sumer***Reconstruction and comparative evidence**

Kroonen gives the lexeme as **sumara-*, and Ringe and Taylor likewise use **sumaraz* ‘summer’, while Orel preserves an alternate **sumeraz* ‘summer’ (?; ?; ?). The derivational input **súmaraz* ‘summer’ follows the **a* vocalism that underlies the regular development represented here.

Old English evidence

Clark Hall gives *sumor* *m.*, *gs.* *sumeres*, *ds.* *sumera*, *sumere*, and Bright likewise lists *sumor* (*sumer*) with genitive *sumeres* (?; ?). The tradition therefore preserves both *sumor* and *sumer*, with the oblique forms strongly supporting second-syllable *e*.

Development to Old English

From **súmaraz* ‘summer’, loss of final *-az* is followed by fronting and merger in the unstressed second syllable, yielding *sumer*. The form compared here is the regularized *e*-form, while the common citation form *sumor* remains part of the attested OE tradition (?).

Form note

The entry does not deny *sumor*. It represents *sumer* as the regular outcome chosen here, while *sumor* remains a common headword spelling and *sumeres/sumere* show that the *e*-vocalism was also real in Old English (?; ?).

sunder — OE *sundrian*

Derivation: **súndrōjaną* > *sundrian* (regular).

Derivation trace

Proto input: **súndrōjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	<i>*súndrōjan</i>
[no change]	OE Secondary Nasalization	<i>*súndrōjan</i>
Northwest Germanic	OE I Umlaut	<i>*súndrējan</i>
[no change]	OE Unstressed Long Vowel Shortening	<i>*súndrejan</i>
	OE Weak Tail Reduction	<i>*súndrejan</i>
	OE Intervocalic J Vocalization	<i>*súndreian</i>
	OE Unstressed EI Contraction	<i>*súndrian</i>

Old English form: *sundrian*

Reconstruction and comparative evidence

Orel distinguishes three related formations: adverbial **sunþraz* > *sundor*, Class I verbal **sunþrjanan* > *syndrian*, and Class II verbal **sunþrōjanan* > *sundrian* (?). Kluge-Seebold aligns the cognate set with German *sondern* and OE *gesundrian*, so this entry belongs with the Class II verb, not the adverb (?).

Old English evidence

Clark Hall and Bosworth-Toller list *sundrian* and *syndrian* separately from adverbial *sundor*; both also record the prefixed verbal family *ā-sundrian* (?; ?). The target is therefore the weak verb *sundrian*.

Development to Old English

From **súndrōjaną* ‘sunder’, the Class II weak-verb suffix yields regular OE *-ian*, producing *sundrian*. Because this is the **-ōjan-* verb and not the Class I **-jan-* formation, the form represented here does not belong to the umlauted *syndrian* branch.

Form note

The earlier confusion was lexical, not phonological: *sundor* is the separate adverb, and *syndrian* is a related but different verb. The verb treated here is the Class II verb *sundrian* (?; ?).

swallow — OE *swealwe*

Derivation: **swálwōn* > *swealwe* (regular).

Derivation trace

Proto input: **swálwōn*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	* <i>swælwō</i>
[no change]		OE Breaking	* <i>swealwō</i>
Northwest Germanic		OE Unstressed Long Vowel Shortening	* <i>swealwæ</i>
NWGmc N Stem N	* <i>swálwō</i>	OE Unstressed AE Merger	* <i>swealwe</i>
Loss			

Old English form: *swealwe*

Reconstruction and comparative evidence

Kroonen gives the bird name as **swalwōn*-, and Ringe and Taylor cite the later West Germanic stage **swalwa* ‘swallow’, from which West Saxon *swealwe* and Mercian *swalwe* develop (?; ?). The selected etymological comparison belongs to the swallow-bird family, not to the verb *swelgan*.

Old English evidence

Clark Hall records *swealwe* (*a, o*) as the noun headword (?). Campbell and Brunner also preserve later or oblique-family forms such as *swaluwe*, *swalewan*, and *swealuwe*, but those belong to wider variation around the noun rather than to the citation form represented here (?; ?).

Development to Old English

From **swálwōn* ‘swallow’, brightening yields **swælw-*, and breaking before *lw* gives **swealw-*. The later noun ending develops regularly to *swealwe*. The relevant point is that the bird name has no inherited **g*: that consonant belongs to the separate verb *swelgan* (?; ?).

swine — OE *swīn*

Derivation: citation reconstruction **swīnq*; form followed here **swīnq* > *swīn* (regular).

Derivation trace

Proto input: **swīnq*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	* <i>swīn</i>
[no change]			
Northwest Germanic			
[no change]			

Old English form: *swīn*

Reconstruction and comparative evidence

Kroonen cites the noun as **swina*- ‘pig’ (?). The selected comparative form here is the bare neuter citation cell **swīnq* ‘swine’, which matches the singular noun aimed at in Old English rather than an oblique stem form.

Old English evidence

Clark Hall records *swin* (y) as the ordinary noun headword (?). The target here is that singular citation form *swīn*, not a plural glossed in Modern English as *swine*.

Development to Old English

From derivational input **swīnq* ‘swine’, loss of the final nasal vowel yields *swīn*. The outcome is therefore the regular monosyllabic noun with preserved long root *ī*.

Source note

The derivational input writes stressed long *ī* as **ī*, so comparative **swīnq* ‘swine’ and derivational **swīnq* ‘swine’ represent the same lexical form.

think — OE *þencan*

Derivation: **θánkijanq* > *þencan* (regular).

Derivation trace

Proto input: **θánkijanq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	* <i>θánkijan</i>
[no change]	OE Secondary Nasalization	* <i>θánkijān</i>
Northwest Germanic	Sievers Law Syncope	* <i>θánkjan</i>
[no change]	OE Velar Palatalization	* <i>θánċċjan</i>
	OE I Umlaut	* <i>θentċċjan</i>
	OE Weak Tail Reduction	* <i>θentċjan</i>
	OE J Loss After Heavy	* <i>θentċan</i>

Old English form: *þencan*

Reconstruction and comparative evidence

Kroonen gives the verb as **þankjan*- ‘to think’, and Ringe and Taylor cite fully inflected **þanki-janq* ‘think’ beside OE *þencan* (?; ?). The noun **þankaz* ‘think’ belongs only to the wider derivational background.

Old English evidence

Bosworth-Toller preserves the verb under *þencan/geþencan*, and the citation form here is the ordinary infinitive *þencan* (?).

Development to Old English

From **θánkijanq* ‘think’, palatalization before **j* and i-umlaut produce *þencan*. The infinitive is therefore a straightforward weak-verb outcome.

Lexical note

Campbell's assibilation discussion uses the same verb *þencan*; the class-III relic *hycgan* is a different lexeme (?; ?).

thorn — OE *þorn*

Derivation: **θúrnaz* > *þorn* (regular).

Derivation trace

Proto input: **θúrnaz*

Earlier Germanic changes		Old English changes
West Germanic		PWGmc Final Bare A * <i>θórn</i>
[no change]		Loss
Northwest Germanic		
NWGmc U Lowering	* <i>θórnaz</i>	
PGmc Final Z Deletion	* <i>θórna</i>	

Old English form: *þorn*

Reconstruction and comparative evidence

Kroonen gives **þurna-* ‘thorn, briar’, while Orel preserves the masculine pair **þurnuz* **þurnaz* ‘thorn’ (?; ?). The derivational input **θúrnaz* ‘thorn’ belongs to that same comparative family.

Old English evidence

Bright lists *þorn*, m., and Clark Hall likewise treats *þorn* as the ordinary noun headword (?; ?).

Development to Old English

The inherited stem shows regular lowering of *u* to *o* before *r*, and final loss yields *þorn*. The noun is therefore a regular Old English continuation of the Proto-Germanic thorn-family.

Source note

The comparative sources preserve more than one stem formation, but the Old English target itself is simply the citation form *þorn*.

tide — OE *tīd*

Derivation: citation reconstruction **tídiz*; form followed here **tídiz* > *tīd* (regular).

Derivation trace

Proto input: **tídiz*

Earlier Germanic changes		Old English changes
West Germanic		OE High Vowel Apocope * <i>tíd</i>
[no change]		
Northwest Germanic		
PGmc Final Z Deletion	* <i>tídi</i>	

Old English form: *tīd*

Reconstruction and comparative evidence

Kroonen gives **tīdi-* ‘time’, and Orel’s **tīdiz* ‘tide’ points to the same feminine noun (?; ?). The related verb *tīdan* is separate.

Old English evidence

Bright records *tīd* with singular *tīde* and plural *tīda*, and Clark Hall treats *tīd* as the ordinary noun ‘time, period, season’ (?; ?).

Development to Old English

From **tīdiz* ‘tide’, final *z* is lost and the high final vowel drops, leaving *tīd*. The development is straightforward for a feminine *i*-stem.

Lexical note

English *tide* can also lead to the separate weak verb *tīdan*; the noun is *tīd*.

token — OE *tācn*

Derivation: **táiknq* > *tācn* (regular).

Derivation trace

Proto input: **táiknq*

Earlier Germanic changes	Old English changes
West Germanic	OE Heavy Syllable Nasal Apocope
PWGmc <i>Ai</i>	<i>*tākn</i>
Monophthongization	
Northwest Germanic	
[no change]	

Old English form: *tācn*

Reconstruction and comparative evidence

Kroonen cites **taikna-* and Orel **taiknan* ‘token’ for the noun ‘sign, token’ (?; ?). The derivational input **táiknq* ‘token’ is the simple citation-form noun used for the derivation.

Old English evidence

Campbell and Sievers-Brunner preserve both unbroken *tācn* and broken *tācen*, with oblique *tācnes* remaining unbroken (?; ?).

Development to Old English

Monophthongization of *ai* yields *ā*, and loss of the final nasal vowel leaves *tācn*. The unbroken form is therefore a regular Old English outcome.

Form note

tācn is the attested unbroken citation form selected here. Later West Saxon prose often prefers *tācen*, but that does not displace the older unbroken form (?; ?).

town — OE *tūn*

Derivation: **tūnq* > *tūn* (regular).

Derivation trace

Proto input: **tūnq*

Earlier Germanic changes	Old English changes
West Germanic	OE Heavy Syllable Nasal Apocope
[no change]	<i>*tūn</i>
Northwest Germanic	
[no change]	

Old English form: *tūn*

Reconstruction and comparative evidence

Kroonen cites **tūna-* ‘fenced area’, while Orel gives **tūnan* **tūnaz* ‘town’ (?; ?). The derivational input **tūnq* ‘town’ is the simple citation-form noun used in the derivation.

Old English evidence

Clark Hall records *tūn* as the ordinary headword ‘enclosure, yard, village, town’ (?).

Development to Old English

The inherited long *ū* is preserved, and loss of the final nasal vowel yields *tūn* regularly (?).

Source note

The comparative headwords vary, but the Old English target here is the direct citation form *tūn*, not an oblique **tūnǣn* ‘town’.

wade — OE *wadan*

Derivation: **wādanq* > *wadan* (regular).

Derivation trace

Proto input: **wādanq*

Earlier Germanic changes	Old English changes
West Germanic	Anglo Frisian Brightening
[no change]	OE A Restoration
Northwest Germanic	OE Heavy Syllable Nasal Apocope
[no change]	OE Secondary Nasalization
	OE Weak Tail Reduction

Old English form: *wadan*

Reconstruction and comparative evidence

Campbell and Ringe and Taylor describe A-restoration before a following back vowel, and Luick explicitly includes *wadan* among the standard open-syllable examples (?; ?; ?).

Old English evidence

Clark Hall gives *wadan* as the verb ‘to go, move, stride, advance’, and Bright lists the same infinitive in the strong-verb paradigm (?; ?).

Development to Old English

From **wádanq* ‘wade’, Anglo-Frisian brightening first gives **wædanq* ‘wade’. A-restoration before the back-vocalic infinitive ending then returns *a*, and later reduction yields *wadan* (?; ?).

Development note

The infinitive belongs to the A-restoration class. The citation form is therefore *wadan*, not a fronted *wæden*-type output.

warp — OE *weorpan*

Derivation: **wérpanq* > *weorpan* (regular).

Derivation trace

Proto input: **wérpanq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	<i>*wéorpanq</i>
[no change]	OE Heavy Syllable Nasal Apocope	<i>*wéorpan</i>
Northwest Germanic	OE Secondary Nasalization	<i>*wéorpan</i>
[no change]	OE Weak Tail Reduction	<i>*wéorpan</i>

Old English form: *weorpan*

Reconstruction and comparative evidence

Ringe and Taylor distinguish preterite **warp* ‘warp’ from infinitive **werpana* ‘warp’, and the derivational input here is the verbal form **wérpanq* ‘warp’ (?).

Old English evidence

Clark Hall records *weorpan* as the strong verb headword and separately lists *wearp* as both noun and preterite. Bright gives the paradigm *weorpan*, *wearp*, *wurpon*, *worpen* (?; ?).

Development to Old English

Breaking before *r* + *C* yields *weor-*, and the infinitive develops regularly to *weorpan* (?; ?).

Lexical note

English *warp* also points to related *wearp* material. The target is the infinitive *weorpan*.

wash — OE *wascan*

Derivation: **wáskanq* > *wascan* (regular).

Derivation traceProto input: **wáskanq*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	<i>*wæskanq</i>
[no change]	OE A Restoration	<i>*waskanq</i>
Northwest Germanic	OE Heavy Syllable Nasal Apocope	<i>*waskan</i>
[no change]	OE Secondary Nasalization	<i>*waskǫn</i>
	OE Weak Tail Reduction	<i>*waskan</i>

Old English form: *wascan***Reconstruction and comparative evidence**

Kroonen cites **waskan-*, Orel **waskanān* ‘wash’, and Ringe and Taylor likewise derive Old English *wascan* from the same verb family (?; ?; ?).

Old English evidence

Clark Hall heads the verb as *wascan*, while Sievers-Brunner also notes the variant *wæscan* (?; ?).

Development to Old English

From **wáskanq* ‘wash’, brightening gives **wæskanq* ‘wash’. A-restoration before the *sC* cluster restores *a*, and medial *sc* remains unpalatalized before the following back vowel, yielding *wascan* (?; ?).

Form note

The conservative citation form *wascan* is selected here. Spellings such as *wæscan* or *wascan* belong to variant or normalized background rather than to the target of this entry.

wax — OE *weaxan*

Derivation: **wáxsanq* > *weaxan* (regular).

Derivation traceProto input: **wáxsanq*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	<i>*wæxsanq</i>
[no change]	OE Breaking	<i>*weaxsanq</i>
Northwest Germanic	OE Heavy Syllable Nasal Apocope	<i>*weaxsan</i>
[no change]	OE Secondary Nasalization	<i>*weaxsqn</i>
	OE Weak Tail Reduction	<i>*weaxsan</i>
	OE Xs Merge	<i>*weaxSan</i>

Old English form: *weaxan***Reconstruction and comparative evidence**

Kroonen cites the verb as **wahs(j)an-*, Orel as **waxsanān* ‘wax’, and Ringe and Taylor discuss the prehistory of Old English *weaxan* within the same verbal family (?; ?; ?).

Old English evidence

Clark Hall gives *weaxan* as the verb headword and separately records bare *wax* as a preterite form; Bright likewise treats *weaxan* as the infinitive (?; ?).

Development to Old English

From **wáxsanq* ‘wax’, brightening and breaking yield *weax-*, and the infinitive develops regularly to *weaxan*. The cluster is preserved here, since *xs > s* belongs to forms where another consonant follows, such as *wæstm*, not to the infinitive itself (?; ?).

Lexical note

The target here is the infinitive *weaxan*. Noun *weax* and preterite *wax/wēox* belong to different lexical or paradigm slots.

way — OE *weg*

Derivation: **wégaz > weg* (regular).

Derivation trace

Proto input: **wégaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	<i>*wég</i>
[no change]		OE Velar Palatalization	<i>*wéǵ</i>
Northwest Germanic			
PGmc Final Z Deletion	<i>*wéga</i>		

Old English form: *weg*

Reconstruction and comparative evidence

Kroonen cites the noun as **wega-* ‘way, road’, while the selected derivational form here is nominative-singular **wégaz* ‘way’ (?). Campbell, Hogg, and Ringe and Taylor use the same word as the standard contrast between singular palatal *weg* and inflected *wegas/wegum* (?; ?; ?).

Old English evidence

The Old English singular is the ordinary noun *weg*, here normalized as *weg* to show the palatal final. The contrasting plural and oblique forms *wegas*, *wegum* keep a velar stop before the following back vowel (?; ?).

Development to Old English

From **wégaz* ‘way’, final **z* is lost and the weak tail apocopates, leaving word-final **g* after a front vowel. In that environment Old English palatalization yields *wéǵ*, whereas *wegas* remains velar because the following *a* blocks the same outcome (?; ?; ?).

Form note

Normalized *weg* and dictionary *weg* represent the same noun. *wē* is not supported in the checked Old English evidence for ‘way’.

weapon — OE *wāpn*

Derivation: **wēpną* > *wāpn* (regular).

Derivation trace

Proto input: **wēpną*

Earlier Germanic changes	Old English changes
West Germanic	OE Heavy Syllable Nasal Apocope * <i>wāpn</i>
[no change]	
Northwest Germanic	
NWGmc Long E Lowering * <i>wāpną</i>	

Old English form: *wāpn*

Reconstruction and comparative evidence

Kroonen reconstructs a double-stem noun **wēbna-* ~ **wēpna-* and cites OE *wāpn* among its reflexes (?). The derivational input **wēpną* ‘weapon’ represents the unbroken citation-form noun rather than the later broken simplex.

Old English evidence

Campbell’s cluster-noun discussion preserves unbroken *wēpn* beside broken *wēpen*-type forms (?; ?). Bright contrasts broken nominative *wāpen/wapen* with unbroken oblique *wāpnes*, while Clark Hall lemmatizes the noun under *wapen* and also preserves unbroken forms in compounds and related spellings (?; ?).

Development to Old English

Northwest Germanic lowering gives *wāpn*, and loss of the final nasal vowel leaves the unbroken cluster word-finally. The Old English form here is the attested unbroken form *wāpn*.

Form note

The ordinary late West Saxon simplex headword is *wāpen*, but Campbell’s noun-class discussion also preserves unbroken *wēpn* beside broken *wēpen*-type forms (?; ?). *wāpnes* remains the regular unbroken oblique comparator (?; ?).

will — OE *willa*

Derivation: **wéljô* > *willa* (regular).

Derivation trace

Proto input: **wéljô*

Earlier Germanic changes	Old English changes
West Germanic	OE I Umlaut * <i>willjô</i>
PWGmc J * <i>wélljô</i>	OE Unstressed Long Vowel Shortening * <i>willja</i>
Gemination	OE J Loss After Heavy * <i>willa</i>
Northwest Germanic	
[no change]	

Old English form: *willa*

Reconstruction and comparative evidence

Kroonen separates noun **weljan-* 2 ‘will, wish’ from verb **weljan-* 1 ‘to want’, while Orel and Kluge represent the noun as **weljōn* ‘will’ (?; ?; ?). The selected derivational form **wéljô* ‘will’ is the noun-side input used for this row.

Old English evidence

Clark Hall lemmatizes noun *willa* *m.* separately from verb *willan* (?). The Old English form here is the noun citation form, not the related verb.

Development to Old English

From **wéljô* ‘will’, j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa*. The noun is therefore a regular weak masculine outcome.

Lexical note

The target here is the noun *willa* ‘will, wish’. Related verb *willan* belongs to a separate lexeme and should not be substituted for the noun row (?; ?).

wind — OE *windan*

Derivation: **wíndanq* > *windan* (regular).

Derivation trace

Proto input: **wíndanq*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	<i>*wíndan</i>
[no change]	OE Secondary Nasalization	<i>*wíndqn</i>
Northwest Germanic	OE Weak Tail Reduction	<i>*wíndan</i>
[no change]		

Old English form: *windan*

Reconstruction and comparative evidence

Kroonen distinguishes noun **winda-* from verb **windan-*, and the present row belongs to the verb (?). Fulk and Ringe and Taylor derive the dental directly from PIE **wendh-*, not from a Verner alternant (?; ?).

Old English evidence

Clark Hall and Bosworth-Toller record *windan* as the verb headword (?; ?). The Old English form here is the ordinary infinitive of the strong verb.

Development to Old English

The derivational input **wíndanq* ‘wind’ yields the regular infinitive *windan* by ordinary heavy-syllable apocope and weak-tail reduction. The form is therefore a straightforward strong-verb outcome.

Lexical note

English *wind* also names the noun. This row represents the class-III verb, not the noun (?; ?).

wold — OE *weald*

Derivation: **wálθuz* > *weald* (regular).

Derivation trace

Proto input: **wálθuz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	* <i>wældu</i>
PWGmc L Th Voicing	* <i>wálduz</i>	OE Breaking	* <i>wealdu</i>
Northwest Germanic		OE High Vowel Apocope	* <i>weald</i>
PGmc Final Z Deletion	* <i>wáldu</i>		

Old English form: *weald*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **walpu-* and gives OE *weald* beside other West Germanic *wald* forms (?). The derivational input **wálθuz* ‘wold’ is the nominative singular used for the derivation.

Old English evidence

Clark Hall makes *weald* the main noun headword and cross-refers *wald* and *wold* to it (?). The Anglian-looking *wald* therefore remains variant background rather than the main target.

Development to Old English

**lþ* voices to *ld*, Anglo-Frisian brightening yields *wæld-*, and breaking before the cluster gives *weald-*; apocope then yields *weald*. The noun is therefore a regular breaking outcome.

Dialect note

wald survives as an Anglian-type variant in the same family. The Old English form here is normalized *weald*, not the variant form (?; ?).

yarn — OE *ġearn*

Derivation: **gárna* > *ġearn* (regular).

Derivation trace

Proto input: **gárna*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	* <i>gærna</i>
[no change]		OE Breaking	* <i>ġearnā</i>
Northwest Germanic		OE Heavy Syllable Nasal Apocope	* <i>ġearn</i>
[no change]		OE Velar Palatalization	* <i>ġearn</i>

Old English form: *ġearn*

Reconstruction and comparative evidence

Kroonen cites the noun as **garna-*, and Ringe and Taylor give the early chain **garna* > **geern* > **gearn* > OE *gearn* (?; ?). The derivational input **gárna* ‘yarn’ is the nominal citation form used here, while oblique **garnǎn* ‘yarn’ belongs only to comparative background.

Old English evidence

Clark Hall records *gearn* (*e*) *n.* ‘yarn, spun wool’, and Bosworth-Toller glosses *gearn* as *filatum* (?; ?).

Development to Old English

From **gárna* ‘yarn’, brightening and breaking before *rn* yield *gearn*; palatalization of initial *g* before the resulting front-vocalic sequence gives normalized *ġearn*. The derivation is regular.

Form note

Dictionary *gearn* and normalized *ġearn* refer to the same noun. The comparative stem **garna-* and oblique **garnǎn* ‘yarn’ do not replace the derivational input **gárna* ‘yarn’.

3.6 Attested variants and comparison forms

Here the Old English comparator belongs to a documented set of variants. The variation is part of the evidence and not an inconvenience to be normalized away.

cud — OE *cwedu*

Derivation: citation reconstruction **kwíθuz*; form followed here **kwéðuz* > *cwedu* (attested variant).

Derivation trace

Proto input: **kwéðuz*

Earlier Germanic changes		Old English changes
West Germanic		[no change]
PWGmc Dental Hardening	<i>*kwéðuz</i>	
Northwest Germanic		
PGmc Final Z Deletion	<i>*kwédu</i>	

Old English form: *cwedu*

Reconstruction and comparative evidence

Kroonen reconstructs the resin word as **kwedu-2* ‘cud’ and gives Old English variants *cwidu*, *cweodu*, and *c(w)udu* (?). Orel likewise lists *cwidu* under the cognate set (?). The derivational input **kwéðuz* ‘cud’ therefore represents the older e-grade, voiced-dental form behind the chosen variant *cwedu*.

Old English evidence

The Old English word survives in a wider variant set than one dictionary headword suggests. Ringe and Taylor discuss *cwidu* > *cwudu* > *cudu* and also note late West Saxon *cweodu*; Clark Hall gives *cwudu*, *cweodu*, and *cudu* (?; ?). Attested *cwedu* is treated here as the conservative variant within that set.

Development to Old English

From **kwéðuz* ‘cud’, the West Germanic voiced dental hardens in the expected way and the regular Old English development yields *cwedu*. The other Old English spellings belong to the same lexical family, but reflect later leveling, back-umlaut, or further reduction rather than a need to replace the selected input.

Variant comparison

Variant type	Old English form	Comment
conservative target	<i>cwedu</i>	selected attested variant represented here
leveled i-grade form	<i>cwidu</i>	common lexical variant in the same family
back-umlauted forms	<i>cweodu</i> , <i>cwudu</i>	later developments within the same OE tradition
reduced form	<i>cudu</i>	further reduced member of the same variant set

ten — OE *tēon*

Derivation: **téxun* > *tēon* (attested variant).

Derivation trace

Proto input: **téxun*

Earlier Germanic changes	Old English changes	
West Germanic	OE Med Unstressed U Lowering	<i>*téxon</i>
[no change]	OE Breaking	<i>*téoxon</i>
Northwest Germanic	OE H Loss	<i>*téoon</i>
[no change]	OE Contraction	<i>*téon</i>

Old English form: *tēon*

Reconstruction and comparative evidence

**tēon* ‘ten’ is the bare cardinal form. Fulk states that Old English *tien* shows umlaut from the inflected forms, whereas the uninflected form without umlaut is reflected in *tēon* (?). Brunner gives the same contrast more broadly: **tēon* ‘ten’ develops from **tēhun* ‘ten’, while West Saxon *tien*, *tȳn* belong to a different, umlauted branch of the numeral history (? Anm. 6, 234).

The comparison form *tēon* therefore represents the bare cardinal’s un-umlauted line, not the later umlauted simplex tradition.

Old English evidence

The attested simplex forms are varied. Campbell gives *tien*, north-western West Saxon *tēn*, and late Northumbrian *tēo*, *tēa* (?). Brunner likewise lists West Saxon *tien*, *tȳn* beside *tēn*, *tēo*, *tēa* in other dialects (?).

Exact simplex *tēon* is weaker as a directly cited headword than those spellings. The un-umlauted stem is, however, explicit in *tēoða* and *tēontig* (?; ?). The comparison form *tēon* is therefore a normalized spelling of that un-umlauted base.

Development to Old English

From **téxun* ‘ten’, lowering of medial unstressed *u* gives **téxon* ‘ten’, breaking gives **téoxon* ‘ten’, loss of intervocalic *h/x* yields **téoon* ‘ten’, and contraction produces **téon* ‘ten’, written *tēon*. This is the regular bare-cardinal path.

The umlauted forms *tien* / *tīen* belong to a different branch, created when the numeral was levelled from inflected forms with a front-vocalic trigger (?; ? Anm. 6).

Variant comparison

The comparison below sets the relevant forms side by side. It distinguishes the normalized un-umlauted base from the attested simplex variants.

Form or branch	Status	Relevance to this entry
<i>tēon</i>	normalized un-umlauted comparison form; trace-supported	Old English form here
<i>tien</i> / <i>tīen</i>	attested West Saxon umlauted simplex forms	genuine OE variants, but not the bare-cardinal line modeled here